When Careers Stall: Early-Career Job Insecurity and Fertility over the Life Cycle

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Abstract

Can greater exposure to job insecurity at labor-market entry have lasting effects on fertility? We study Spain's 1984 reform, which sharply expanded fixed-term hiring for new entrants while preserving protection for permanent workers. Using differences in reform exposure across nearby birth cohorts of low- and high-skilled women, we find that the reform increased years of non-permanent employment and lowered contribution-based income among low-skilled women, without a detectable change in days worked. Exposed low-skilled cohorts experienced a 15.2% decline in births per woman at the cohort level and delayed first motherhood by about five months. Mechanism evidence suggests that employment insecurity constrains the realization of fertility plans, with exposed low-skilled women more likely to have fewer children than they desire. We build and estimate a life-cycle model linking early-career employment risk to fertility decisions over the reproductive horizon. Within the model, post-entry employment instability and accumulated insecurity account for most of the fertility decline, while reducing insecurity permits some births to be recovered by age 40. Together, the evidence shows that reforms increasing early-career job insecurity can delay motherhood and reduce realized fertility.

Keywords: temporary employment, fertility, infant health, life-cycle model
JEL Codes: I12, I14, I18, J13, J16, J24

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1. INTRODUCTION

Fertility has fallen across most high-income countries in recent decades, with explanations emphasizing changes in economic constraints, gender norms, and the challenges of combining careers and children (Kearney, Levine and Pardue 2022; Goldin 2025; Doepke et al. 2023).[1] Rising childlessness, later first births, and smaller completed families have accompanied women's increasing attachment to paid work, raising questions about how the employment conditions women face shape their decisions about parenthood (Kearney and Levine 2025). Spain illustrates these parallel changes: its total fertility rate fell from 2.77 births per woman in 1976 to 1.10 in 2024, while the female activity rate at ages 16 and above rose from 28.5% in 1976 to 54.4% in 2025 (Instituto Nacional de Estadística INE,I,I).[2] Although women's growing attachment to the labor market has received considerable attention, less is understood about how the contractual terms and remuneration of the jobs through which they begin their careers shape fertility over the life cycle. This paper asks whether greater exposure to job insecurity at labor-market entry has lasting effects on fertility and infant health.

Economic theory suggests that the relationship between women's employment and fertility depends on both the remuneration and the stability of their jobs. Higher wages and stronger career opportunities can raise the opportunity cost of time devoted to children, but they also relax household budget constraints and make desired children easier to afford (Becker 1960, 1965; Schultz 1985; Hotz and Miller 1988; Heckman and Walker 1990; Black et al. 2013). Fixed-term jobs may accelerate entry into paid work while offering shorter tenure and weaker earnings growth, potentially limiting credit access and employment security around childbirth. Existing evidence shows that fertility responds to employment uncertainty, gender-specific labor demand, and both partners' labor-market prospects, with effects that vary by gender, education, and the type of insecurity (Adserà 2011; Del Bono, Weber and Winter-Ebmer 2012; Prifti and Vuri 2013; Clark and Lepinteur 2022; Auer and Danzer 2015; Vignoli, Mencarini and Alderotti 2020; Vignoli, Drefahl and De Santis 2012; Huttunen and Kellokumpu 2016; Alderotti et al. 2021; Schaller 2016; Autor, Dorn and Hanson 2019). Recent work also emphasizes changing norms, household bargaining, and the interaction between careers and family life (Goldin 2021; Doepke and Kindermann 2019; Doepke et al. 2023; Kearney and Levine 2025). What remains less clear is whether entering the labor market through non-permanent and lower-paid employment has lasting consequences for fertility, affecting both when women have children and how many they ultimately have.

[1] The decline in total fertility rates across many OECD countries began in the mid-1980s (Goldin 2025); Appendix Figures A1.1 and A1.2 document its persistence.

[2] The INE provisionally estimates 321,164 births in 2025, 1.0% more than in 2024, but has not yet published a final 2025 total fertility rate (Instituto Nacional de Estadística INE).

Contractual instability can affect both the number and timing of births, as women may postpone childbearing while seeking a more secure job and a more predictable income. When employment conditions improve, women may be able to realize these postponed fertility plans; when insecurity persists, however, some intended births may never occur. A finite reproductive horizon can make delays difficult to recover, even when desired family size remains unchanged. The consequences of insecure employment at career entry may therefore extend beyond an initial delay in motherhood to a lasting gap between desired and realized fertility.

Because workers select into temporary employment, differences in fertility by observed contract status need not reflect the causal effects of contractual instability. To study these effects, we exploit a labor-market reform that sharply expanded employers' ability to hire workers on fixed-term contracts while preserving protection for incumbent permanent employees. Implemented in Spain in 1984, the reform changed the contractual conditions facing new entrants and was followed by a rapid expansion of temporary employment, which accounted for 2.2% of employees aged 15–64 in 1983 and 29.9% in 1990.

The design exploits differences in expected labor-market entry and exposure to temporary employment across skill groups. Low-skilled women were particularly exposed to the expansion of fixed-term hiring, making their early careers more sensitive to the contractual conditions facing new entrants. We assign expected labor-market entry at age 16 for low-skilled women and age 18 for high-skilled women, reflecting differences in schooling and the timing of entry into paid work.[3] Using birth-cohort variation in assigned reform exposure within each group, we compare changes in fertility outcomes among low-skilled women with the corresponding changes among high-skilled women. The high-skilled group therefore provides a comparison for assessing the differential consequences of reform exposure, rather than an entirely unaffected control group. A causal interpretation requires that, absent the reform, the difference in outcomes between the two groups would have evolved smoothly across the assigned exposure cutoffs, conditional on the specified cohort trends and geographic controls. The difference-in-differences design follows the labor-market-entry logic of Oreopoulos, von Wachter and Heisz (2012) and the Spanish reform design of García-Pérez, Marinescu and Vall Castelló (2019).

Our primary outcomes are births per woman at the cohort level and the timing of motherhood. We also examine whether women have fewer children than they desire, helping distinguish constraints on realizing fertility plans from changes in desired family size. We then study perinatal outcomes among observed births as potential downstream consequences of changes in childbearing, since maternal age and birth spacing are associated with perinatal risks (Conde-Agudelo, Rosas-Bermúdez and Kafury-Goeta 2006; Lean et al.

[3] Appendices A and B provide context and economic intuition; Appendices C and D document data and definitions.

2017). Because employment conditions can affect whether and when women give birth, these estimates may also reflect changes in the composition of observed births.

Three administrative and survey sources play distinct roles. Social Security employment histories identify the reduced-form labor-market first stage. Birth records from Spain in 1980–2017 provide births per woman at the cohort level, maternal age at birth, first-delivery age, and perinatal outcomes among recorded deliveries. The 1999 and 2018 Fertility Surveys provide a child-count robustness check and evidence on potential mechanisms involving desired fertility, household circumstances, and cultural variables.[4] The World Values Survey (WVS) and the International Social Survey Programme (ISSP) provide descriptive cultural context.

The labor-market first stage shows that reform exposure increased contractual instability and lowered contribution-based income among women with meaningful early-career employment attachment. Exposure increases time spent in non-permanent employment during the first decade of working life by 0.415 years and reduces the real contribution-based income proxy by 535.821 euros, without a statistically detectable change in days worked per potential year. Complementary entry-age evidence shows that early recorded affiliation follows a more positive cohort trend after the cutoff among low-skilled women.[5]

The main reduced-form finding is a sizable decline in births per woman at the cohort level.[6] The DD estimate implies 0.257 fewer births per woman, or a 15.2% decline relative to the treated pre-reform mean. Annual event studies and an outcome-fitted synthetic comparison trace the post-reform dynamics and provide complementary magnitude checks. We interpret these estimates together with the pre-period diagnostics. Appendix F reports a respondent-level child-count robustness check using the Fertility Survey.

The birth records also show later motherhood. Exposure increases maternal age at birth by 0.526 years, or 1.9% of the treated group's pre-reform mean, and first-delivery age by 0.441 years, or 1.7%. Births shift away from ages 20 or below and toward ages 35 and above, including at the first-birth margin. Both conditional-trimming and joint-normal point ranges are positive for the timing outcomes.[7] The mechanism evidence points most clearly to constraints on realizing desired fertility. Exposure raises the probability that desired children exceed realized biological children by 27.9 percentage points. Estimates for marriage, homeownership, financial strain, childcare use, religious affiliation, and egalitarian attitudes are imprecise.

[4] Births per woman at the cohort level is a cumulative birth-intensity measure: registered births in a cohort-quarter × province × skill cell divided by an estimated female cohort population: historical provincial birth-cohort counts multiplied by the national female birth share and respondent-weighted skill shares from the 1999 Fertility Survey; it is not completed fertility for unique women. For maternal-age and perinatal fields observed only among realized births, we also report conditional-trimming and joint-normal selection sensitivities (Lee 2009; Tauchmann 2014).

[5] Appendix E provides supporting first-stage evidence; Appendix F collects reduced-form robustness checks.

[6] The outcome scales registered births by an estimated female cohort-by-skill denominator, constructed from historical female birth-cohort counts and survey-weighted skill shares. It therefore measures cohort-level birth intensity rather than completed fertility for individually identified women.

[7] The 2018 Fertility Survey provides separate reference selection losses for biological motherhood and for a first biological birth during 1980–2017 at maternal ages 15–55. The calculations assume weakly decreasing selection and common proportional effects across the survey and register populations; Appendix F.2 describes the implementation.

We find no evidence of a general deterioration in perinatal health among realized births. In sensitivity analyses allowing for selection into observed births, the C-section increase is the only perinatal effect that remains positive across both the conditional-trimming and joint-normal sensitivity ranges.

The reduced-form evidence shows fewer births per woman at the cohort level and later motherhood, but cannot by itself establish how much delayed fertility is subsequently recovered or separate the mechanisms behind these responses. We therefore build and estimate a life-cycle model linking employment conditions to fertility choices over the reproductive horizon. Employment histories discipline labor-market transitions and contribution-resource profiles, while the 2018 Fertility Survey disciplines completed fertility, childlessness, and first-birth timing. The birth-record estimate of births per woman at the cohort level remains reduced-form evidence rather than an individual model target. Under the historical reform mapping, the estimated model predicts changes of -0.034 children per woman by age 40, -1.20 percentage points in motherhood, and $+0.072$ years in first-birth age.[8]

The first contribution of this paper is to research on low fertility in high-income countries. The paper shows that an institutional shift toward non-permanent entry employment reduced realized fertility and delayed motherhood. This evidence complements work emphasizing economic constraints, household bargaining, gender norms, and the difficulty of combining careers and children (Goldin 2021, 2025; Doepke and Kindermann 2019; Doepke et al. 2023; Kearney, Levine and Pardue 2022; Kearney and Levine 2025). The results distinguish the timing and quantity margins: exposed women began motherhood later, had fewer realized births, and were more likely to report that desired children exceeded realized biological children. The father-side mechanism evidence shows that the decline in births per woman at the cohort level is concentrated among births for which both the mother and the recorded father are low-skilled, whereas delays in maternal age are tied more closely to the mother's skill group. This pattern is consistent with fertility quantity being sensitive to both partners' labor-market prospects, while the timing of motherhood is more closely associated with the mother's employment conditions.

The second contribution is to the literatures on labor-market duality, fixed-term contracts, and early-career scarring. Studies of partial deregulation show that liberalizing fixed-term hiring while preserving permanent-contract protection can create segmented careers and persistent losses for entrants (Blanchard and Landier 2002; Dolado, García-Serrano and Jimeno 2002; Bentolila and Dolado 1994; Aguirregabiria and Alonso-Borrego 2014; García-Pérez, Marinescu and Vall Castelló 2019). We connect this institutional structure to family formation by using reform exposure at labor-market entry rather than workers' selected contract status. The labor-market first stage distinguishes access to work from job quality: exposed cohorts

[8] Appendices G, H, and I document the model, estimation, and sensitivity, respectively.

have a more positive slope in early recorded affiliation, more years of non-permanent employment, and lower contribution-based income, without a statistically detectable change in days worked. Combining evidence on these cohorts from birth records and fertility surveys, we show that the consequences extend well beyond the initial job to fertility quantity and timing, the gap between desired and realized fertility, and perinatal outcomes among realized births. The paper therefore broadens the study of labor-market duality from career outcomes to its longer-run consequences for family formation and the realization of fertility plans.

Third, we quantify how early-career employment conditions shape fertility over the life cycle. The reduced-form design identifies the effect of reform exposure, but it cannot determine how much delayed fertility is later recovered or separate the roles of entry conditions and persistent employment insecurity. We therefore estimate a parsimonious life-cycle model of forward-looking fertility choice under employment risk and a finite reproductive horizon, with age-dependent birth-success probabilities. The model allows desired family size, religiosity, and gender norms to shift fertility preferences, and quantifies the labor-market channels conditional on this heterogeneity and the model's assumptions. It reproduces the direction of the reform-induced fertility responses, with a smaller timing delay than in the data, and points to post-entry employment instability and accumulated insecurity as the main drivers within the model. The counterfactuals show that reducing early-career insecurity can recover some forgone fertility and bring motherhood forward, while the model's accounting indicates that most of its fertility decline reflects fewer attempts before age 40, including opportunities lost through postponement.

The findings contribute to understanding low fertility in high-income countries by showing how employment instability at career entry can have lasting consequences for family formation. As women spend more of their lives in paid work, the stability, remuneration, and prospects for progression offered by their jobs may shape both the timing of motherhood and their ability to realize fertility plans. Labor-market policies that alter these conditions may therefore have consequences for fertility that extend well beyond their immediate effects on employment and earnings.

2. BACKGROUND

2.1. *Income, career risk, and the timing of childbearing*

Fixed-term contracts can affect fertility through income, opportunity-cost, and uncertainty channels. In standard fertility models, parents weigh household resources, child-rearing costs, and time away from work (Becker 1960, 1965; Schultz 1985; Hotz and Miller 1988; Black et al. 2013). Lower earnings reduce the resources available for children, while weaker career prospects reduce the opportunity cost of time and could instead raise fertility. The sign is therefore theoretically ambiguous. Contractual instability adds a dynamic

margin: short job horizons, limited maternity-related protection, and restricted access to credit can make the resources available during pregnancy and early childhood harder to secure.

Three mechanisms can turn this instability into lower realized fertility (Ranjan 1999; Sommer 2016). First, uncertain earnings raise the risk of committing to persistent childcare costs, while childcare availability and part-time opportunities also shape these decisions (Del Boca 2002). Second, women may postpone childbearing while seeking a more secure career path (Rica and Iza 2005; Adda, Dustmann and Stevens 2017); if instability persists, biological time limits can convert postponement into fewer births. Third, repeated fixed-term jobs can scar earnings and expectations after the current contract ends, making accumulated exposure more relevant than current employment status alone. Partner resources can partly insure these risks, so fertility quantity may depend on the household's joint employment position even when timing is especially sensitive to the mother's career path.[9]

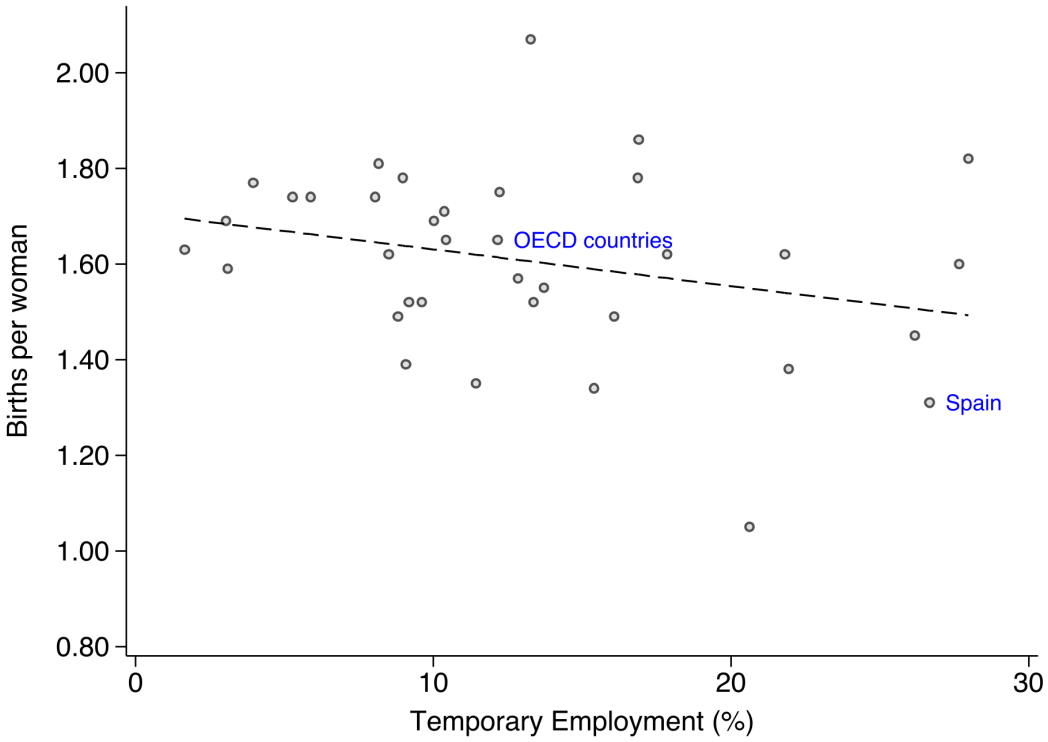
Figure 1 provides descriptive context, showing a negative cross-country association between temporary employment and fertility across OECD countries in 2017.

2.2. *Existing Evidence and the Identification Challenge*

Existing evidence is consistent with several of these mechanisms, although the direction and magnitude of the fertility response vary across settings. Job loss and employment uncertainty can delay motherhood (Del Bono, Weber and Winter-Ebmer 2012; Prifti and Vuri 2013); Clark and Lepinteur (2022) instead find smaller families without a change in the probability of parenthood, while other studies find small, positive, or heterogeneous responses (Kreyenfeld 2009; Santarelli 2011; Bhaumik and Nugent 2011; Kind and Kleibrink 2013). Associations between temporary work and delayed first birth appear in Finland, Germany, and Spain (Sutela 2012; Auer and Danzer 2015; Rica and Iza 2005), but observed contract type is jointly determined with career preferences, partner labor-market prospects, and local labor-market conditions. The empirical challenge is therefore to shift contract exposure independently of those fertility determinants.

[9] Appendix B develops this economic intuition and maps it into the estimated life-cycle model.

Figure 1. Cross-country association between temporary employment and fertility



Notes: The points show 35 countries and the OECD aggregate with matched 2017 observations. The horizontal axis is temporary employment as a percentage of dependent employment, for all persons and ages; the vertical axis is the period total fertility rate. The dashed line is an unweighted fit through the 36 plotted observations. The negative cross-country association is descriptive and is not interpreted causally. *Source:* OECD statistics.

The 1984 reform provides the policy-induced shift in contract exposure used for identification, while the richness of the Spanish data allows us to examine the mechanisms through which that exposure affects fertility. We study desired fertility, religiosity, and gender attitudes to distinguish labor-market mechanisms from slower-moving differences in preferences (Goldin 2021, 2025; Kearney and Levine 2025). The policy relevance is reinforced by Spain's 2021 labor-market reform, which moved in the opposite direction by sharply restricting the use and renewal of fixed-term contracts and tightening limits on successive temporary employment. This later reform motivates the model's counterfactual exercises asking how fertility responds to reductions in early-career employment insecurity.[10]

Contractual instability may also affect perinatal outcomes among realized births through parental resources, stress, and time investments (Heckman 2012; Reeves, McKee and Stuckler 2014; Rohde et al. 2016; Grigoriadis et al. 2018; Lepinteur 2021). Because the reform changes whether and when births occur, these outcomes cannot be interpreted as unconditional effects on all potential mothers. The empirical

[10] The 2021 reform abolished the widely used contract for work or service, restricted fixed-term hiring to specified temporary needs, tightened duration and renewal rules, and strengthened limits on successive temporary contracts. These changes were followed by a substantial decline in temporary employment (OECD 2025; Instituto Nacional de Estadística INE).

analysis accordingly treats gestational age, preterm birth, birthweight, and c-sections as secondary outcomes among recorded births and reports selection bounds for their observation.

2.3. *The Spanish Labor Market*

Spain entered the democratic period with a highly regulated labor market. The 1980 Workers' Statute modernized labor relations but retained permanent employment as the standard contract and restricted temporary hiring largely to seasonal work.[11]

The 1984 reform, approved in August and implemented in October, allowed fixed-term contracts to be used for regular activities rather than only seasonal work.[12] Firms gained access to short-duration contracts with lower expected dismissal costs, while protection for incumbent permanent workers remained largely unchanged. Temporary contracts carried limited duration and lower severance obligations; permanent contracts had no fixed end date and required substantially higher dismissal payments. This asymmetric liberalization reduced firms' incentives to convert temporary positions into permanent jobs and produced the characteristic Spanish dual labor market (Aguirregabiria and Alonso-Borrego 2014). Appendix Table A2.1 reports the contract terms in detail.

Temporary employment rose from 2.2% in 1983 to roughly 33% in the late 1990s, without a comparable break in the already high unemployment rate (Figure 2). The expansion was concentrated among entrants, particularly young, female, and low-skilled workers: by 1998, 73% of employees aged 15–24 held temporary contracts. Between 1987 and 1996, only 14% of temporary workers converted to permanent employment in an average year.[13]

[11] Law 8/1980 (*Estatuto de los Trabajadores*), March 10, is available at https://www.boe.es/buscar/doc.php?id=BOE-A-1980-5683.

[12] The reform formed part of the Economic and Social Agreement signed in October 1984 after negotiations with labor and business organizations.

[13] Appendix Table A2.3 reports one-year transitions from temporary employment.

Figure 2. Total share of temporary employment and unemployment in Spain (1981-2018)



Notes: This figure shows the evolution of temporary employment and unemployment rates in Spain since records are available. Temporary employment rises rapidly after the 1984 reform, from 2.2% in 1983 to 14.6% in 1985. *Sources:* Own elaboration using data to construct the temporary-employment share and unemployment rate from the Spanish Labor Force Survey (LFS) and Spanish Chamber of Commerce for years 1988–2018, and data from the Central Balance (Bank of Spain), Spanish Working Conditions Surveys (SWCS), Economy and Labor Ministry sources, the World Bank, and newspaper series for years 1981–1987. The vertical red line marks the 1984 labor market reform implementation period (October 1984).

3. DATA

Three primary data sources identify distinct parts of the reduced-form argument. Administrative Social Security histories establish how the reform changed entry, contract type, days worked, and earnings. Birth records provide births per woman at the cohort level, fertility timing, and perinatal outcomes among realized deliveries. The Fertility Surveys provide the education-based child-count robustness check and mechanism evidence on temporary employment, desired fertility, household circumstances, and cultural variables. For the life-cycle model, the employment histories discipline labor-state and earnings processes, while the 2018 Fertility Survey supplies the completed-fertility distribution and preference moments. World Values Survey and ISSP measures are used as descriptive cultural context.

3.1. *Employment histories and the labor-market first stage*

The labor-market first stage uses the 2006 Continuous Sample of Working Lives, an administrative Social Security dataset containing affiliation spells, contract information, contribution bases, schooling, and demographic characteristics. We use these retrospective records to reconstruct employment histories over the first ten years of potential labor-market experience: ages 16–25 for low-skilled women and 18–27 for high-skilled women. The early-affiliation and coverage criteria described below focus the analysis on women whose records allow us to characterize this period in Spain's labor market.[14] The treated cohort window is 1965Q1–1972Q4 and the comparison cohort window is 1962Q1–1969Q4, matching the skill-specific entry-age cutoffs used throughout the reduced-form analysis.

We construct two types of labor-market outcomes. First, entry by ages 18 and 19 is measured from the first recorded Social Security affiliation, which need not coincide with the first job. These outcomes use the broader low-skilled entry sample without requiring subsequent attachment. Second, we measure employment stability, work intensity, and contribution-based income over the ten-year window described above. This career sample requires at least four years of positive registered affiliation and retains the source sample's exclusions for own-account and specified nonemployment main affiliations. To capture careers near their beginning, we exclude women first recorded before age 16 or after age 20 for low-skilled women and age 22 for high-skilled women.[15] These criteria capture early careers without requiring identical entry ages across skill groups.

We select three complementary early-career outcomes to distinguish employment stability, employee work intensity, and contribution-based resources. *Years of non-permanent employment* counts years in which the annual main affiliation is employee work and is classified as temporary or belongs to the agricultural, domestic-service, maritime, or mining employee regimes.[16] *Days worked per year* is the average annual number of employee employment days over the ten-year window. Overlapping jobs count once, annual totals are capped at 365, and years without employee work contribute zero to this ten-year average; benefit and own-account records are excluded. *Real contribution-based income* is each woman's average of annual sums of monthly contribution bases associated with employee work and benefit receipt over fully observed years

[14] The sample includes Spanish-born or Spanish-national women. Skill groups use each woman's highest recorded schooling consistently across all first-stage tables and figures. The higher-skilled group includes school-leaving, EGB/ESO and higher qualifications; it therefore does not uniformly represent upper-secondary completion. The lower-skilled group includes lower recorded schooling and the source's unknown-schooling category. Missing schooling outside that category remains unclassified.

[15] Eligibility uses registered affiliation, which can include benefit records, and does not depend on motherhood. The sample retains the source panel's historical income support and requires observed regional controls. Career regressions cover cohorts born in 1965Q1–1972Q3 for low-skilled women and 1962Q2–1969Q4 for high-skilled women. Outcome-specific availability yields 27,191, 27,026, and 26,382 observations for non-permanent employment, days worked, and income, respectively. These histories are conditional on inclusion in the 2006 MCVL and subsequent affiliation support.

[16] To cover Spain's sectoral employee affiliations, we count years in the specified special regimes regardless of recorded contract duration. Missing contract status is assigned using a duration-and-continuity proxy, documented in Appendix E; it is not independently verified temporary-contract status.

within the same ten-year window, in 2011 euros per year.[17] Figure 4 and Appendix Figure E2 use these same three outcomes; Table 2 uses the broader entry-age sample. Appendix E documents their construction. Employment histories also supply model inputs, whose separate estimation is described in Appendix H.

3.2. *Birth records, fertility timing, and perinatal outcomes*

The birth-record source is the Childbirth Statistics Bulletin, supplied by the Spanish National Statistics Office. We use 1980–2017, the complete harmonized birth-record window available for observing realized births among the treatment and comparison cohorts. Using this complete window provides consistent coverage of these maternal cohorts over their childbearing years. The source records the birth, perinatal outcomes, and parents' demographic and occupational characteristics through the mandatory document completed to register a birth with the Civil Registry.[18]

The birth-record analysis covers births registered in Spain during 1980–2017 to mothers of all nationalities within the selected maternal cohorts. Because education is not consistently recorded, we classify maternal skill using the harmonized occupation and activity-status categories in Table D1. We apply this classification consistently across all years. The high-skilled group includes professional and technical workers, executives and directors, clerical workers, and traders and salesworkers. The low-skilled group includes service workers; agricultural, forestry and fishing workers; production workers, transport operators and laborers; professional armed-forces personnel; and people engaged in household work. We exclude records in which the mother is a student, retired, a pensioner or rentier, or has missing or unclassified occupation/status information.

We focus on prospective job uncertainty at entry by assigning treatment based on whether a mother's expected labor-market entry occurred before or after the 1984 liberalization of fixed-term contracts. For low-skilled women–who are most likely to enter at the earliest legal age–we use age 16 as the expected labor-market entry age.[19] Our main analysis window covers low-skilled cohorts born 1965Q1–1972Q4. The cohort born in 1968Q3 is treated as a transition cohort near the reform date, while the cohort born in 1968Q4 is the first fully exposed cohort under our baseline coding.

For high-skilled women, expected labor-market entry is age 18, corresponding to high-school completion. We compare low-skilled women turning 16 before and after the reform (born in 1965Q1–1972Q4, with the first fully exposed cutoff at 1968Q4) with high-skilled women turning 18 before and after the reform (born in

[17] We deliberately include both employment- and benefit-related contribution bases to capture resources associated with work and benefit receipt in Spain's Social Security records. Contribution bases provide an administrative income proxy under statutory rules and limits; they are distinct from actual wage or benefit payments and do not cover all household income. A year is used only when every eligible account-month has an observed nonzero base; missing bases and years without eligible affiliations are not assigned zero. The mean is therefore conditional on observed contribution-year support.

[18] Appendix C.1 describes the source and the INE microdata portal.

[19] The minimum legal age in Spain for working individuals was 14 before 1980 and 16 since then. However, as we can see in García-Pérez, Marinescu and Vall Castelló (2019), in 1977, there was already effective legislation that increased the age to 16 years old so that 16 was the binding minimum working age since 1977 in Spain.

1962Q1–1969Q4, with the corresponding cutoff at 1966Q1). Nearby birth-cohort quarters therefore reached the relevant labor-market entry age under different contractual regimes. We use the same cohort windows and reform cutoffs across birth-record outcomes, while sample sizes differ according to the availability of each outcome.

We measure births per woman at the cohort level by counting registered infants in each mother's birth-cohort quarter (c), recorded administrative province (p), and skill group (s), denoted B_{cps}. The denominator combines historical female cohort counts with respondent-weighted skill shares from the 1999 Fertility Survey, including childless women. Appendix C.2 documents the construction. The resulting ratio is a cohort-level birth measure, not completed fertility for unique women. Ceuta and Melilla lack the required historical denominator and are excluded from quantity regressions; this does not restrict eligibility for infant-level outcomes.

The timing outcomes from birth records are maternal age at birth and first-delivery age, both taken from the verified merged maternal-age-at-delivery field. Invalid maternal ages below 15 are unavailable for timing; first-delivery age also excludes ages above 55. First births mean first deliveries, retaining every infant from twin and other multiple first deliveries. Before 2007, the live-child count and year-specific plurality information identify deliveries with no previous live-born child. From 2007 onward, the official count of live-born children in previous deliveries identifies first delivery when that count equals zero. No within-delivery infant order determines eligibility. These outcomes describe the timing of realized births. Perinatal outcomes use birth weight, gestational age and C-section status when the respective field is available and valid. An unavailable field remains missing, never zero; for example, C-section is available from 2007. Appendix C and the table notes report outcome-specific support and observation counts.

3.3. *Fertility Surveys and completed fertility*

The Spanish Fertility Survey (FS) waves 1999 and 2018 provide nationally representative microdata on women's retrospective fertility histories, desired fertility, household circumstances, labor-market status, and attitudes.[20] In the reduced-form analysis, we use these surveys primarily to study mechanisms: desired and unmet fertility, current employment and contract type, family and household circumstances, financial strain, childcare constraints, religiosity, and gender norms. We also retain one appendix robustness check using the survey number of biological children per woman, estimated with education-based skill groups; the fertility-outcome analysis in the main tables relies on birth records. Both survey waves include women of all nationalities, using the same skill-specific cohort windows and birth-cohort-quarter exposure definitions.

[20] Appendix C.4 provides more details on the two Spanish fertility surveys, including a link where the microdata can be readily downloaded.

Descriptive statistics use each wave's respondent weights; genuinely childless women have zero biological children and remain in the denominator. The descriptive child-count sample includes all eligible women, including those with no biological children, with one observation per respondent. The 2018 mechanism regressions use the original respondent file and include women of all nationalities, whether or not they have children. General household and preference outcomes impose no restrictions on children's nationality or birth years. Including childless women avoids conditioning these mechanisms on motherhood, which may itself respond to the reform, and is consistent with the cohort-level fertility analysis. Child-specific questions retain their eligible population. Because occupational status in the FS has substantial missingness, we define low-skill as high-school dropouts and high-skill as women with at least a high-school diploma; the alternative occupation mapping is described in Table D3. Eligible respondents were aged 25–37 in the 1999 survey and 45–55 in the 2018 survey. The earlier wave therefore need not measure completed fertility. We report the pooled child-count estimate as a retrospective survey check and retain births per woman at the cohort level from birth records as the main fertility outcome.[21]

The 2018 wave also includes detailed economic, demographic, and cultural indicators that we use to investigate mechanisms linking temporary employment to fertility decisions. The economic and household variables include current employment, temporary-contract status among employees with an observed contract, marital status, homeownership, difficulty making ends meet, monthly net income of at least 1,000 euros, disability or chronic illness, ever experiencing long-term unemployment, age at first job, age at economic independence, age at completion of studies, and the need for flexible childcare to consider having a child. The fertility-preference measure is *unmet desired fertility*, an indicator equal to one when the respondent's desired number of children exceeds her realized number of biological children. Appendix Figures C4.1 and C4.2 provide descriptive benchmarks for desired fertility by sex, age, and skill; their stated preferences are distinct from realized births and the unmet-fertility indicator. The support and cultural variables include using no childcare among parents with a living child under 14, having no religious affiliation, and an egalitarian-views index. The childcare alternatives cover formal and informal arrangements. Respondents who prefer not to answer the religion question are excluded from the affiliation outcome. The index averages seventeen recoded items on gender roles, caregiving, partnership norms, and work-family trade-offs on a 0–1 scale, where higher values indicate more egalitarian views.[22]

For the quantitative model, we measure desired family size, religiosity, and gender norms jointly in the 2018 Fertility Survey. The desired-family-size variable is the INE-derived all-respondent measure, so it

[21] The respondent-weighted pooled child-count estimate is -0.217 children (s.e. 0.061), using 5,621 women, including genuinely childless respondents. Region and survey-wave fixed effects are included. The earlier wave observes some women before the end of their childbearing years.

[22] Table D2 reports the exact source-variable names, response scale, and item formulas.

includes women with zero children. Religiosity combines affiliation and practice intensity. The traditional-norm index averages seven work–family statements, reverse-coding egalitarian items and requiring at least four valid responses. The estimator targets their joint partial gradients in completed children and first-birth age, conditional on skill, age, and region. They enter as fixed preference correlates rather than causal effects of beliefs or outcomes changed by the reform. Appendix Figures A1.3 and A1.4 document Spanish attitudes toward maternal employment and differences in traditional norms across skill proxies; Figure A1.5 places norms alongside female labor-force participation across countries, and Figure A1.6 describes Spanish religiosity by skill. These World Values Survey and ISSP comparisons provide descriptive context, not reform effects or structural estimation targets.

3.4. *Sample composition and descriptive statistics*

Table 1 summarizes outcomes and background characteristics by skill group. Panel A describes the birth-record sample: low-skilled mothers have births and first deliveries at younger ages on average, with less favorable average perinatal indicators. These differences are descriptive. Panel B pools the 1999 and 2018 Fertility Surveys, retains every eligible woman once including childless respondents, sets biological children to zero for genuinely childless women, and uses each wave's respondent weights. Birth-age statistics are conditional on recorded biological-child histories or motherhood; childless women are not assigned a birth age. Panel C describes the 2018 mechanism sample of all eligible women; each outcome uses its eligible population and valid responses. Births per woman at the cohort level in Panel A and the respondent-level survey child count in Panel B are distinct measures; Appendix C.2 documents the estimated female cohort denominator.

Figure 3 plots the administrative cohort paths by maternal skill: births per woman at the cohort level, maternal age at birth and first-delivery age. Births-per-woman points weight the same province-cell outcome by the estimated female cohort population, as in Table 1; regressions retain birth-count analytical weights. Individual timing outcomes use observed valid ages, retaining every infant from multiple first deliveries. The figure uses the full skill-specific cohort windows. Within each row, panels have equal vertical spans and tick spacing but different origins, making changes in the descriptive trends visible without changing their absolute scale. No skill-specific outcome normalization is applied. The fitted lines allow a level and slope change at the skill-specific reform cutoff; they are descriptive fits, distinct from DD estimates and annual event-study coefficients. Appendix C.3 checks the underlying register: Figure C3.1 and Table C3.1 compare full-register coverage with the analysis sample, Figures C3.2 and C3.3 show maternal-cohort and birth-year support, Figure C3.4 reports the raw and log numerators, and Figure C3.5 documents the first-delivery age-threshold profiles.

Table 1. Descriptive statistics by skill group

	Low-skilled women		High-skilled women	
Variables	Mean	SD	Mean	SD
Panel A: Administrative outcomes (1980–2017)				
Births per woman (cohort-level)	1.443	0.425	1.112	0.480
Mother's age at birth	28.068	6.151	31.576	4.829
Mother's age at first delivery	26.031	5.880	30.232	4.796
C-section	0.292	0.455	0.361	0.480
Birth weight (grams)	3224.097	518.306	3234.059	506.092
Low birth weight (<2,500g)	0.068	0.252	0.063	0.244
High birth weight (>4,000g)	0.048	0.213	0.046	0.209
Gestational age (weeks)	38.988	2.135	39.067	2.005
Preterm birth (<37 weeks)	0.098	0.298	0.077	0.266
Eligible cells (infant records)	1,600 (2,058,507)		1,600 (1,325,611)	
Panel B: Fertility Surveys (1999/2018), all women				
Woman's age at survey	37.919	9.828	44.209	9.619
Mother's age at any birth	26.185	5.412	29.764	5.277
Mother's age at first birth	24.428	4.938	28.274	5.155
Number of biological children	1.319	1.013	1.158	1.012
Childless (share)	0.259	0.438	0.333	0.471
Women	2,307		3,314	

Notes: Panel A uses registered births in Spain, 1980–2017, all nationalities, and the cohort and skill definitions in Appendix D. Births per woman uses the cohort-population denominator in Appendix C.2; its mean and SD are population-weighted cell statistics, not completed fertility for unique women. Quantity excludes Ceuta/Melilla; parentheses count eligible infants. Other rows give infant-level means over valid observations. First deliveries retain all multiple-birth infants; C-section is available from 2007. Panel B pools the 1999/2018 surveys with wave-specific respondent weights, including childless women with zero children; birth ages condition on observed biological births. *Sources:* INE birth register, historical cohort counts and Fertility Surveys.

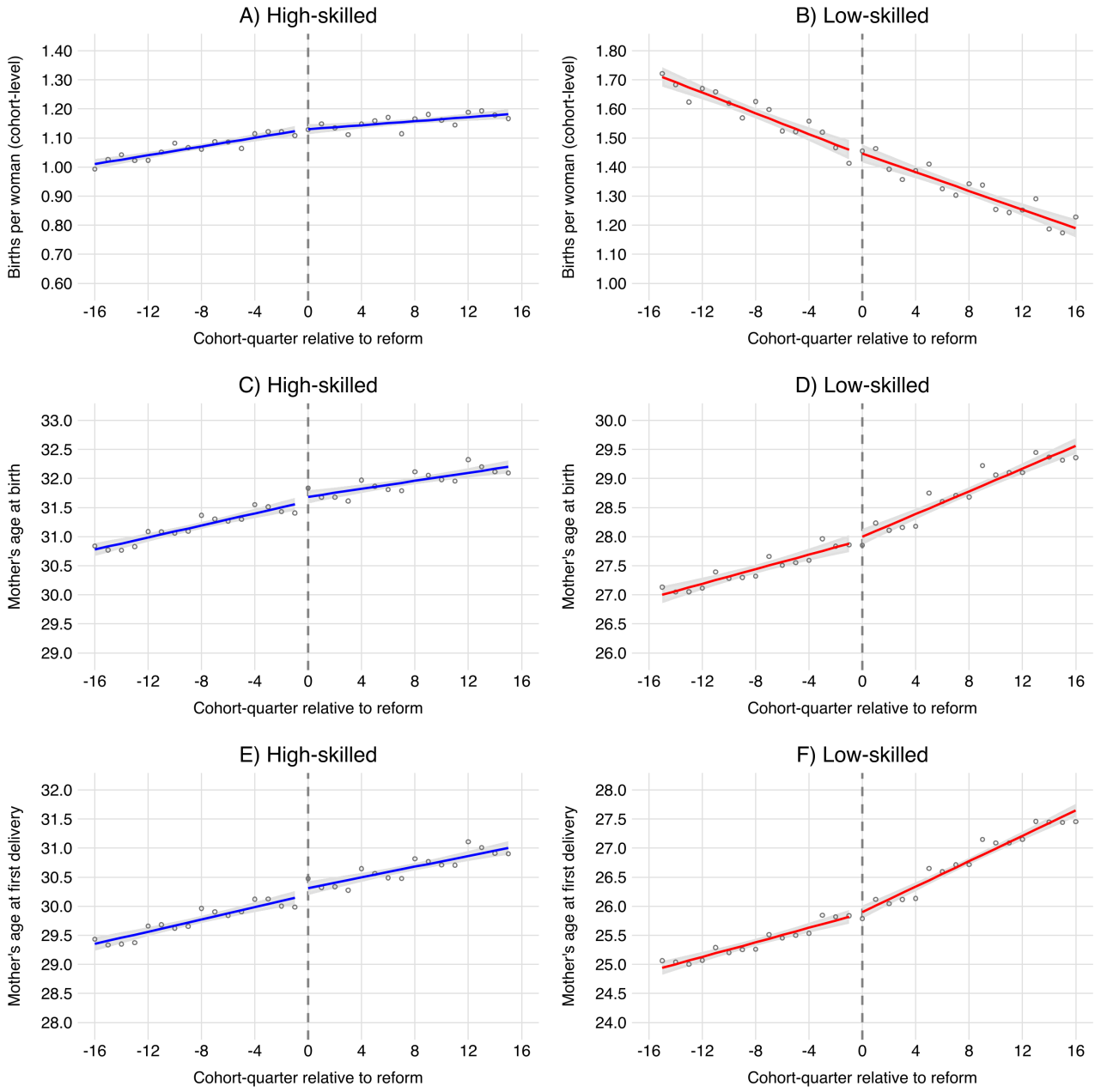
García-Pérez, Marinescu and Vall Castelló (2019) show that the 1984 liberalization changed the early careers of male high-school dropouts: entry occurred earlier, but affected men accumulated fewer days worked and lower earnings over time. Appendix Figure F4 points in the same direction for women in the 2018 Fertility Survey, with exposed low-skilled cohorts less likely to hold permanent jobs later in life. Section 5.1 establishes the corresponding labor-market first stage for women using administrative Social Security histories.

Table 1 continued: 2018 Fertility Survey characteristics

	Low-skilled women		High-skilled women	
Variables	Mean	SD	Mean	SD
Panel C: Fertility Survey (2018), mechanism sample				
Woman's age at survey	49.033	2.280	51.565	2.228
Temporary contract (current employees)	0.280	0.449	0.149	0.356
Currently working	0.486	0.500	0.735	0.441
Monthly net income ≥1,000 EUR	0.144	0.351	0.517	0.500
Married	0.718	0.450	0.682	0.466
Difficulty making ends meet	0.655	0.475	0.412	0.492
Any disability	0.077	0.266	0.060	0.238
Homeownership	0.766	0.424	0.815	0.388
Ever long-term unemployed	0.649	0.478	0.563	0.496
Age at first job	23.245	9.082	24.009	6.932
Age at economic independence	21.730	4.335	23.340	3.872
Age finished studies	20.160	5.673	23.363	5.761
Egalitarian views (index)	0.687	0.131	0.737	0.127
Unmet desired fertility	0.338	0.473	0.415	0.493
No childcare used (children under 14)	0.298	0.458	0.265	0.442
Women	1,282		2,329	

Notes: Panel C uses the 2018 respondent mechanism sample and original survey weights. Women of all nationalities are included, whether or not they have children. General outcomes impose no restrictions on children's nationality or birth years. Each row uses valid responses; contract status conditions on employees reporting a contract. Unmet fertility means desired children exceed biological children. No childcare refers to parents with a living child under 14 and valid responses about formal and informal care. *Source:* INE 2018 Fertility Survey.

Figure 3. Cohort-level births and maternal ages by mother's cohort and skill level



Notes: Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Dots are cohort-quarter means. A)–B) show births per woman (cohort-level), using estimated female-cohort denominator weights; C)–D) show unweighted maternal age at birth; E)–F) show unweighted first-delivery age, including every infant from multiple first deliveries. The quantity denominator combines historical female births and weighted skill shares from eligible 1999 survey women born 1965–1969, including childless women; it does not measure unique-woman completed fertility. Lines are descriptive piecewise-linear cohort fits with a level and slope change at the cutoff. Shading gives 95% confidence bands for fitted means, conditional on the estimated denominator. These are not DD estimates or event-study coefficients. Each row uses equal vertical spans and tick spacing across skills, with different origins; no display-specific normalization is applied. *Sources:* INE birth register, historical births and yearbook sex shares, and 1999 Fertility Survey.

4. EMPIRICAL STRATEGY

Identification comes from differences in reform exposure at labor-market entry across maternal birth cohorts and skill groups. A mother's birth-cohort quarter determines whether she reached the relevant entry age before or after fixed-term hiring was liberalized. The cohort-within difference-in-differences (DD) specification compares the change at the low-skilled entry cutoff with the corresponding change among nearby high-skilled cohorts.

For low-skilled women, the sample covers 1965Q1–1972Q4 and assigns 1968Q4 as the first *fully exposed* cohort, because these women turn 16 after implementation in late 1984. For high-skilled women, the sample covers 1962Q1–1969Q4 and assigns 1966Q1 as the first fully exposed cohort, because these women turn 18 after the reform and can have completed high school. Earlier cohorts eventually encountered the new regime, but only after entering the labor market. The design therefore relies on evidence that entry conditions have especially persistent effects (Kahn 2010; Oreopoulos, von Wachter and Heisz 2012) and that Spanish fixed-term and apprenticeship contracting changed sharply for young entrants around the reform (García-Pérez, Marinescu and Vall Castelló 2019).[23] Appendix F reports cohort-position, trend and bandwidth checks.

We collapse the birth records at the level of mothers' quarter of birth cohort (c), province (p), and skill group (s), as in Oreopoulos, von Wachter and Heisz (2012) and García-Pérez, Marinescu and Vall Castelló (2019).[24] This collapse enables us to count births by maternal cohort while controlling for province fixed effects and a common linear cohort-quarter trend with a common post-cutoff slope change. The appendix Fertility Survey child-count and mechanism specifications are estimated at the individual level and include region and survey-wave fixed effects where relevant. In the collapsed birth-record specifications, regressions are estimated using cell-size analytic weights.

The maintained within-cohort DD model is:

$$\begin{aligned} y_{ucpsw} &= \alpha + \beta_1 r_{cs} + \beta_2\big(r_{cs} \times Post_{cs}\big) + \beta_3 Post_{cs} \\ &\quad + \beta_4 Treated_s + \beta_5\big(Treated_s \times Post_{cs}\big) + \eta_p + \lambda_w + \varepsilon_{ucpsw}\,. \end{aligned} \tag{1}$$

where y_{ucpsw} denotes the outcome for unit u: a cohort–province–skill cell for birth-record quantity, an infant record for birth-record timing and health, a respondent including childless women for the pooled survey child count, or a woman for the 2018 survey mechanisms. The indices are maternal or respondent birth-cohort quarter c, location p, skill group s, and, when applicable, survey wave w. The running variable

[23] Figure A2.1 reports apprenticeship contracts relative to employees in non-agricultural private sectors. Following the reform, the share rose sharply, especially among workers aged 16–19.

[24] The Fertility Survey number-of-children estimate is reported in Appendix Table F1 as an individual-level robustness check using education-based skill groups. It is reported as a robustness check and is not interpreted as a labor-market first stage.

is $r_{cs} \equiv BirthTrim_c - C_s$, measured in quarters relative to the skill-specific cutoff C_s.[25] Hence, $r_{cs} = 0$ is the first fully exposed cohort under our baseline coding, $Post_{cs} = \mathbb{1}\{r_{cs} \geq 0\}$, and $Treated_s = 1$ for low-skilled women in the relevant source-specific sample. The interaction $r_{cs} \times Post_{cs}$ allows the slope of the running variable to differ after the cutoff. DD regressions using birth records include province fixed effects. Pooled all-women Fertility Survey regressions include region and survey-wave fixed effects, with λ_w representing the latter. The single-wave 2018 all-women mechanism regressions instead include region and respondent birth-year fixed effects; they retain the common post, skill, post-by-skill, linear cohort-quarter trend and post-by-trend terms. Both survey designs use the original respondent weights and cluster errors by respondent birth-cohort quarter. These source-specific fixed effects are preserved in related diagnostics; a collinear annual contrast is reported as unidentified rather than made estimable by dropping a maintained fixed effect.[26] In collapsed birth-record specifications, we use cell-size analytic weights. We report standard errors clustered by the mother birth-cohort quarter.[27] Our parameter of interest is β_5, which captures the post-reform differential change among low-skilled women relative to high-skilled women at the skill-specific cutoff, within each outcome's stated sample.

Two threats are especially important. First, the reform may have changed employment itself rather than only contract type. The administrative first stage therefore estimates entry, days worked, contract composition, and earnings separately; the clearest changes are in non-permanent employment and earnings, not days worked. Second, anticipation could invalidate the entry cutoff. The reform was approved in August 1984 and implemented in October 1984 under an accelerated legislative process (Bentolila, Dolado and Jimeno 2008), leaving little scope for advance adjustment. Event-study leads and formal joint tests provide the corresponding empirical check. The annual pre-period tests are reported with the event-study results; Appendix F provides additional credibility exercises.

For maternal-age and perinatal-health outcomes, we assess selection using conditional empirical trimming and a joint-normal sensitivity calculation (Lee 2009). Both use the 2018 Fertility Survey, official respondent weights, and women of all nationalities, including those without children. Maternal age across all births uses the estimated 5.204% overall proportional reduction in biological motherhood. First-delivery age and first-delivery perinatal health use the 4.813% overall reduction in having a first biological birth during 1980–2017 at maternal ages 15–55. These calendar and age limits define the reference selection event; women whose first birth falls outside them remain in the denominator. Birth country is unrestricted in both calibrations.

[25] In the reported tables, this term is the cohort running variable around the skill-specific cutoff.

[26] In the Fertility Surveys, we observe the respondent's region of residence (19 Autonomous Communities), whereas the birth-record regressions use the recorded delivery or registration province.

[27] Clustering at the cohort level allows arbitrary correlation across records and provinces within the same maternal birth-cohort quarter.

Conditional trimming estimates selection separately by maternal birth quarter and considers the lowest and highest attainable outcome means after retaining each quarter's calibrated share. The joint-normal calculation uses the overall selection rate, adds a distributional restriction and allows the outcome–selection correlation to span $[-1, 1]$. Both refit the birth-record DD specification. Health outcomes retain their original samples and availability periods, including C-section records from 2007–2017. Appendix F.2 describes the survey-to-register assumptions and uncertainty calculations. The tables report point endpoints rather than confidence intervals.

The event-study specification is:

$$\begin{aligned} y_{ucpsw} &= \alpha + \eta_p + \lambda_w + \theta\, Treated_s \\ &\quad + \sum_{m \in \mathcal{M},\, m \neq -1} \rho_m \mathbb{1}\{a_{cs} = m\} \;+ \sum_{m \in \mathcal{M},\, m \notin \{-1,4\}} \beta_m \big[\mathbb{1}\{a_{cs} = m\} \times Treated_s\big] \;+\; \varepsilon_{ucpsw}, \end{aligned} \tag{2}$$

where $a_{cs} = \lfloor r_{cs}/4 \rfloor$ groups relative cohort quarters into annual event bins, $\mathcal{M} = \{-4, \ldots, 4\}$, and $m = -1$ is the omitted annual reference bin. The final low-skilled quarter lies in bin 4 without high-skilled support. Main birth-record DD and annual regressions retain this quarter, with its bin indicator absorbing its level; no bin-4 interaction is reported. The father-side annual comparisons use only the jointly supported bins, -4 through 3. The coefficients β_m trace the treated-versus-comparison differential relative to the last pre-reform annual bin. Birth-record event studies include province fixed effects. Survey event studies include region and wave fixed effects for the pooled all-women sample, and region and respondent birth-year fixed effects for the single-wave mechanism sample. We use these event studies primarily as diagnostic tools for pre-trends and to describe dynamic patterns, and we report pre-period diagnostics alongside the annual results and in Appendix F.

For the main birth-record fertility outcome, we summarize the event-study evidence in the main table by averaging the post-reform coefficients from (2). Specifically, the event-study column reports

$$\widehat{\Delta}^{ES} = \frac{1}{4} \sum_{m=0}^{3} \widehat{\beta}_m, \tag{3}$$

with clustered inference computed from the covariance matrix of the event-study regression. The reported average therefore summarizes the post-reform dynamics within the same specification used in the plotted event study.

We also report an outcome-fitted synthetic-control diagnostic in Figure 5 and the note to Table 3. For each province, we construct a high-skilled outcome path. We combine these paths to reproduce the low-skilled outcome in the four pre-reform annual bins, then keep the same weights after the cutoff. Thus,

provinces supply comparison paths; the exercise does not compare provinces' treatment exposure. Let Y_{Lt} denote the low-skilled path and Y_{jt} the high-skilled path in province j. The implementation selects the minimum-squared-weight solution satisfying the pre-period fit and adding-up constraints:

$$\begin{aligned} \widehat{w} &= \arg\min_{w\in\mathbb{R}^J} \sum_j w_j^2, \\ \text{subject to} \quad & \sum_j w_j = 1, \qquad \sum_j w_j Y_{jt} = Y_{Lt}, \quad t = -4, \ldots, -1. \end{aligned} \tag{4}$$

Weights may be negative: this is a linear outcome-fitted comparison, rather than a convex average of provinces. Only pre-reform outcomes determine the weights. The normalized synthetic-control gap is

$$\widehat{\Delta}_t^{SC} = \left(Y_{Lt} - \sum_j \widehat{w}_j Y_{jt} \right) - \left(Y_{L,-1} - \sum_j \widehat{w}_j Y_{j,-1} \right), \tag{5}$$

and Figure 5 follows this gap over event time. The exercise asks whether the fertility decline also appears relative to a comparison path fitted before the reform. It provides a useful robustness perspective alongside the DD and event-study estimates.

5. RESULTS

5.1. *Contractual instability, days worked, and contribution-based income*

The administrative employment histories provide evidence on differences in the early careers of exposed women. The first stage distinguishes entry from subsequent job stability and resources: the lower-schooling group is assigned the 1968Q4 cutoff at age 16, while the higher-schooling group is assigned the 1966Q1 cutoff at age 18.

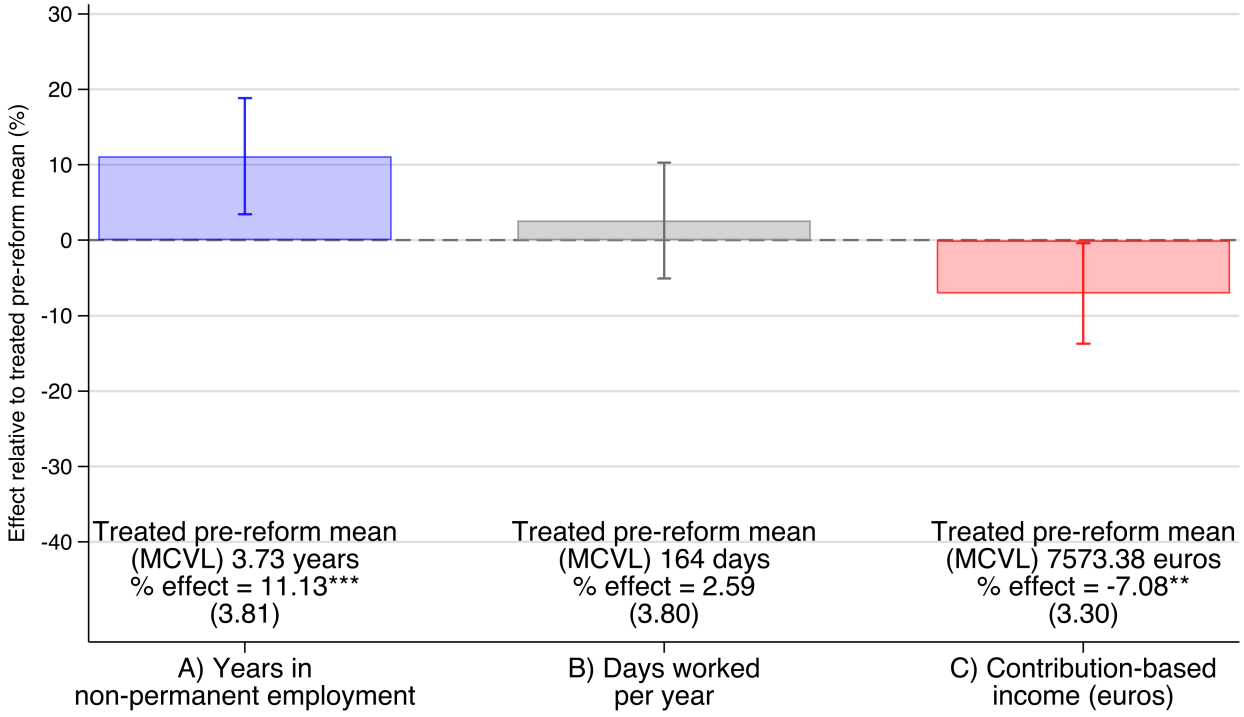
For the early-career outcomes, we estimate the person-level specification:

$$\begin{aligned} Y_i = \alpha + \beta\,(Post_c \times Treated_c) + \theta Treated_c + \delta Post_c + \eta Trend_c + \mu\,(Post_c \times Trend_c) \\ + \rho\,(Treated_c \times Trend_c) + \tau\,(Treated_c \times Post_c \times Trend_c) + \kappa U_i + \lambda_r + \pi_q + \varepsilon_i, \end{aligned} \tag{6}$$

where i indexes women, c indexes birth-cohort quarters, r is the autonomous community of first affiliation, and q is quarter of birth. $Post_c$ and $Trend_c$ are defined relative to the skill-specific exposure cutoff. The interactions with $Treated_c \times Trend_c$ allow the two education groups to have different linear cohort slopes before and after their cutoffs. λ_r are entry-region fixed effects, and π_q are birth-quarter fixed effects. The career sample retains event quarters -15 to 15 around the 1968Q4 and 1966Q1 cutoffs. U_i is regional unemployment assigned to the first retained affiliation year and its recorded spell-start quarter.[28] Standard errors are clustered by birth-cohort quarter.

[28] This timing need not equal first labor-market entry and may reflect post-reform choices. The estimates condition on this control and on the affiliation-support restrictions; they should not be read as unconditional effects for all women.

Figure 4. Effects of the 1984 reform on women's labor outcomes relative to treated pre-reform means



Notes: Person-level DD estimates as percentages of the outcome-specific low-skilled pre-reform mean. A) Years of non-permanent employment includes temporary employee main affiliations and agricultural, domestic-service, maritime and mining employee regimes; missing contract status uses the duration proxy in Appendix E. B) Average annual employee days: overlapping spells count once, annual totals are capped at 365 days, and the sum across the ten-year window is divided by ten, including years without employee work. C) Annual sums of monthly employment- and benefit-related contribution bases, averaged over complete observed years in the ten-year window (2011 euros per year), as an income proxy. Regressions include skill-specific cohort trends and post-cutoff slope changes, first-affiliation-region and birth-quarter effects, and the regional unemployment control defined in the text. Sample sizes: 27,191, 27,026, 26,382. Whiskers are 95% confidence intervals with birth-cohort-quarter clustering; parentheses give rescaled standard errors. Stars denote $p < 0.10, 0.05, 0.01$. *Source:* MCVL 2006.

All three outcomes use the career eligibility definition described in the data section, with outcome-specific availability. Years of non-permanent employment counts the number of years in which a woman's main employee affiliation is temporary or belongs to one of the employee special regimes classified as non-permanent in our measure. Days worked per year counts unique employee days over ten potential years, while real contribution-based income averages complete observed contribution years. We combine employment- and benefit-related contribution bases to proxy resources associated with both work and benefit receipt.

Table 2. Probability of first recorded affiliation by age 18 or 19 among low-skilled women

	(1)	(2)	(3)	(4)
	By age 18		By age 19	
	Baseline	Transition	Baseline	Transition
After	0.035	0.035	0.024	0.050*
	(0.033)	(0.042)	(0.027)	(0.029)
Trend	−0.012***	−0.012***	−0.004	−0.006***
	(0.002)	(0.003)	(0.002)	(0.002)
After × Trend	0.023***	0.025***	0.012***	0.014***
	(0.003)	(0.004)	(0.003)	(0.003)
Mean dep. var.	0.611	0.611	0.816	0.816
Pre-reform mean	0.590	0.590	0.782	0.782
R-squared	0.020	0.020	0.015	0.016
Observations	4,474	4,474	4,474	4,474
Birth-quarter controls	Yes	Yes	Yes	Yes
Transition-cohort indicators	No	Yes	No	Yes

Notes: Standard errors in parentheses, clustered by birth-cohort quarter. The sample is Spanish-born or Spanish-national low-skilled women in the 1965q1–1972q4 cohort window, excludes first recorded affiliation before age 16, and requires first observation by age 20. This entry-age analysis does not condition on four subsequent positive-affiliation years. The baseline specification includes the post-reform cohort indicator, a linear cohort trend, its post-reform interaction, and birth-quarter controls. Outcomes equal one when first recorded affiliation age is at most 18 or 19, respectively. Source: MCVL 2006. Transition specifications additionally include indicators for 1968q3 and 1968q4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4 rescales the DD estimates by the treated pre-reform means of the same regression outcomes. Exposure increases years of non-permanent employment by 0.415 years (s.e. 0.142, $p = 0.006$), or 11.1% of the treated pre-reform mean of 3.729 years. The days-worked estimate is 4.260 days per potential year (s.e. 6.244) and is not statistically distinguishable from zero. The contribution-based income estimate is -535.821 euros per year (s.e. 249.859, $p = 0.038$), or -7.1% of its treated pre-reform mean. These estimates indicate greater exposure under the definition of years of non-permanent employment and a lower contribution-based income proxy in the selected career sample. Appendix Figure E2 reports the corresponding cohort-quarter means separately by skill group using the same outcome-specific samples and definitions as Figure 4. These descriptive paths make the source of the relative estimates transparent; formal inference is provided by the DD estimates in Figure 4.

Table 2 reports the complementary entry-age evidence for low-skilled women. The post-cutoff slope change is positive for recorded affiliation by age 18 and by age 19 in both the baseline and transition-cohort specifications. The level shift is less precisely estimated, so the evidence is strongest for a progressive increase in early labor-market insertion across exposed cohorts. The entry-age estimates describe changes in cohort slopes; the career DD estimates indicate greater measured non-permanent employment and lower contribution-based income. Appendix Figure E1 shows the underlying entry-age pattern.

5.2. *Fewer births and later motherhood*

Births per woman (cohort-level)—Entry exposure to the fixed-term-contract regime reduces births per woman at the cohort level by 0.257, a 15.2% decline relative to the treated pre-reform mean of 1.693. The DD standard error is 0.018, based on 3,200 cohort-quarter × province × skill cells. The outcome divides registered births in each cell by a female cohort population denominator.[29] It therefore measures cohort-level birth intensity rather than completed fertility for individually identified women. Table 3 compares estimands using the same outcome definition.

The annual event study also points to a decline in fertility. Averaging the post-reform coefficients relative to the last pre-reform year gives an estimate of -0.157 (s.e. 0.031), compared with the DD effect of -0.257. Figure 5 provides a complementary synthetic comparison, combining high-skilled province paths to reproduce the low-skilled pre-reform trajectory and then holding those weights fixed after the reform. The average post-reform gap is -0.106, equivalent to -6.3% of the treated pre-reform mean. The pre-period lead test rejects equality of the event-study coefficients ($p = 0.000120$), so we interpret the post-reform dynamics together with the observed pre-reform movement. Appendix F, Table F1, reports additional birth-count and survey child-count checks.

[29] The denominator combines historical female birth-cohort counts with survey-weighted skill shares and is held fixed across the main specifications.

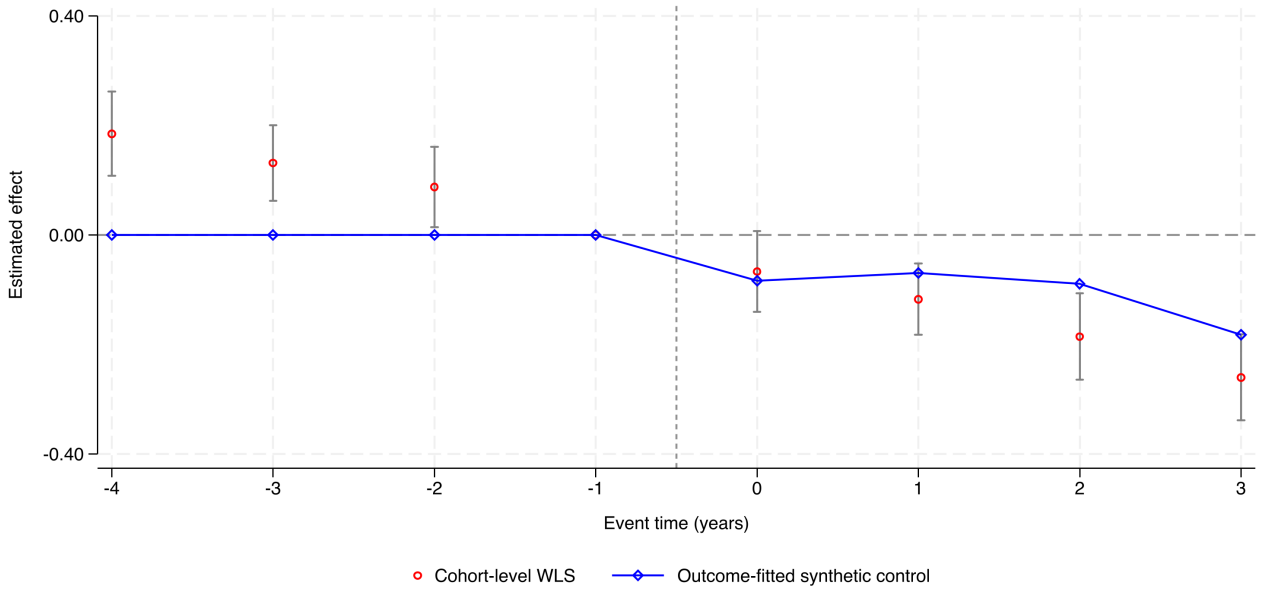
Table 3. Effects of reform exposure on Births per woman (cohort-level)

	DD	Event-study
Estimated coefficient	-0.257***	-0.157***
	(0.018)	(0.031)
Level effect (% of pre mean)	-15.2	-9.3
Observations	3,200	3,200
R-squared	0.456	0.459

Notes: Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. The event-study column averages annual interactions 0–3 relative to omitted bin -1, using their full covariance. Both specifications retain all eligible cohorts; the unsupported final-bin contrast is not reported. Percentages use the corresponding treated pre-reform mean. One observation is a cohort-quarter by province by skill cell. The cohort measure uses a historical female-cohort denominator allocated with weighted 1999 survey skill shares, including childless women; it is not completed fertility. Descriptive means use denominator weights; regressions use birth-count weights. Percentages use the regression-weighted treated pre-reform mean. Standard errors condition on estimated shares. Ceuta/Melilla lack denominator support. Sources: INE birth register, historical births and sex shares, and 1999 Fertility Survey. Figure 5 reports a mean post-reform normalized synthetic gap of -0.106 (-6.3 percent of the treated pre-reform mean), a magnitude diagnostic. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). Stars denote p below .10, .05, .01.

Figure 5 compares the annual WLS event-study estimates with an outcome-fitted synthetic benchmark. The event-study series traces the differential evolution across skill groups, while the synthetic comparison combines high-skilled province paths to closely reproduce the low-skilled pre-reform trajectory and then holds those weights fixed after the reform. The resulting post-reform gap provides a complementary magnitude check and points to the same decline in fertility. Appendix Figures F1 and F2 show that the pattern also appears in raw and log birth counts.

Figure 5. Annual event study for births per woman (cohort-level) with outcome-fitted synthetic control



Notes: The outcome is births per woman (cohort-level). One observation is a maternal cohort-quarter by province by skill cell. Cohort-level WLS denotes the annual event-study regression with eligible birth-count analytical weights. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Annual skill interactions and province fixed effects are included. Intervals are 95% confidence intervals clustered by maternal cohort-quarter; bin -1 is omitted and the unsupported final-bin contrast is not plotted. Symmetric axes retain all intervals, with common limits within same-unit panel pairs. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). The denominator uses historical female cohorts and weighted 1999 survey skill shares, including childless women; it is not completed fertility. Inference conditions on estimated shares. Ceuta/Melilla lack denominator support. Additional sources: INE historical births and sex shares, and 1999 Fertility Survey. Weights combine high-skilled province paths to match the low-skilled pre-reform path in bins -4 through -1; they sum to one and may be negative. The gap is normalized at -1. Its post-reform path is a design-based magnitude diagnostic; pre-period fit is imposed.

Appendix Table F1 reports the Fertility Survey number-of-children estimate in the same table as the birth-record numerator checks. Including childless women, counting each respondent once and applying wave-specific respondent weights gives -0.217 children (s.e. 0.061), consistent in sign with the birth-record estimates in Table 3 and Figure 5. Because the survey waves observe women at different ages, we read this as supporting evidence rather than the main result. The birth-record evidence remains the basis for the main fertility interpretation because it is based on the universe of birth records and the cohort-level denominator; the Fertility Survey estimate is an individual, education-based child-count robustness check from a smaller retrospective sample.

Table 4. Maternal-age outcomes: DD estimates and selection sensitivities

	Maternal age at birth	First-delivery age
Panel A: DD estimates		
Post × low-skilled mother	0.526***	0.441***
	(0.067)	(0.060)
Treated pre-reform mean	27.428	25.368
Effect (% of treated pre-reform mean)	1.9	1.7
Observations	3,382,191	1,880,215
R-squared	0.107	0.154
Panel B: Selection sensitivity		
Conditional: lower bound	0.055	0.036
Conditional: upper bound	0.982	0.877
Joint-normal: lower bound	0.165	0.109
Joint-normal: upper bound	0.885	0.773

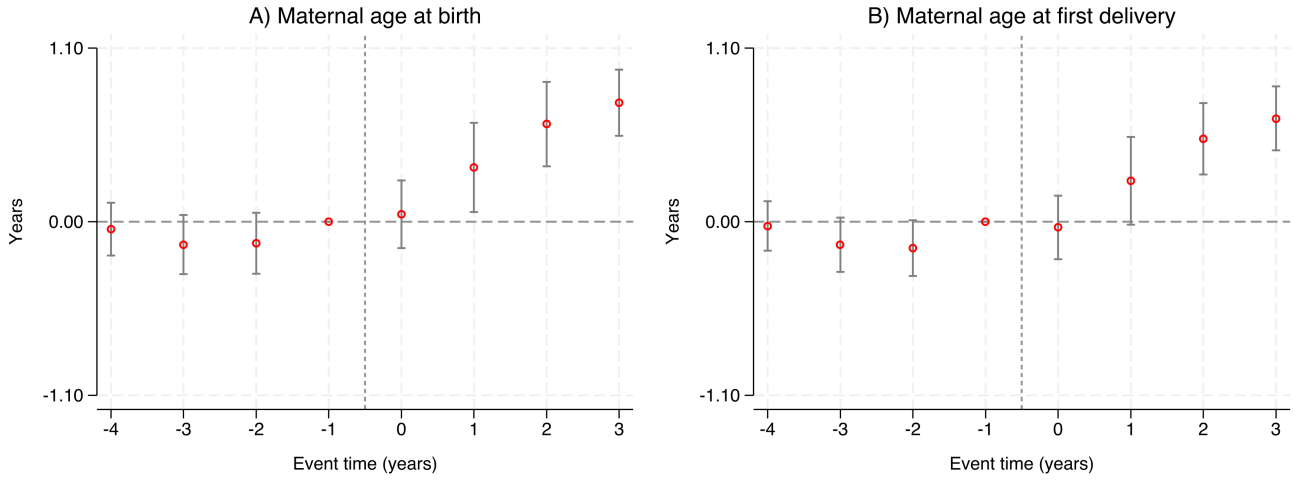
Notes: DD includes skill and exposure indicators, their interaction, cohort and post-cutoff trends, and province effects. Each infant receives equal baseline weight; standard errors in parentheses cluster by maternal cohort quarter. Sensitivity ranges are point endpoints, not confidence intervals. Selection probabilities are estimated using the 2018 Fertility Survey, official respondent weights and women of all nationalities, including childless women. The overall proportional selection loss is 5.204% for maternal age across all births and 4.813% for first-delivery age. The former uses biological motherhood; the latter uses a first biological birth during 1980–2017 at maternal ages 15–55. Ages are in years; every infant from a multiple first delivery is retained. Conditional trimming estimates separate selection rates by maternal birth quarter; joint-normal sensitivity uses the overall rate. Sources: Childbirth Statistics Bulletin, 1980–2017, and Fertility Survey, INE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ apply to DD estimates only.

Timing and Maternal Age at Birth—Exposure increases maternal age at birth by 0.526 years, or 1.9% of the treated pre-reform mean, and first-delivery age by 0.441 years, or 1.7% (Table 4). These estimates correspond to economically meaningful delays in motherhood and are precisely estimated in the birth records. Because birth timing is observed among women who give birth, we complement the baseline estimates with selection-sensitivity exercises that allow reform exposure to change selection into motherhood. Both the conditional Lee-style and joint-normal exercises retain positive timing effects across their reported point-estimate ranges,[30] reinforcing the evidence that reform exposure shifted motherhood to later ages.

The annual paths in Figure 6 trace how these timing differences evolve across cohorts relative to the last pre-reform year.

[30] The exercises use the change in motherhood measured in the 2018 Fertility Survey to calibrate selection into the birth-record sample. The conditional Lee-style approach assumes weakly decreasing selection into motherhood, while the joint-normal approach additionally imposes a joint-normal structure on outcomes and selection.

Figure 6. Annual event studies for maternal age at birth and first-delivery age



Notes: A) maternal age at birth; B) first-delivery age. One observation is an infant record; each infant receives equal weight. Maternal age requires at least 15 years; first-delivery age additionally requires at most 55 years and retains every infant from multiple first deliveries. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Annual skill interactions and province fixed effects are included. Intervals are 95% confidence intervals clustered by maternal cohort-quarter; bin -1 is omitted and the unsupported final-bin contrast is not plotted. Symmetric axes retain all intervals, with common limits within same-unit panel pairs. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

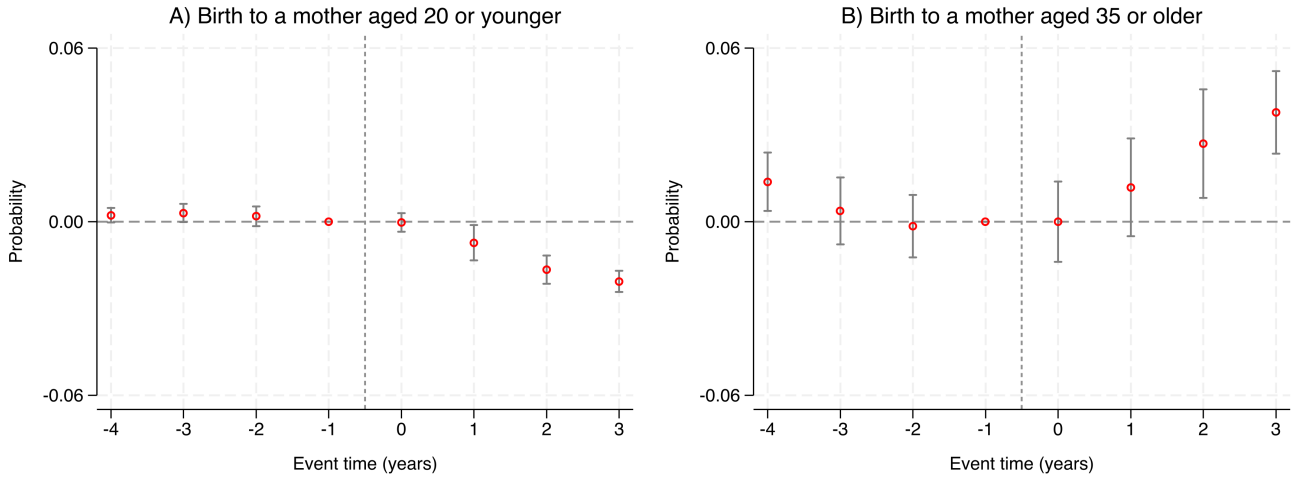
The birth-record evidence carries greater weight here because the birth records are vastly larger, measure realized deliveries directly, and are not affected by survey-wave composition or retrospective sampling in the same way as the Fertility Survey. The birth-record results therefore point to a reallocation of realized births toward older maternal ages. In Table 4, the conditional-trimming ranges are [0.055, 0.982] years for maternal age at birth and [0.036, 0.877] years for first-delivery age. The corresponding joint-normal ranges are [0.165, 0.885] and [0.109, 0.773] years. Both conditional-trimming and joint-normal point ranges are positive for the timing outcomes. These are sensitivity endpoints rather than confidence intervals.

Figure 6 presents the dynamic counterparts to the timing DD estimates. The supported annual post-period averages are 0.440 years (s.e. 0.088) for maternal age at birth and 0.350 years (s.e. 0.082) for first-delivery age. These are averages of annual interaction coefficients relative to bin -1, rather than the DD level effects. The pre-period joint test does not reject for maternal age ($p = 0.204$), but rejects for first-delivery age ($p = 0.025$). Appendix Table F3 reports the corresponding DD and annual average estimates.

Figure 7 describes movements in the maternal-age distribution. The DD estimates reduce the probability that a realized birth occurs to a mother aged 20 or younger and increase the probability that it occurs at age 35 or above. These estimates decompose the age composition of realized births. [31]

[31] Appendix Table F4 reports the generated threshold DD estimates, including age 40 or above, and the figures show the corresponding annual paths.

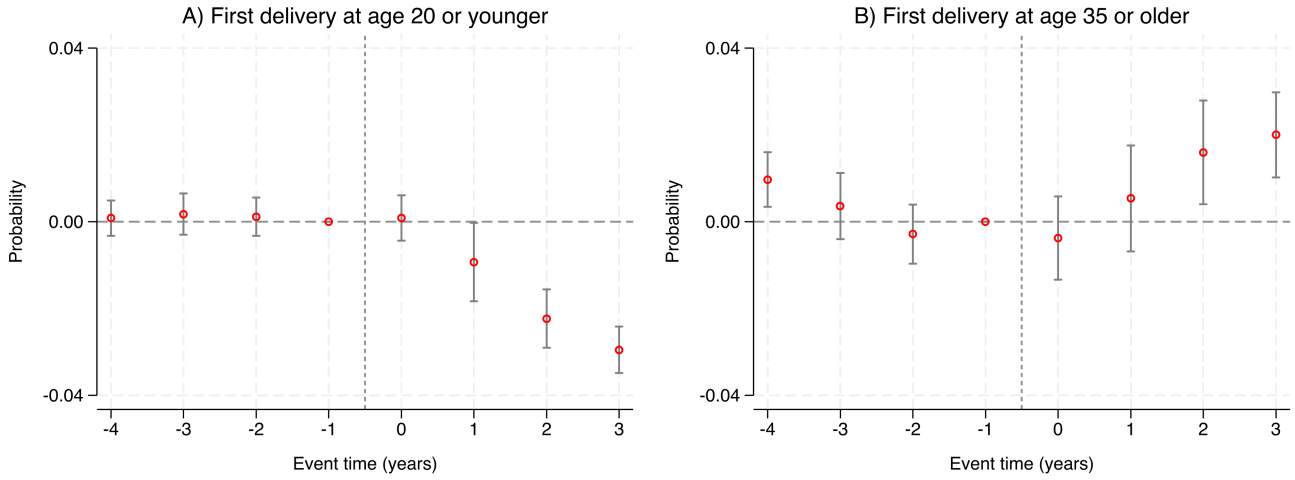
Figure 7. Annual event studies for maternal-age thresholds



Notes: A) birth to a mother aged 20 or younger; B) birth to a mother aged 35 or older. One observation is an infant record with maternal age of at least 15 years; each infant receives equal weight. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Annual skill interactions and province fixed effects are included. Intervals are 95% confidence intervals clustered by maternal cohort-quarter; bin -1 is omitted and the unsupported final-bin contrast is not plotted. Symmetric axes retain all intervals, with common limits within same-unit panel pairs. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

Figure 8 applies the same thresholds to first deliveries, retaining every infant from a multiple first delivery. The DD estimates again indicate fewer first deliveries at age 20 or younger and more at age 35 or above. Table F4 uses the same valid first-delivery-age sample as Table 4; observed missingness and age validity determine its sample size. The annual paths describe the timing pattern and should be interpreted together with the pre-period diagnostics.

Figure 8. Annual event studies for first-delivery-age thresholds



Notes: A) first delivery to a mother aged 20 or younger; B) first delivery to a mother aged 35 or older. One observation is an infant from a first delivery with maternal age between 15 and 55 years, including all infants from multiple first deliveries; each infant receives equal weight. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Annual skill interactions and province fixed effects are included. Intervals are 95% confidence intervals clustered by maternal cohort-quarter; bin -1 is omitted and the unsupported final-bin contrast is not plotted. Symmetric axes retain all intervals, with common limits within same-unit panel pairs. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

5.3. *Secondary perinatal outcomes among observed births*

The DD estimates in Table 5 describe realized first deliveries and give a mixed picture of perinatal outcomes. Birthweight increases by 3.788 grams (s.e. 1.585), and gestational age rises by 0.066 weeks (s.e. 0.008). Preterm birth falls by 1.22 percentage points. These outcomes use available fields throughout 1980–2017. C-sections increase by 4.38 percentage points, but that field is available only in 2007–2017, when these maternal cohorts are older. Its coefficient consequently describes a different calendar-support sample. The evidence does not support a general adverse perinatal effect, and differences across these outcomes cannot be attributed to maternal age alone.

Table 5. Perinatal outcomes at first delivery: DD estimates and selection sensitivities

	Birthweight	Low BW	High BW	Gest. age	Preterm	C-section
Panel A: First deliveries: DD level effects						
Exposed cohort × low-skilled	3.788**	-0.001	0.001**	0.066***	-0.012***	0.044***
	(1.585)	(0.001)	(0.000)	(0.008)	(0.001)	(0.009)
Treated pre mean	3196.912	0.071	0.038	38.931	0.118	0.451
Mean dep. var.	3189.830	0.075	0.038	39.020	0.098	0.408
Effect (% of treated pre mean)	0.1	-1.8	3.1	0.2	-10.3	9.7
Observations (millions)	1.71	1.71	1.71	1.48	1.48	0.08
R-squared	0.004	0.001	0.001	0.007	0.017	0.023
Panel B: Selection sensitivity						
Conditional: lower bound	-49.670	-0.005	-0.001	-0.182	-0.016	0.023
Conditional: upper bound	46.095	0.036	0.022	0.184	0.026	0.069
Joint-normal: lower bound	-32.754	-0.005	-0.001	-0.079	-0.016	0.021
Joint-normal: upper bound	40.338	0.045	0.037	0.211	0.034	0.071

Notes: DD includes skill and exposure indicators, their interaction, cohort and post-cutoff trends, and province effects. Each infant receives equal baseline weight; standard errors in parentheses cluster by maternal cohort quarter. Sensitivity ranges are point endpoints, not confidence intervals. Selection probabilities are estimated using the 2018 Fertility Survey, official respondent weights and women of all nationalities, including childless women. All first-delivery outcomes use the 4.813% overall reference selection loss estimated for a first biological birth during 1980–2017 at maternal ages 15–55. Health samples retain their own outcome availability; C-section is observed in 2007–2017. Every infant from a multiple first delivery is retained. Birthweight is in grams; low/high birthweight is below 2500/above 4000 grams; gestation is in weeks; preterm is below 37 weeks. Binary normal sensitivities use latent thresholds with unrestricted cell correlations. Conditional trimming estimates separate selection rates by maternal birth quarter; joint-normal sensitivity uses the overall rate. Birth country is unrestricted in the survey calibration. Appendix F.2 describes both methods and their assumptions. Sources: Childbirth Statistics Bulletin, 1980–2017, and Fertility Survey, INE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ apply to DD estimates only.

Panel B reports conditional-trimming and joint-normal selection sensitivities for the same first-delivery outcomes, using the 4.813% overall reference selection loss and birth-quarter-specific conditional rates. For gestational age, the ranges are [-0.182, 0.184] and [-0.079, 0.211] weeks, respectively. The perinatal ranges should be read separately by outcome and calendar support, under the assumptions in Appendix F.2.

5.4. *Contractual instability and unmet desired fertility*

Economic outcomes. —The survey provides suggestive evidence of persistent temporary employment. Among employees with an observed contract, exposure raised the probability of holding a temporary job by 15.7 percentage points ($p = 0.0331$), or 63.7% of the treated pre-reform mean (Table 6). Marriage, homeownership, difficulty making ends meet, and the probability of monthly net income of at least 1,000

euros have imprecisely estimated responses. Temporary contracts can make tenure, earnings, credit access, and maternity-related employment protection less secure, raising the cost of committing to births. These outcomes describe women interviewed in 2018 and use their original respondent weights. Appendix F, Table F6, reports multiple-testing adjustments separately within the five economic outcomes in Table 6 and the five preference and support outcomes in Table 9. The temporary-contract estimate has a within-table BH-adjusted p-value of 0.1656.

Household structure.—Table 7 describes family composition among realized births. Missing father information means that at least one recorded paternal demographic or occupation field is absent; it does not necessarily mean that the father is absent from the household. The other outcome identifies unmarried mothers who report not cohabiting. Cohabitation information is available only from 2007, so earlier records do not enter that outcome's regression. The DD for missing father information is close to zero among all births. Noncohabiting single motherhood rises by 2.19 percentage points among all births observed in 2007–2017 and by 1.85 percentage points among first deliveries, with the latter estimate less precise. These are household characteristics at delivery, not estimates of changes in partnership stability over women's lives.

Table 6. Economic outcomes and household resources (DD), Fertility Survey 2018

	(a) Married	*(b)* Home ownership	*(c)* Difficulty making ends meet	*(d)* Temporary contract	*(e)* Monthly income ≥1,000 EUR
Exposed cohort × low-skilled	0.018	−0.094	0.065	0.157**	0.103
	(0.101)	(0.070)	(0.096)	(0.071)	(0.096)
Mean dep. var.	0.695	0.798	0.497	0.177	0.387
Treated pre mean	0.744	0.792	0.671	0.247	0.131
Effect (% of pre mean)	2.4	-11.8	9.7	63.7	78.7
R^2	0.026	0.033	0.080	0.041	0.149
Observations	3,611	3,611	3,611	1,812	3,611

Notes: One observation per woman, with valid responses for each outcome. Women of all nationalities are included, whether or not they have children. General outcomes impose no restrictions on children's nationality or birth years. Low-skilled women (education below high-school completion) are born in 1965Q1–1972Q4, with cutoff 1968Q4; high-skilled women (high-school completion or above) are born in 1962Q1–1969Q4, with cutoff 1966Q1. All regressions include Post, Treat, a common linear cohort-quarter trend and its post-cutoff interaction, region and maternal birth-year fixed effects, original 2018 respondent weights, and standard errors clustered by maternal calendar birth-cohort quarter. Means and percentage effects use each regression's sample and respondent weights. Temporary-contract status conditions on employees with an observed contract and excludes self-employed women and nonworkers. The income indicator is net monthly income of at least 1,000 EUR; it is not a poverty or minimum-wage measure. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. *Source:* 2018 Fertility Survey, INE.

Household economic insecurity. —The household economic insecurity analysis in Table 8 asks whether fertility quantity and timing responses differ with the skill of the father recorded at birth. Couple-level fertility evidence motivates considering both partners' employment stability (Vignoli, Drefahl and De Santis 2012; Huttunen and Kellokumpu 2016). Father characteristics are observed only at realized births, so this is a mechanism and heterogeneity exercise within the mother-cohort design. The principal regression excludes missing or unclassified father occupation and compares the two mutually exclusive low-skilled-mother groups with all high-skilled mothers with classified recorded father skill, pooling low- and high-skilled fathers in that comparison.

Table 7. Family structure at birth

	All births		First deliveries	
	Missing father	Single mother	Missing father	Single mother
Exposed cohort × low-skilled	-0.000	0.022***	-0.001**	0.018*
	(0.000)	(0.008)	(0.000)	(0.010)
Treated pre mean	0.021	0.083	0.032	0.127
Mean dep. var.	0.015	0.083	0.023	0.131
Mean dep. var. (treat=1)	0.020	0.079	0.030	0.123
Effect (% of treated pre mean)	-0.1	26.3	-2.9	14.6
Observations (millions)	3.38	0.23	1.88	0.08
R-squared	0.005	0.013	0.007	0.017

Notes: Standard errors in parentheses; observations in millions. One observation is an infant record. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. First deliveries retain every infant from multiple deliveries. The appendix discussion following Table F3 reports sensitivity to conflicting parity information. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. Regressions are unweighted. Missing father information means at least one of father age, occupation, birth month or birth year is missing; it is not the father-skill classification indicator. Noncohabiting single motherhood means unmarried and not cohabiting, observed only in 2007–2017; married mothers are zero. Missing cohabitation responses among unmarried mothers remain missing. Outcome availability changes counts, not eligibility. Means and percentage scaling use each regression sample. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (a) allocates births per woman at the cohort level across father categories using their shares of eligible realized births in each mother cohort-quarter × province × skill cell. Missing-father births remain in this accounting total, but their component is excluded from the principal regression. The high-skilled-mother components are pooled across classified father categories; the two low-skilled-mother components remain separate. The denominator and original cell-size analytical weights are unchanged. Columns (b)–(c) use unweighted infant records with valid maternal age and first-delivery age. We estimate

$$\begin{aligned} Y_{icp} = \alpha_p + \gamma_{LL}\mathbf{1}\{M_L, F_L\} + \gamma_{LH}\mathbf{1}\{M_L, F_H\} + \delta Post_i \\ + \beta_{LL}Post_i \times \mathbf{1}\{M_L, F_L\} + \beta_{LH}Post_i \times \mathbf{1}\{M_L, F_H\} + \theta r_i + \kappa Post_i r_i + \varepsilon_{icp}, \end{aligned} \tag{7}$$

where M_L denotes a low-skilled mother and F_L and F_H denote classified low- and high-skilled fathers. The omitted group pools high-skilled mothers across both classified father categories. Exposure and the running variable use the mother's cohort and skill-specific entry cutoff; province fixed effects are included and errors cluster by maternal birth-cohort quarter. Both parents use the occupation mapping in Table D1. The birth-quantity coefficients are -0.182 for low-skilled mothers with low-skilled fathers and -0.047 for low-skilled mothers with high-skilled fathers. Maternal-age effects are 0.406 and 0.995 years, and first-delivery-age

Table 8. Father-side skill composition and household economic insecurity

	(a) Births per woman (cohort-level)	*(b)* Maternal age at birth	*(c)* First-delivery age
Panel A. Mother-side benchmark			
Post × low-skilled mother	-0.257***	0.526***	0.441***
	(0.018)	(0.067)	(0.060)
Panel B. Father-side skill among observed births			
Post × low-skilled mother × low-skilled father	-0.182***	0.406***	0.331***
	(0.013)	(0.064)	(0.058)
Post × low-skilled mother × high-skilled father	-0.047***	0.995***	0.938***
	(0.014)	(0.085)	(0.084)
Treated pre mean: low-skilled father	1.197	27.023	25.021
Observations	4,800	3,148,416	1,737,176
R-squared	0.736	0.121	0.167

Notes: Panel A gives mother-side benchmarks; Panel B gives mutually exclusive father-skill interactions. Panel B compares each low-skilled-mother category with high-skilled mothers pooled across classified father skills; missing/unclassified father occupation is excluded. Its summary statistics refer to its own samples. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. Quantity uses couple-group components of the same mother-cohort measure, with birth-count weights; missing-father births remain in the accounting denominator. Timing uses unweighted infant records with outcome-specific validity. First deliveries retain every infant from multiple deliveries. The appendix discussion following Table F3 reports sensitivity to conflicting parity information. Percentage effects use category-specific treated pre-reform means. Stars denote p below .10, .05 and .01. *Sources:* INE birth register, historical births and sex shares, and 1999 Fertility Survey.

effects are 0.331 and 0.938 years. Thus the quantity decline is larger when both parents are low-skilled, while timing delays occur in both groups and are larger with high-skilled fathers. These are father-category components of mother-cohort fertility, not fertility rates for populations of fathers.

Relative to the preceding specification that kept the two classified high-skilled-mother categories separate, the pooled comparison preserves the signs and ordering of the results but changes their magnitudes. On the same classified-father sample, the quantity coefficients change from -0.154 and -0.020 to -0.182 and -0.047. The latter change is 141.6 percent of its preceding magnitude, so simplification is consequential for that subgroup even though the qualitative interpretation remains. Appendix Table F2a reports missing-father diagnostics separately. Appendix Table F2 and Figure F3 apply the same classified-father comparison to annual event studies.

Preferences, norms, and childcare support.—Table 9 reports the preference and support margins. The estimated responses in childcare use, egalitarian views, religious affiliation, and age at economic inde-

Table 9. Preferences, norms, and support margins (DD), Fertility Survey 2018

	(a) Age at economic independence	*(b)* No childcare used (under 14)	*(c)* Unmet desired fertility	*(d)* Egalitarian attitudes	*(e)* No religious affiliation
Exposed cohort × low-skilled	0.760	−0.205	0.279**	0.029	0.019
	(1.224)	(0.231)	(0.106)	(0.031)	(0.070)
Mean dep. var.	22.802	0.278	0.388	0.720	0.174
Treated pre mean	21.874	0.396	0.294	0.691	0.132
Effect (% of pre mean)	3.5	-51.8	95.1	4.3	14.5
R^2	0.047	0.069	0.021	0.057	0.036
Observations	3,057	629	3,611	3,611	3,265

Notes: One observation per woman, with valid responses for each outcome. Women of all nationalities are included, whether or not they have children. General outcomes impose no restrictions on children's nationality or birth years. Low-skilled women (education below high-school completion) are born in 1965Q1–1972Q4, with cutoff 1968Q4; high-skilled women (high-school completion or above) are born in 1962Q1–1969Q4, with cutoff 1966Q1. All regressions include Post, Treat, a common linear cohort-quarter trend and its post-cutoff interaction, region and maternal birth-year fixed effects, original 2018 respondent weights, and standard errors clustered by maternal calendar birth-cohort quarter. Means and percentage effects use each regression's sample and respondent weights. No childcare used equals zero days of all listed formal and informal childcare arrangements (question 6.5), among parents with a living child under 14 and valid responses. Women outside that population are excluded from this column. Unmet desired fertility equals desired children exceeding biological children. The egalitarian index averages observed recoded items on a 0–1 scale. No religious affiliation equals religion `6` (none); `7` (prefer not to answer) is missing, never treated as an absence of belief. This is affiliation at interview, not a change in observance. The economic-independence age retains source winsorization. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. *Source:* 2018 Fertility Survey, INE.

pendence are imprecise. Religious nonresponse is excluded from the affiliation outcome, and this measure describes affiliation at interview rather than a change in religious observance. The notable fertility-preference margin is unmet desired fertility: the probability that desired children exceed realized biological children rises by 27.9 percentage points ($p = 0.0116$), or 95.1% of the treated pre-reform mean. Its within-table BH-adjusted p-value is 0.0579. This larger gap is consistent with constrained realization of fertility plans. Together with the birth-record estimates and the temporary-contract response, it supports contractual instability as a mechanism behind lower realized fertility. The father-skill estimates provide separate household heterogeneity evidence among realized births while preserving the mother-cohort fertility denominator.

6. ROBUSTNESS CHECKS

The identifying interpretation faces four principal threats: contemporaneous changes in schooling or family policy, treatment misclassification, differential cohort trends, and selection into motherhood. The main text reports what the corresponding checks establish; Appendix F contains the specification tables, cohort-position diagnostics, and multiple-testing corrections. Annual pre-trend findings are reported with the event-study results. Conditional-trimming and joint-normal selection sensitivities appear with the timing and perinatal results. Appendix Table F5 reports inference for the headline estimates.

Contemporaneous schooling, employment, and family-policy changes—The reform could affect fertility through schooling or broad employment changes rather than through the contractual conditions of entry. Appendix Table F7 reports imprecisely estimated changes in current paid work, long-term unemployment, age at first job, age at leaving education, and the stated need for flexible childcare to consider having a child. The administrative first stage separately shows that days worked did not change detectably, while non-permanent employment rose and contribution-based income fell. Together, these results do not point to withdrawal from work as the main mechanism; they do not imply that earnings were unaffected.

Family-policy timing provides a second check. Political control was stable during the central treatment window, and the major maternity- and paternity-leave changes discussed in Appendix Table A2.4 occurred later and did not vary by the skill-specific entry cutoffs. The stated need for flexible childcare to consider having a child also shows no differential change in Appendix Table F7. These checks do not detect differential changes in the observed schooling and childcare measures.

Treatment measurement and sample construction—The birth records do not consistently report education, so the maintained analysis classifies skill using occupation or status at delivery and uses historical female cohort counts allocated with weighted 1999 survey skill shares. Appendix D documents this mapping. A broader control-group definition that includes mothers still in education leaves the sign of all three headline outcomes unchanged: the effect on births per woman at the cohort level remains negative, while maternal age at birth and age at first birth remain positive (Appendix Table F8). The education-based Fertility Survey child-count estimate also points in the same direction. Together, these checks reduce the concern that the results are an artifact of the maintained birth-record classification.

Exposure gradients, cohort trends, and bandwidths—A cohort-position specification replaces the binary cutoff with interactions between maternal skill and bins of an inverse post-cutoff cohort index (Appendix Table F9 and Figure F5). This index locates cohorts relative to the entry cutoff; it does not measure contract duration or individual exposure intensity. Later post-cutoff cohorts show larger negative quantity coefficients and larger positive timing coefficients. The earlier post-cutoff bins show smaller responses, and the first-delivery-age coefficient in the earliest bin is imprecise. These are differences across cohort positions, not a dose-response estimate based on observed contractual histories.

Adding quadratic and cubic cohort terms preserves the negative quantity coefficient and positive timing coefficients (Appendix Table F10). A shorter window of event quarters -8 through $+7$ also preserves these signs, with smaller magnitudes; its first-delivery-age effect is not statistically distinguishable from zero at the five-percent level (Appendix Table F11). The sensitivity of magnitudes and precision matters when interpreting the maintained DD estimates. These checks assess particular cohort-trend and bandwidth

choices and cannot rule out every cohort-specific shock. Appendix Table F12 reports the corresponding expanded-cohort-window check.

7. A PARSIMONIOUS LIFE-CYCLE MODEL OF EMPLOYMENT RISK AND FERTILITY

The reduced-form evidence shows that reform exposure increased years of non-permanent employment and lowered contribution-based income early in low-skilled women's careers, while reducing births per woman at the cohort level and delaying first motherhood. These results establish a persistent link between early-career employment conditions and fertility, but they do not reveal how much delayed fertility is eventually recovered or which labor-market channel drives the response. We therefore build and estimate a life-cycle model that links changes in entry conditions, subsequent employment stability, and contribution-based income to fertility decisions over the reproductive life cycle. The model allows us to separate these channels, distinguish delayed births from births not realized by age 40, and evaluate how fertility responds to counterfactual changes in early-career employment insecurity.

The model follows a woman through the reproductive life cycle as she decides each year whether to attempt a birth given her current employment situation, accumulated temporary-employment exposure, contribution-based income, and remaining reproductive time. Waiting may improve future employment stability and income, but it also shortens the horizon over which postponed births can be realized. The 1984 reform shifts the labor-market conditions women face at entry and later in their careers, and the model maps these changes into birth timing and completed fertility. Our model builds on dynamic models of fertility choice (Hotz and Miller 1988; Sommer 2016), income and employment risk (Ranjan 1999; Low, Meghir and Pistaferri 2010), and career costs of childbearing (Adda, Dustmann and Stevens 2017); its application to labor-market duality is closest in spirit to Guner, Kaya and Sánchez-Marcos (2024). de la Croix and Pommeret (2021) study the option value of postponement under income risk and a biological clock; here, the employment shock is disciplined by Spain's reform and observed labor histories.[32]

7.1. *Economic environment and fertility choice*

Women differ by skill, $s \in \{L, H\}$, where L and H denote low- and high-skilled women, and make annual fertility decisions from ages 16 through 40. Their demographic state records parity, $n \in \{0, 1, 2, 3\}$, where the terminal parity of three is compared with the empirical three-or-more category, and years since the last birth, $g \in \{0, 1, 2, 3\}$. The model also tracks accumulated temporary-employment exposure, $h \in \{0, \ldots, 25\}$, which allows past job insecurity to affect current fertility choices even after a woman leaves temporary

[32] We keep the model deliberately parsimonious and focus on the woman's employment history. We do not model household bargaining or endogenous partner employment because the available longitudinal data do not jointly identify those transitions (Doepke and Kindermann 2019; Doepke et al. 2023).

employment. Current labor-market status is $e \in \{O, T, P\}$, where O denotes an annual main affiliation outside covered dependent wage employment, T temporary employment, and P permanent employment. The outside state includes schooling and nonparticipation and is therefore not equivalent to unemployment. Potential labor-market entry is set at age 16 for low-skilled women and age 18 for high-skilled women, following the reduced-form design; actual covered employment can begin later. The distribution across labor states at entry and the subsequent age- and skill-specific transition probabilities are estimated from the MCVL histories of pre-cutoff potential-entry cohorts. Together, skill, labor status, parity, birth spacing, and accumulated temporary-employment exposure form the state x_a that a woman observes when making her fertility decision at age a.[33]

Each year, the woman first observes her current state and decides whether to try to have a child. A birth then occurs with an age-dependent probability, and her employment status evolves before the next period. We summarize this sequence as

$$x_a = (s, e_a, n_a, g_a, h_a), \qquad d_a \in \mathcal{D}(n_a), \qquad B_a = d_a C_a, \quad C_a \sim \text{Bernoulli}\{q(a)\}, \quad e_{a+1} \sim P_{s,a}(e_a, \cdot). \tag{8}$$

At age a, x_a denotes the woman's state, containing skill (s), current employment status (e_a), parity (n_a), years since the last birth (g_a), and accumulated temporary-employment exposure (h_a). The action set is $\mathcal{D}(n) = \{0, 1\}$ for $n < 3$ and $\mathcal{D}(3) = \{0\}$: $d_a = 1$ denotes an attempt and $d_a = 0$ waiting. No further birth is possible at parity three. Conditional on an attempt, C_a indicates whether it results in a live birth, with probability $q(a)$, so that $B_a = d_a C_a$ is the realized-birth indicator. Employment then evolves according to the age- and skill-specific transition matrix $P_{s,a}$, which gives the distribution of next period's employment state e_{a+1} conditional on the current state e_a. A realized birth increases parity and resets years since the last birth, while temporary employment increases accumulated temporary-employment exposure. Fertility choices are therefore endogenous, while employment transitions and the age-specific birth-success probability are taken as inputs to the fertility decision.[34]

The fertility decision is forward looking because waiting can improve future employment prospects and contribution-based income, but it also reduces the time available to realize postponed births. We estimate the parameters governing this trade-off by solving the following Bellman problem:

[33] High-skilled women are assigned to the outside-employment state (O) at ages 16–17 under the model's potential-entry convention, while fertility choices remain available at those ages. This is a normalization around expected labor-market entry, not an assumption that no high-skilled woman works before age 18. Appendix Figure H1 shows the observed and model-implied labor-state profiles. The MCVL histories condition on eventual recorded affiliation, so their mapping to the Fertility Survey population is a maintained sampling assumption.

[34] The spacing state g_a is used to track birth intervals and does not carry a separate utility cost. The birth-success schedule $q(a)$ is calibrated from external demographic evidence, with age-specific values anchored to Leridon (2004) (Figure G2 shows the complete schedule). The employment transition matrices $P_{s,a}$ are estimated from the MCVL histories and are described in Appendix G.

$$V_a(x) = E_\varepsilon\left[\max_{d\in\mathcal{D}(n)}\{v_a^d(x)+\varepsilon_{ad}\}\right], \qquad v_a^d(x) = u_a^d(x) + \beta E[V_{a+1}(x') \mid x, d], \tag{9}$$

where $V_a(x)$ is the expected lifetime value at age a in state x, and $d \in \mathcal{D}(n)$ denotes the feasible choice to wait or attempt a birth. The choice-specific value $v_a^d(x)$ combines the current payoff, $u_a^d(x)$, with the expected value of future decisions, $V_{a+1}(x')$, where x' is next period's state. The discount factor β determines how much weight the woman places on future utility, while ε_{ad} captures choice-specific factors observed by the woman but not by the researcher. This recursive structure gives waiting an option value: delaying a birth may improve future labor-market conditions, but it also leaves fewer periods in which postponed fertility can be recovered.[35]

To make the current payoff in the Bellman problem explicit, we write the payoff from a successful birth as

$$\begin{aligned}\Delta u_a(x; D_i, R_i, G_i) = {} & \alpha_0 + \alpha_2 1\{n=1\} + \alpha_3 1\{n=2\} + \alpha_D D_i + \alpha_R R_i + \alpha_G G_i \\ & - c_A(a) - c_H 1\{s=H\}\omega_H(a) - c_T \mathcal{I}_{s,a}(e,h) + \log c_a^1(x) - \log c_a^0(x).\end{aligned} \tag{10}$$

The birth-order component is governed by the baseline value of a birth, α_0, together with the second- and third-birth adjustments, α_2 and α_3. Observed preference heterogeneity enters through desired family size, religiosity, and traditional gender norms, with coefficients α_D, α_R, and α_G; the corresponding measures D_i, R_i, and G_i are described below. The non-resource costs capture age, career, and employment insecurity. The age-related cost $c_A(a)$ allows the payoff from childbearing to vary over the reproductive life cycle, while c_H governs the additional early-career opportunity cost for high-skilled women and $\omega_H(a)$ determines how that cost varies with age. The parameter c_T scales expected insecurity during early parenting, $\mathcal{I}_{s,a}(e,h)$. This cost combines the full accumulated history $\log(1+h)$ with the chance of remaining outside permanent employment, so waiting for a more stable job can make parenthood less costly without erasing earlier exposure. Appendix Equation (G1) defines the expectation and the accumulation of history. Finally, $\log c_a^1(x) - \log c_a^0(x)$ captures the change in available resources associated with an additional child.[36]

The preference terms in Equation 10 come from the 2018 Spanish Fertility Survey. Desired family size (D_i), religiosity (R_i), and traditional gender norms (G_i) capture observed differences across women in the value they attach to having a child. Desired family size is especially important for the model because it allows women to want the same number of children but realize different fertility outcomes depending on their employment history, contribution-based income, birth success, and remaining reproductive time. We

[35] We impose $V_{41}(x) = 0$ as the terminal condition. The shocks ε_{ad} are Type-I extreme value, which gives the standard logit choice probability. Birth-success risk enters both current and continuation values through the age-specific probability $q(a)$ defined in Equation 8. Appendix G provides the full solution and numerical implementation.

[36] The age-related cost $c_A(a)$ is governed by separate early- and late-age parameters, c_E and c_L, with its fixed age weights specified in Appendix Equation (G6). Appendix G details the career-cost profile $\omega_H(a)$ and the construction of resources $c_a^j(x)$.

use the corresponding relationships between these measures and observed fertility in the survey to discipline the preference parameters α_D, α_R, and α_G.[37]

We jointly estimate ten parameters governing the value of children, fertility timing, career and insecurity costs, and observed preference heterogeneity. Employment transitions, contribution-based income profiles, the reform-induced labor-market changes, the discount factor, and the age-specific birth-success schedule are estimated or calibrated separately and enter the woman's decision problem as given. Given these inputs, the model determines birth attempts, realized births, and completed fertility over the life cycle.

7.2. *Labor-market inputs and structural estimation*

Before estimating fertility preferences, we use MCVL histories of pre-cutoff potential-entry cohorts to determine the age- and skill-specific probabilities of moving across the outside-employment (O), temporary (T), and permanent (P) states, together with median contribution-based income in temporary and permanent employment; resources outside covered employment are calibrated as described in Appendix G. These histories include women who enter employment after the potential entry age and years spent outside covered employment. Appendix Figure H1 compares the resulting labor-state profiles with those observed in the MCVL, while Appendix Tables G2 and G4 report contribution-based income across labor states and transitions out of temporary employment.[38]

The 1984 reform changes this baseline labor process through entry conditions, subsequent contract persistence, and contribution-based income. The reference regime describes the employment histories of earlier potential-entry cohorts, including their subsequent careers. For low-skilled women, these changes are represented by

$$\begin{aligned} p^R_{L,16} &= p^0_{L,16} + (-\Delta_{ET} - \Delta_{EP}, \Delta_{ET}, \Delta_{EP}), \\ P^R_{L,a}(e,e') &\propto \widetilde{P}^{\delta}_{L,a}(e,e') \exp\{\rho 1[a \le 25, e' = T] + \eta 1[a \le 18, e = O, e' \neq O]\}, \\ y^R_{L,a,e} &= y_{L,a,e} \exp\{\Delta_y 1[a \le 25, e \in \{T,P\}]\}. \end{aligned} \tag{11}$$

Here $p^0_{L,16}$ is the baseline distribution of low-skilled women across labor states at age 16 and $p^R_{L,16}$ is the corresponding distribution under reform exposure. The entry-stock changes Δ_{ET} and Δ_{EP} are estimated using the model's annual labor states and skill groups: -0.038 percentage points in temporary employment and +2.024 in permanent employment. A separate early-entry adjustment, η, shifts transitions from O

[37] Desired family size is observed for all respondents, including childless women, and is capped at three to match the model's parity state. The cultural measures are standardized, and the survey relationships condition jointly on the three preference measures, skill, age, and region. These relationships are used as conditional associations rather than causal effects of changing preferences or beliefs. Appendix G documents their construction and empirical counterparts.

[38] Some age-by-skill transition cells contain few observations. We use a one-half Jeffreys correction before normalizing each transition row, which prevents mechanically zero transition probabilities in sparse cells. Appendix G provides the full construction.

to covered employment through age 19, preserving the conditional temporary/permanent destination mix before the temporary-destination adjustment. It matches a +13.244 percentage-point increase in covered entry by 19 in the same structural labor sample. The parameter δ changes temporary persistence relative to permanent employment, while ρ shifts early-career destinations toward temporary employment through age 25. Finally, Δ_y shifts employed contribution-based resources during the first decade.[39]

The parameters governing the post-entry labor process are disciplined by the same labor-market margins documented in the reduced-form analysis. The first-decade increase of 0.415 years of non-permanent employment disciplines early exposure, while later MCVL histories discipline temporary-employment persistence and transitions into permanent work. The temporary-contract share observed in the 2018 Fertility Survey provides an additional later-career employment moment. Contribution-based income is disciplined by the first-stage decline of €535.82 relative to its corresponding pre-reform mean. Appendix Table G3 reports these empirical labor estimates together with the fitted labor-process parameters, and Appendix Table G4 reports the underlying transitions from temporary employment.[40]

We then estimate the ten preference parameters using 18 fertility moments.[41] Twelve moments describe fertility levels and timing across skill groups, three capture the conditional relationships of desired family size, religiosity, and gender norms with fertility, and three capture the fertility response to reform exposure. The reform moments are the Fertility Survey effects on capped biological children and motherhood through age 40 and the birth-record first-delivery-age effect within ages 16–40. The published reduced-form estimates retain their original support; Appendix H explains the horizon-matched structural counterparts. First-birth timing is summarized using births up to age 25, ages 26–31, and age 32 or older, together with mean first-birth age. Appendix Table H2 reports numerical local parameter sensitivity to the reform targets, while Appendix Table H4 reports all 18 empirical targets and their model counterparts.[42]

[39] These structural first-step regressions use the frozen MCVL histories. The published conditional first-affiliation-by-19 DD of +14.30 percentage points concerns a different conditioning event and remains a separate source diagnostic. It is no longer interpreted as an age-16 temporary-employment stock effect. High-skilled potential entry remains at 18 and its labor process is held fixed in the differential reform experiment. This implements the identified relative contrast with the high-skill response normalized to zero; the resulting effects are relative to that reference.

[40] The annual $T \to P$ transition records a move from temporary employment in one year to permanent employment in the next and may therefore involve a change of employer. It should not be interpreted as a within-firm promotion. In addition, the reduced-form measure of years of non-permanent employment includes specified employee special regimes. A job recorded as permanent within one of these regimes remains in state P in the structural model. Appendix G describes how the observed non-permanent-employment measure is mapped into the structural $O/T/P$ states. Estimation uses women's complete observed labor histories rather than treating each annual observation as independent.

[41] The structural Fertility Survey targets include women of all nationalities, as does the reduced-form survey analysis. Appendix H specifies the structural sample and its link to the birth-record timing target.

[42] The birth-record outcome, births per woman at the cohort level, is a cohort-level measure of birth intensity rather than completed fertility for individually identified women. It therefore remains reduced-form evidence and is not used as a structural completed-fertility target. The more detailed first-birth age cells are reported as specification checks in Appendix Table H4.

Let θ denote the vector of ten preference parameters, $\widehat{m}$ the vector of 18 empirical moments, and $m(\theta)$ their model counterparts. We estimate θ by covariance-weighted minimum distance,

$$\widehat{\theta} = \arg\min_{\theta \in \Theta} [m(\theta) - \widehat{m}]' \widehat{W} [m(\theta) - \widehat{m}], \tag{12}$$

where $\widehat{W}$ weights discrepancies between the model and data according to their sampling uncertainty. The parameters are therefore chosen jointly to bring the model's fertility levels, timing, preference gradients, and reform responses as close as possible to their empirical counterparts. Appendix Table H3 reports the numerical diagnostics for the estimator, and Appendix Table H2 reports the parameter-to-moment sensitivity analysis following Andrews, Gentzkow and Shapiro (2017).[43]

Table 10 reports the estimated preference parameters that govern fertility choices and therefore determine the model's mechanism and counterfactual responses. The child-value coefficients α_0, α_2, and α_3 govern entry into motherhood and progression across parities. The age parameters underlying $c_A(a)$ govern how the cost of childbearing changes over the reproductive life cycle, while c_H governs the additional early-career opportunity cost for high-skilled women. The temporary-exposure parameter is estimated at $c_T = 0.998$ with a standard error of 0.303. Because it enters the birth payoff as $-c_T \mathcal{I}_{s,a}(e,h)$, greater accumulated temporary-employment exposure lowers the value of a birth when early parenting is expected to occur outside permanent employment. Permanent employment relieves the current component of this burden, while future job loss can reactivate the effect of retained history. Finally, α_D, α_R, and α_G govern heterogeneity associated with desired family size, religiosity, and traditional gender norms. Appendix Table H2 shows the empirical variation that disciplines each of these coefficients, while Appendix Table H4 shows how the resulting parameterization fits the moments used in estimation. These estimates provide the preference parameters used in the model-fit and counterfactual exercises that follow.

[43] The covariance calculation accounts for dependence among Fertility Survey moments and propagates uncertainty from the MCVL labor-market first step and first-birth timing from birth records. Estimation uses eight dispersed starting values, and the reported solution has a full-rank numerical Jacobian with parameter bounds and constrained stationarity checked explicitly. Appendix H provides the covariance construction, regularization, bootstrap checks, and numerical implementation.

Table 10. Estimated preference parameters

Parameter	Estimate	Standard error
Birth value	-0.875	0.243
Second-birth shift	1.089	0.160
Third-birth shift	-2.711	0.293
Early-age utility shift	0.872	0.174
Late-age cost	1.337	0.129
High-skill career cost	2.260	0.118
Prospective accumulated-insecurity cost	0.998	0.303
Desired-family-size value	1.598	0.107
Religiosity value	0.085	0.035
Traditional gender-norm value	0.598	0.080

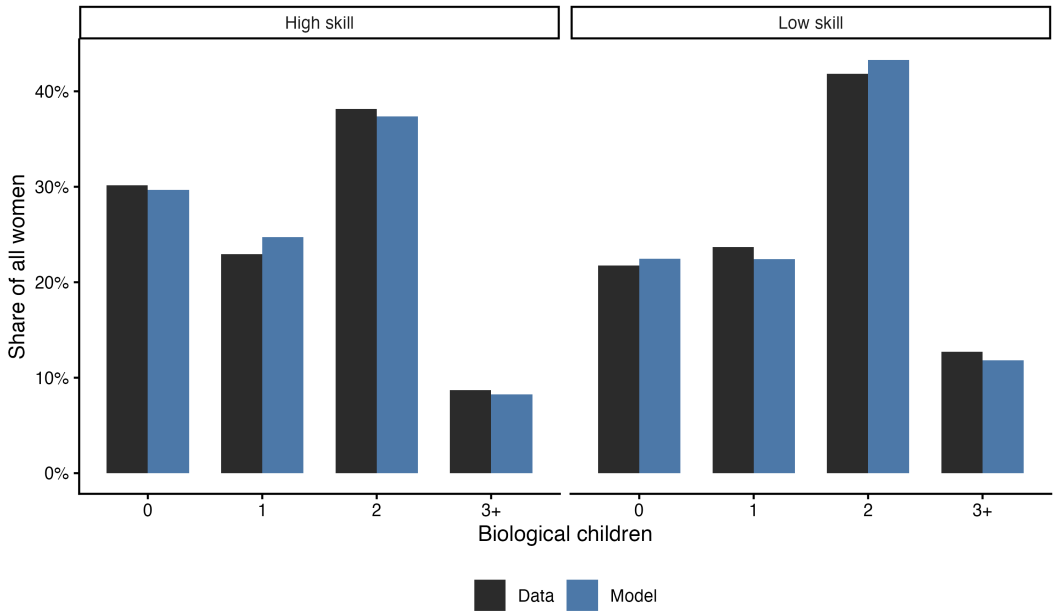
Notes: Minimum-distance estimates of the ten preference parameters governing Equation 10. Standard errors propagate uncertainty from the Fertility Survey moments, the MCVL labor-market first step, and first-birth timing from birth records. Utility coefficients are reported in Type-I extreme-value scale units. Appendix Table H2 reports the parameter-to-moment identification map, Appendix Table H3 reports numerical estimation diagnostics, and Appendix Table H4 reports the complete set of targeted moments and model counterparts.

7.3. *Model fit*

We first assess whether the model reproduces the main fertility patterns that discipline women's life-cycle choices. Figure 9 compares the completed-fertility distribution in the data and the model for high- and low-skilled women. The model closely reproduces the distribution across zero, one, two, and three or more children, as well as the higher childlessness rate among high-skilled women. For example, the model predicts childlessness of 29.7 percent for high-skilled women and 22.5 percent for low-skilled women, compared with 30.2 and 21.8 percent in the data. It also captures the two-child category as the most common family size for both skill groups. Appendix Table H4 reports the corresponding targeted moments and standardized differences.

Figure 10 examines the timing of entry into motherhood. Among women who become mothers, the model reproduces the main difference across skill groups, with high-skilled women starting motherhood later than low-skilled women. The comparison follows the same three age groups used in estimation: first birth by age 25, ages 26–31, and age 32 or older. The model is particularly close for low-skilled women in the 26–31 group and captures the shift of high-skilled first births toward later ages. Mean first-birth age is 28.99 years in the

Figure 9. Completed-fertility distribution: data and model



Notes: Survey-weighted shares of births realized through age 40 for women surveyed at ages 45–55. Children are grouped into zero, one, two, and three or more biological children. The model and data use the same capped parity definition.

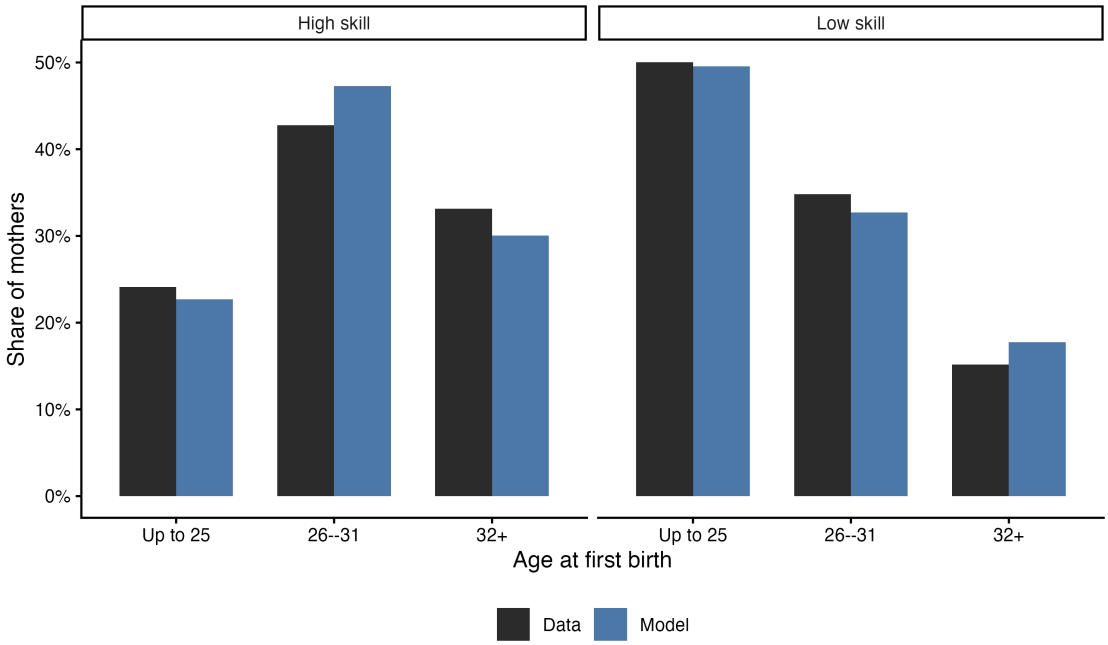
model and 28.99 in the data for high-skilled women, and 25.70 and 25.79 years, respectively, for low-skilled women.[44]

The final fit comparison asks whether the same parameterization can reproduce the fertility response to the 1984 reform as well as the cross-sectional profiles. Table 11 reports the model's changes of -0.034 children, -1.20 percentage points in motherhood, and +0.076 years under the administrative first-delivery observation equation, alongside targets of −0.108, −4.06 percentage points, and +0.412 years. The model reproduces the direction of all three responses: approximately 31 percent of the matched child-count effect, 30 percent of the motherhood effect, and 18 percent of the timing effect. These comparisons concern the matched structural targets, rather than the broader-support headline reduced-form estimates. The remaining timing gap motivates the sensitivity analysis. Appendix Tables H5 and I4 examine separate moment blocks, alternative preference samples, and first entry versus return to employment, clarifying the sources of this discrepancy. Appendix Table H4 reports every targeted residual and the finer timing checks; these comparisons show which dimensions of the data the model captures most closely.

The model captures the main fertility-profile differences across skill groups, including motherhood, the modal two-child family, and mean first-birth age. Its smaller reform timing response and the high-skilled 26–

[44] Appendix Table H4 reports the complete set of targeted moments and finer age-group checks. These finer categories are useful for diagnosing where the remaining timing differences arise. Among low-skilled mothers, an excess of first births through age 22 offsets a shortfall at 23–25, so agreement in the combined band does not imply agreement within it. These detailed cells are not the age groups shown in Figure 10 or separately targeted in estimation.

Figure 10. Age at first birth: data and model



Notes: Survey-weighted first-birth shares among women who become mothers. The categories correspond to those used in estimation: up to age 25, ages 26–31, and age 32 or older.

Table 11. Key model-fit moments

Moment	Data	Model	Standardized residual
Motherhood share, high skill	0.698	0.703	0.60
Age at first birth, high skill	28.985	28.985	-0.00
Motherhood share, low skill	0.782	0.775	-0.61
Age at first birth, low skill	25.787	25.696	-0.45
Desired–realized fertility gradient	0.583	0.593	0.58
Religiosity–fertility gradient	0.039	0.032	-0.51
Traditional norms–first-birth-age gradient	-0.816	-0.835	-0.19
1984 effect on capped children	-0.108	-0.034	1.16
1984 effect on motherhood	-0.041	-0.012	0.95
1984 effect on first-birth age	0.412	0.076	-6.28

Notes: The table compares selected empirical moments with their model counterparts. Motherhood and completed-fertility moments are unconditional across women; first-birth age is conditional on motherhood. Preference moments are partial associations from the 2018 Fertility Survey.

31 timing cell remain the main quantitative tensions. The counterfactuals therefore describe the contribution of the maintained channels rather than the whole reduced-form effect.

7.4. *Mechanisms and counterfactuals*

We use the model to separate the contributions of the reform-induced changes in early careers to the fertility response and to quantify how fertility would change under a more secure early-career labor market.

Figure 11 shows how the full reform changes low-skilled women's employment and fertility paths. Initial stocks, subsequent entry and temporary persistence affect different parts of the career. Fertility choices respond to the resulting resources, accumulated exposure and expected future employment conditions.

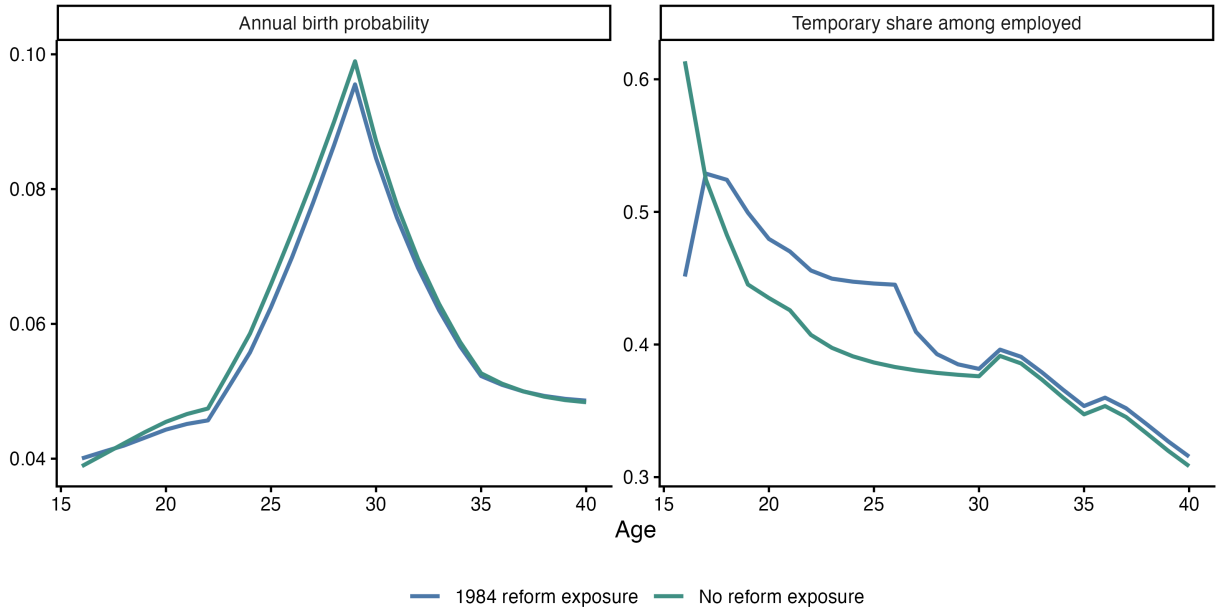
Table 12 then separates the mechanisms behind these paths. The first row reports the model prediction under the full reform. Rows (a)–(c) retain only one part of the reform-induced labor-market change. Row (a) changes labor-market conditions at entry, row (b) changes subsequent employment transitions, and row (c) applies only the observed change in contribution-based income. Rows (d)–(f) instead ask how fertility changes under alternative labor-market environments.

The entry-only experiment in row (a) changes completed fertility by -0.0056 children and first-birth age by $+0.006$ years. The largest child-count response among the separately activated labor channels comes from post-entry employment transitions in row (b). The corresponding changes are -0.0264 and $+0.046$, and the within-state resource-shift changes in row (c) are -0.0002 and $+0.005$. These isolated changes need not add to the full reform effect because fertility choices respond jointly to current conditions and future employment prospects. Entry and transition changes also affect resources by changing employment composition, so the within-state multiplier does not isolate the total resource contribution. Equalizing temporary and permanent resource profiles in row (d) is a separate diagnostic experiment, not an identified contract-reform effect.

Row (e) is a conditional mechanism exercise. It removes $c_T \mathcal{I}_{s,a}(e,h)$ under both employment regimes and recomputes the reform difference, giving $+0.0004$ children and -0.019 years. This quantifies the role of the estimated cost within the maintained model; it does not independently identify a causal history effect. Appendix I also compares re-estimated history and current-status specifications.

The local experiment in row (f) gives changes of $+0.0070$ children, $+0.245$ percentage points in motherhood, and -0.014 years. It changes the early temporary-destination tilt to reduce the ten-year broad insecurity index by 0.1 years, leaving the initial labor distribution, entry-flow adjustment, persistent conversion adjustment and resource profiles at their baseline settings. Thus, the intervention is one specific transition change producing a one-percentage-point reduction in the annual-average index, rather than a generic reduction in the temporary share among employed women.

Figure 11. Reform exposure over the life cycle



Notes: Model paths for low-skilled women with and without exposure to the 1984 reform. The employment panel reports temporary employment among women in covered temporary or permanent employment. The fertility panel reports annual model-implied birth probabilities.

Table 12. Mechanisms and counterfactual fertility effects

Experiment	Δ children	Δ motherhood	Δ first-birth age	Δ broad index
Observed 1984 exposure	-0.034	-0.012	+0.072	+0.042
(a) Entry channel only	-0.006	-0.002	+0.006	+0.002
(b) Post-entry transitions only	-0.026	-0.009	+0.046	+0.039
(c) Within-state resource shift only	-0.000	-0.000	+0.005	+0.000
(d) Equalize temporary-permanent resources	+0.032	+0.010	-0.311	+0.000
(e) 1984 exposure without direct insecurity wedge	+0.000	+0.000	-0.019	+0.042
(f) One-percentage-point reduction in broad insecurity index	+0.007	+0.002	-0.014	-0.010

Notes: Changes are relative to the reference low-skilled labor regime. In the no-insecurity row, the cost is removed in both regimes before taking their difference. Completed children and motherhood are unconditional across women; first-birth age is conditional on becoming a mother. Rows (a)–(c) isolate components of the reform-induced labor-market change. Rows (d)–(f) consider alternative labor-market environments. The final column reports the change in years of non-permanent employment divided by ten, expressed as a proportion of the first decade.

The model also separates the number of attempts by age 40 from their age distribution. Appendix Table I2 attributes −0.0302 children to attempt volume and −0.0037 to age composition, adding to −0.0339. This accounting describes the model-implied response; it does not identify a biological explanation for the entire observed decline. Postponement that leaves fewer opportunities to attempt a birth before 40 enters the volume component, so a small age-composition term does not imply an unimportant finite reproductive horizon.

The estimated mechanism links persistent temporary employment to the cost of early parenting through accumulated insecurity. The local-policy results quantify how this channel connects early-career conditions to fertility; the accompanying sensitivity exercises assess its magnitude across alternative empirical mappings.

8. CONCLUSIONS

This paper shows that the conditions under which young workers enter the labor market can have lasting consequences for family formation. This question is increasingly important as fertility declines and young adults spend more of their early careers in unstable employment. Spain's 1984 reform provides a useful setting to study this question because it generated a sharp shift in the contractual conditions faced by new labor-market entrants. The reform increased exposed women's years of non-permanent employment by 11.1% and reduced contribution-based income by 7.1%, with no detectable change in days worked. These changes in early-career stability were followed by a 15.2% decline in births per woman at the cohort level and a delay in motherhood. The results therefore point to job quality at career entry—not simply access to employment—as an important margin shaping fertility over the life cycle.

These early-career changes translate into economically meaningful fertility responses. Exposed low-skilled cohorts experience a 15.2% decline in births per woman at the cohort level, while births shift to older maternal ages and first motherhood is delayed by about five months. Annual event studies and the outcome-fitted synthetic comparison show the same broad post-reform pattern, and selection-sensitivity exercises account for the fact that timing outcomes are observed only among women who give birth. We find no evidence of a general deterioration in perinatal health among realized births; the only adverse outcome is an increase in C-sections prevalence. The mechanism evidence points to persistent contractual insecurity and difficulty realizing fertility plans. Exposed low-skilled women are about 28 percentage points more likely to report having fewer children than they desire. Estimates for other household resources, childcare, religiosity, and gender attitudes are imprecise. Father-side evidence further suggests that fertility quantity is more sensitive when both partners face weaker labor-market prospects, while the timing response is more closely tied to the mother's career conditions. The evidence is therefore more consistent with constraints on realizing fertility plans than with a shift in desired family size.

We build and estimate a life-cycle model to trace how early-career job insecurity shapes fertility over the reproductive life cycle and to separate the channels behind the reduced-form effects. The model points to post-entry employment instability and accumulated insecurity as the main drivers of its fertility response. The additional resource change within a given employment state has a smaller effect, while movements between jobs also change the resources available to women. It reproduces the main fertility profiles and the direction of the reform effects, while leaving part of the observed timing delay unexplained. The counter-

factuals show that reducing early-career insecurity can recover some forgone fertility and bring motherhood forward. Within the model, most of the decline reflects fewer attempts by age 40, including attempts lost through postponement, with a smaller contribution from shifts toward ages with lower birth success.

Two features of the data are important for interpretation. First, while the birth records provide the universe of registered births, they do not directly identify the number of women in each cohort–province–skill cell. We therefore construct the denominator using historical female birth-cohort counts and survey-weighted skill shares. This denominator does not track migration or mortality over the childbearing years, so the resulting measure need not follow changes in the resident female population. We interpret the estimates alongside the pre-period movement in the event study for births per woman at the cohort level. Several complementary checks are reassuring: alternative birth-count measures and survey child counts point in the same direction, and the outcome-fitted synthetic comparison closely reproduces the low-skilled pre-reform path before showing a post-reform decline. The synthetic fit provides a useful benchmark alongside the DD and other specification checks. Second, the birth records observe fathers only at realized births and do not provide complete partner employment histories, so the father-side results are best interpreted as mechanism and heterogeneity evidence rather than as a separate causal design. Finally, the structural model is intentionally partial equilibrium. It does not model bargaining within the household or firms' endogenous hiring, wage, and contract responses. The counterfactuals therefore quantify how fertility responds to changes in early-career employment conditions while holding these broader equilibrium margins fixed, rather than providing a full welfare or general-equilibrium evaluation. The sensitivity exercises support the direction of the local counterfactuals, while leaving scope to improve timing fit and distinguish accumulated insecurity from current employment status. Quantitative magnitudes remain conditional on child-cost calibration, retrospective preferences, and the mapping of employment histories and populations across datasets.

The policy implication is that labor-market design can shape family formation well beyond the initial job. The reduced-form evidence shows that greater exposure to insecure employment at labor-market entry is followed by lower realized fertility and later motherhood, while the model points to post-entry employment instability and accumulated insecurity as its main channels. The counterfactuals further suggest that reducing early-career insecurity can recover some forgone fertility and bring motherhood forward. Labor-market policies that alter the stability and remuneration of entry jobs should therefore be evaluated not only for their effects on employment and earnings, but also for their consequences for fertility over the life cycle.

REFERENCES

Adam, Paola, and Patrizia Canziani. 1998. "Partial de-regulation: Fixed-term contracts in Italy and Spain." Centre for Economic Performance 388.

Adda, Jérôme, Christian Dustmann, and Katrien Stevens. 2017. "The Career Costs of Children." *Journal of Political Economy*, 125(2): 293–337.

Adserà, Alicia. 2011. "Where are the babies? Labor market conditions and fertility in Europe." *European Journal of Population*, 27(1): 1–32.

Aguirregabiria, Víctor, and César Alonso-Borrego. 2014. "Labor contracts and flexibility: evidence from a labor market reform in Spain." *Economic Inquiry*, 52(2): 930–957.

Alderotti, Giammarco, Daniele Vignoli, Michela Baccini, and Anna Matysiak. 2021. "Employment instability and fertility in Europe: A meta-analysis." *Demography*, 58(3): 871–900.

Andrews, Isaiah, Matthew Gentzkow, and Jesse M. Shapiro. 2017. "Measuring the Sensitivity of Parameter Estimates to Estimation Moments." *The Quarterly Journal of Economics*, 132(4): 1553–1592.

Auer, Wolfgang, and Natalia Danzer. 2015. "Fixed-Term Employment and Fertility: Evidence from German Micro Data." *CESifo Economic Studies*, 62(4): 595–623.

Autor, David, David Dorn, and Gordon Hanson. 2019. "When work disappears: Manufacturing decline and the falling marriage market value of young men." *American Economic Review: Insights*, 1(2): 161–178.

Becker, Gary S. 1960. "An economic analysis of fertility." In *Demographic and economic change in developed countries*. 209–240. Columbia University Press.

Becker, Gary S. 1965. "A Theory of the Allocation of Time." *The Economic Journal*, 75(299): 493–517.

Bentolila, Samuel, and Juan J. Dolado. 1994. "Labour flexibility and wages: lessons from Spain." *Economic Policy*, 9(18): 53–99.

Bentolila, Samuel, Juan J. Dolado, and Juan F. Jimeno. 2008. "Two-tier employment protection reforms: The Spanish experience." *CESifo DICE Report*, 6(4): 49–56.

Bhaumik, Sumon K., and Jeffrey B. Nugent. 2011. "Real options and demographic decisions: empirical evidence from East and West Germany." *Applied Economics*, 43(21): 2739–2749.

Black, Dan A., Natalia Kolesnikova, Seth G. Sanders, and Lowell J. Taylor. 2013. "Are children "normal"?" *The Review of Economics and Statistics*, 95(1): 21–33.

Blanchard, Olivier, and Augustin Landier. 2002. "The perverse effects of partial labour market reform: fixed-term contracts in France." *The Economic Journal*, 112(480): F214–F244.

Clark, Andrew E., and Anthony Lepinteur. 2022. "A natural experiment on job insecurity and fertility in France." *Review of Economics and Statistics*, 104(2): 386–398.

Conde-Agudelo, Agustín, Ana Rosas-Bermúdez, and Adriana C. Kafury-Goeta. 2006. "Birth spacing and risk of adverse perinatal outcomes: a meta-analysis." *JAMA*, 295(15): 1809–1823.

de la Croix, David, and Aude Pommeret. 2021. "Childbearing postponement, its option value, and the biological clock." *Journal of Economic Theory*, 193: 105231.

Del Boca, Daniela. 2002. "The effect of childcare and part time opportunities on participation and fertility decisions in Italy." *Journal of Population Economics*, 15: 549–573.

Del Bono, Emilia, Andrea Weber, and Rudolf Winter-Ebmer. 2012. "Clash of career and family: Fertility decisions after job displacement." *Journal of the European Economic Association*, 10(4): 659–683.

Doepke, Matthias, and Fabian Kindermann. 2019. "Bargaining over Babies: Theory, Evidence, and Policy Implications." *American Economic Review*, 109(9): 3264–3306.

Doepke, Matthias, Anne Hannusch, Fabian Kindermann, and Michèle Tertilt. 2023. "The Economics of Fertility: A New Era." In *Handbook of the Economics of the Family.* , ed. Shelly Lundberg and Alessandra Voena. Amsterdam:North-Holland (Elsevier).

Dolado, Juan J., Carlos García-Serrano, and Juan F. Jimeno. 2002. "Drawing lessons from the boom of temporary jobs in Spain." *The Economic Journal*, 112(480): F270–F295.

Farré, Lídia, and Libertad González. 2019. "Does paternity leave reduce fertility?" *Journal of Public Economics*, 172: 52–66.

García-Pérez, J. Ignacio, Ioana Marinescu, and Judit Vall Castelló. 2019. "Can fixed-term contracts put low skilled youth on a better career path? Evidence from Spain." *The Economic Journal*, 129(620): 1693–1730.

Goldin, Claudia. 2021. *Career and Family: Women's Century-Long Journey toward Equity.* Princeton, NJ:Princeton University Press.

Goldin, Claudia. 2025. "Babies and the Macroeconomy." *Economica*, 93: 1–26.

Grigoriadis, Sophie, Lisa Graves, Miki Peer, Lana Mamisashvili, George Tomlinson, Simone N. Vigod, Cindy-Lee Dennis, Meir Steiner, Cara Brown, Amy Cheung, Hiltrud Dawson, Neil A. Rector, Melanie Guenette, and Margaret Richter. 2018. "Maternal anxiety during pregnancy and the association with adverse perinatal outcomes: systematic review and meta-analysis." *Journal of Clinical Psychiatry*, 79(5): 17r12011.

Guner, Nezih, Ezgi Kaya, and Virginia Sánchez-Marcos. 2024. "Labor market institutions and fertility." *International Economic Review*, 65(3): 1551–1587.

Heckman, James. 2012. "Invest in early childhood development: Reduce deficits, strengthen the economy." *The Heckman Equation, No. 7, pp. 1–2.*

Heckman, James J., and James R. Walker. 1990. "The relationship between wages and income and the timing and spacing of births: Evidence from Swedish longitudinal data." *Econometrica*, 1411–1441.

Hotz, V. Joseph, and Robert A. Miller. 1988. "An empirical analysis of life cycle fertility and female labor supply." *Econometrica*, 91–118.

Huttunen, Kristiina, and Jenni Kellokumpu. 2016. "The effect of job displacement on couples' fertility decisions." *Journal of Labor Economics*, 34(2): 403–442.

Instituto Nacional de Estadística (INE). 2025*a*. "Basic Demographic Indicators: Fertility, National Total, 1975–2024." *INEbase series IDB72160*, Final 2024 data published November 19, 2025.

Instituto Nacional de Estadística (INE). 2025*b*. "Economically Active Population Survey: Activity, Unemployment, and Employment Rates by Sex and Age, 1976–2004." *INEbase table 01011*, Historical series accessed August 17, 2026.

Instituto Nacional de Estadística (INE). 2026*a*. "Economically Active Population Survey: Annual Activity Rates by Sex and Age Group, 2025." *INEbase table 65950*, Final annual averages published February 25, 2026.

Instituto Nacional de Estadística (INE). 2026*b*. "Monthly Births Estimate: Year 2025." Provisional results published February 18, 2026.

Instituto Nacional de Estadística (INE). 2026*c*. "Women and Men in Spain: Employees by Type of Contract, 2009–2025." Annual EAPS averages, accessed August 17, 2026.

Kahn, Lawrence M. 2010. "Employment protection reforms, employment, and the incidence of temporary jobs in Europe: 1996–2001." *Labour Economics*, 17(1): 1–15.

Kearney, Melissa Schettini, and Phillip B Levine. 2025. "Why Is Fertility So Low in High Income Countries?" *National Bureau of Economic Research.*

Kearney, Melissa S., Phillip B. Levine, and Luke Pardue. 2022. "The Puzzle of Falling US Birth Rates since the Great Recession." *Journal of Economic Perspectives*, 36(1): 151–176.

Kind, Michael, and Jörg Kleibrink. 2013. "Sooner or Later – Economic Insecurity and the Timing of First Birth." RWI Essen Ruhr Economic Paper 422.

Kreyenfeld, Michaela. 2009. "Uncertainties in female employment careers and the postponement of parenthood in Germany." *European Sociological Review*, 26(3): 351–366.

Lean, Steven C., Hannah Derricott, Rebecca L. Jones, and Alexander E. Heazell. 2017. "Advanced maternal age and adverse pregnancy outcomes: A systematic review and meta-analysis." *PLoS One*, 12(10): e0186287.

Lee, David S. 2009. "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects." *The Review of Economic Studies*, 76(3): 1071–1102.

Lepinteur, Anthony. 2021. "The asymmetric experience of gains and losses in job security on health." *Health Economics*, 30(9): 2217–2229.

Leridon, Henri. 2004. "Can Assisted Reproduction Technology Compensate for the Natural Decline in Fertility with Age? A Model Assessment." *Human Reproduction*, 19(7): 1548–1553.

Low, Hamish, Costas Meghir, and Luigi Pistaferri. 2010. "Wage Risk and Employment Risk over the Life Cycle." *American Economic Review*, 100(4): 1432–1467.

OECD. 2025. *OECD Economic Surveys: Spain 2025.* Paris:OECD Publishing.

Oreopoulos, Philip, Till von Wachter, and Andrew Heisz. 2012. "The short- and long-term career effects of graduating in a recession." *American Economic Journal: Applied Economics*, 4(1): 1–29.

Prifti, Egle, and Daniela Vuri. 2013. "Employment protection and fertility: Evidence from the 1990 Italian reform." *Labour Economics*, 23: 77–88.

Ranjan, Priya. 1999. "Fertility behaviour under income uncertainty." *European Journal of Population*, 15: 25–43.

Reeves, Aaron, Martin McKee, and David Stuckler. 2014. "Economic suicides in the Great Recession in Europe and North America." *The British Journal of Psychiatry*, 205(3): 246–247.

Rica, Sara de la, and Amaia Iza. 2005. "Career planning in Spain: Do fixed-term contracts delay marriage and parenthood?" *Review of Economics of the Household*, 3: 49–73.

Rohde, Nicholas, K. K. Tang, Lars Osberg, and Prasada Rao. 2016. "The effect of economic insecurity on mental health: Recent evidence from Australian panel data." *Social Science & Medicine*, 151: 250–258.

Santarelli, Enrico. 2011. "Economic resources and the first child in Italy: A focus on income and job stability." *Demographic Research*, 25: 311–336.

Schaller, Jessamyn. 2016. "Booms, busts, and fertility: Testing the Becker model using gender-specific labor demand." *Journal of Human Resources*, 51(1): 1–29.

Schultz, T. Paul. 1985. "Changing world prices, women's wages, and the fertility transition: Sweden, 1860-1910." *Journal of Political Economy*, 93(6): 1126–1154.

Sommer, Kamila. 2016. "Fertility choice in a life cycle model with idiosyncratic uninsurable earnings risk." *Journal of Monetary Economics*, 83: 27–38.

Sutela, Hanna. 2012. "Temporary jobs and first child fertility in Finland." *Community, Work & Family*, 15(4): 425–450.

Tauchmann, Harald. 2014. "Lee (2009) Treatment-Effect Bounds for Nonrandom Sample Selection." *The Stata Journal*, 14(4): 884–894.

Toharia, Luis. 1997. *The labour market in Spain.* Brussels:European Commission.

Vignoli, Daniele, Letizia Mencarini, and Giampiero Alderotti. 2020. "Is the effect of job uncertainty on fertility intentions channeled by subjective well-being?" *Advances in Life Course Research*, 46: 100343.

Vignoli, Daniele, Sven Drefahl, and Gustavo De Santis. 2012. "Whose job instability affects the likelihood of becoming a parent in Italy? A tale of two partners." *Demographic Research*, 26(2): 41–62.

ONLINE APPENDIX

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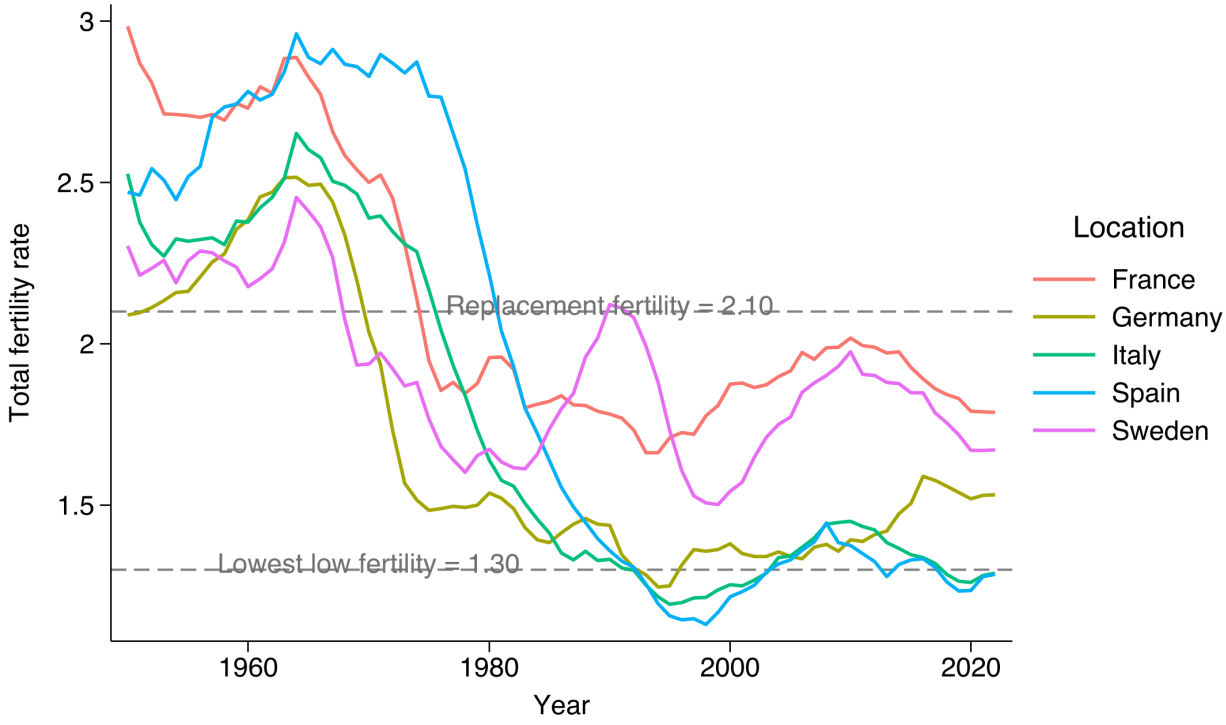
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A. INSTITUTIONAL BACKGROUND

A.1. *The Spanish Demographic Context*

Figure A1.1 plots total fertility rates since 1950 for France, Germany, Italy, Spain, and Sweden against the replacement-fertility and "lowest-low" thresholds. The sustained declines in Italy and Spain are especially pronounced, while the cross-country comparison also shows meaningful heterogeneity in levels and partial rebounds.

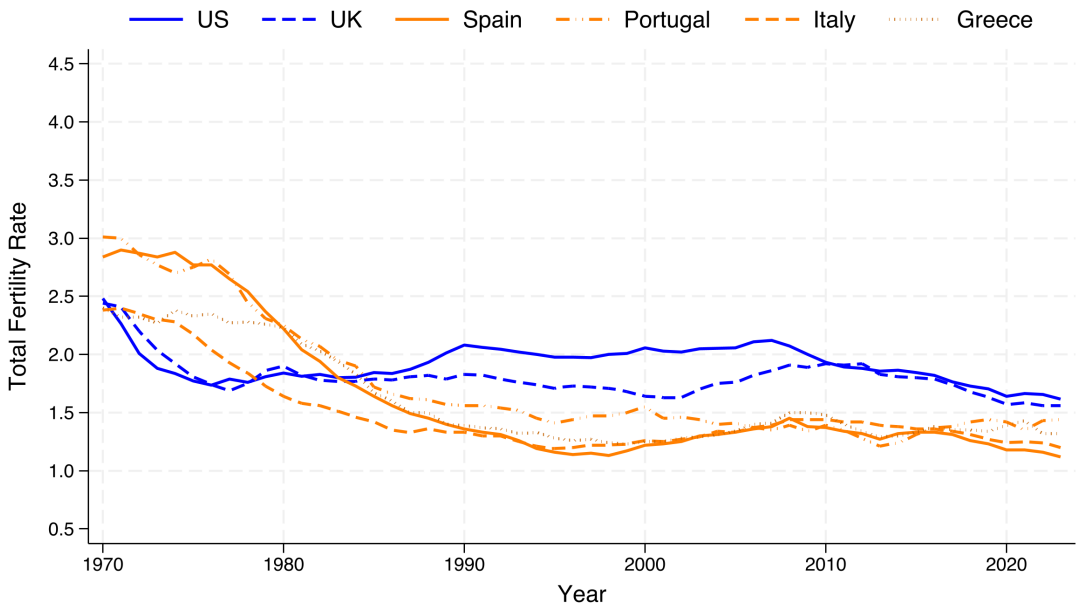
Figure A1.1. Total Fertility Rate, Selected Countries (1950–2022)



Notes: Annual period total fertility rates for France, Germany, Italy, Spain and Sweden. The horizontal reference lines mark replacement fertility (2.10) and the "lowest-low" threshold (1.30). The series includes the 2022 medium projection after historical estimates through 2021. *Source:* United Nations, World Population Prospects 2022, original medium-variant demographic-indicators file.

Figure A1.2 highlights a reversal in fertility patterns: Spain and the other Southern European countries began with higher fertility than the United States and United Kingdom but subsequently fell below them. Goldin (2025) links this divergence to rapid economic change that expanded women's opportunities while family expectations and the division of care adjusted more slowly. This perspective motivates our focus on whether the employment conditions women face at career entry make it harder to combine work and motherhood.

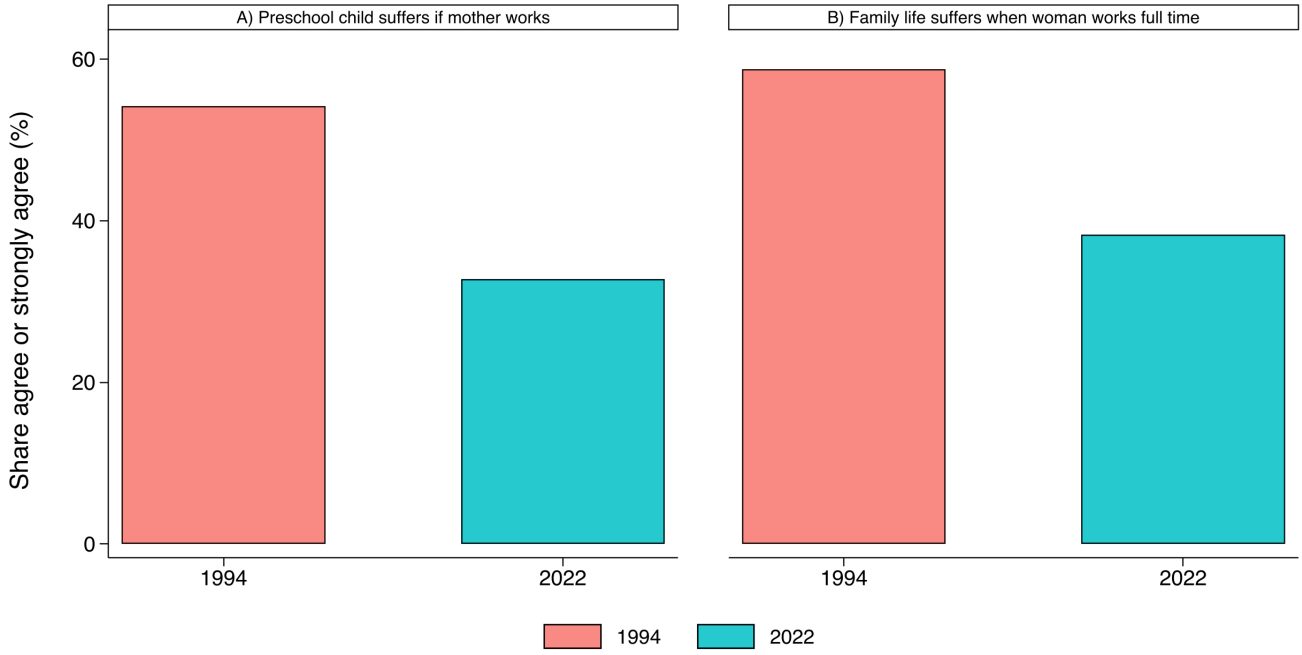
Figure A1.2. Total Fertility Rates for Selected OECD Nations (1970–2023)



Notes: Blue lines denote the United States and United Kingdom; orange lines denote Spain, Portugal, Italy and Greece. The total fertility rate represents the number of children that would be born to a woman if she were to live to the end of her childbearing years and bear children in accordance with the age-specific fertility rates of the specified year. *Sources:* World Bank Group, World Development Indicators, 1970–2023. Accessed November 14, 2025.

Figure A1.3 compares Spanish attitudes in 1994 versus 2022 regarding whether (i) a preschool child is likely to suffer if the mother works and (ii) family life suffers when a woman has a full-time job. The figure reports the share who agree or strongly agree with each statement. The decline in agreement over time indicates a substantial shift toward more supportive views of maternal employment and away from traditional assumptions that maternal work necessarily harms children or family life.

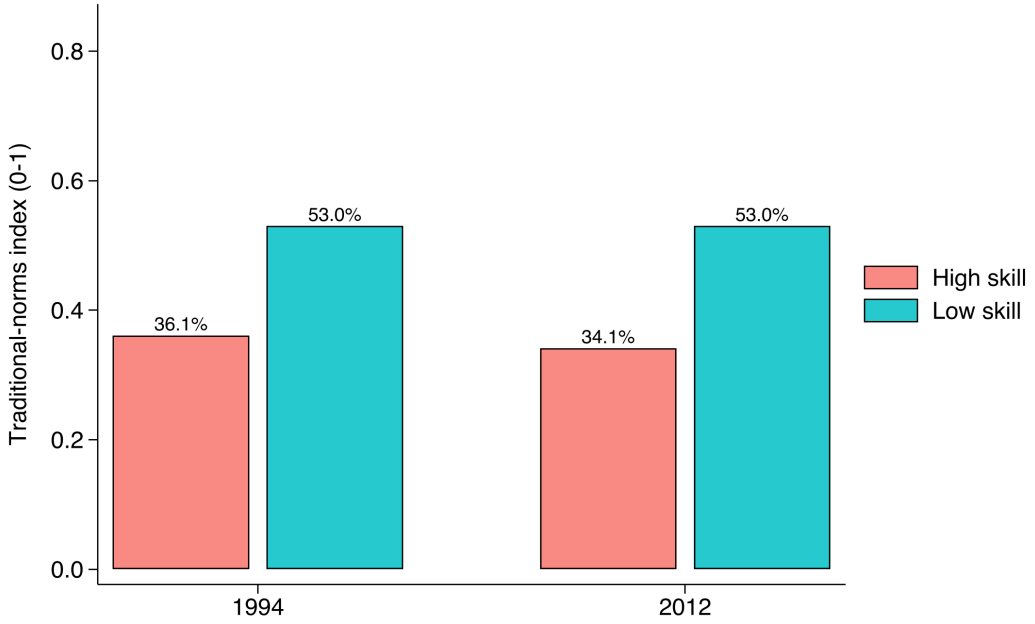
Figure A1.3. Spain: changes in attitudes toward maternal employment, 1994 vs 2022



Notes: Bars report the share of respondents who *agree* or *strongly agree* with each statement. Panel A) refers to the belief that a preschool child is likely to suffer if the mother works; Panel B) refers to the belief that family life suffers when a woman has a full-time job. Shares are unweighted among Spanish respondents with a valid item response (codes 1–5); agreement combines responses 1 and 2. Item-specific response counts are exported with the figure. These descriptive comparisons do not impose the administrative cohort restrictions. *Sources:* ISSP Family and Changing Gender Roles, 1994 (ZA2620, v5 and v6) and 2022 (ZA10000 version 2.0.0, v2 and v3).

Figure A1.4 reports an ISSP gender-norms index for Spain by skill group in 1994 and 2012, defined as the share who agree or strongly agree with a set of traditional statements. The bars provide descriptive comparisons; differences across waves also reflect changes in skill proxies and response scales. This figure provides institutional context but is not an estimation target. The model's distinct traditional-norm shifter is constructed from women in the 2018 Fertility Survey and is disciplined by a joint partial first-birth-age gradient that also conditions on desired fertility, religiosity, skill, age, and region.

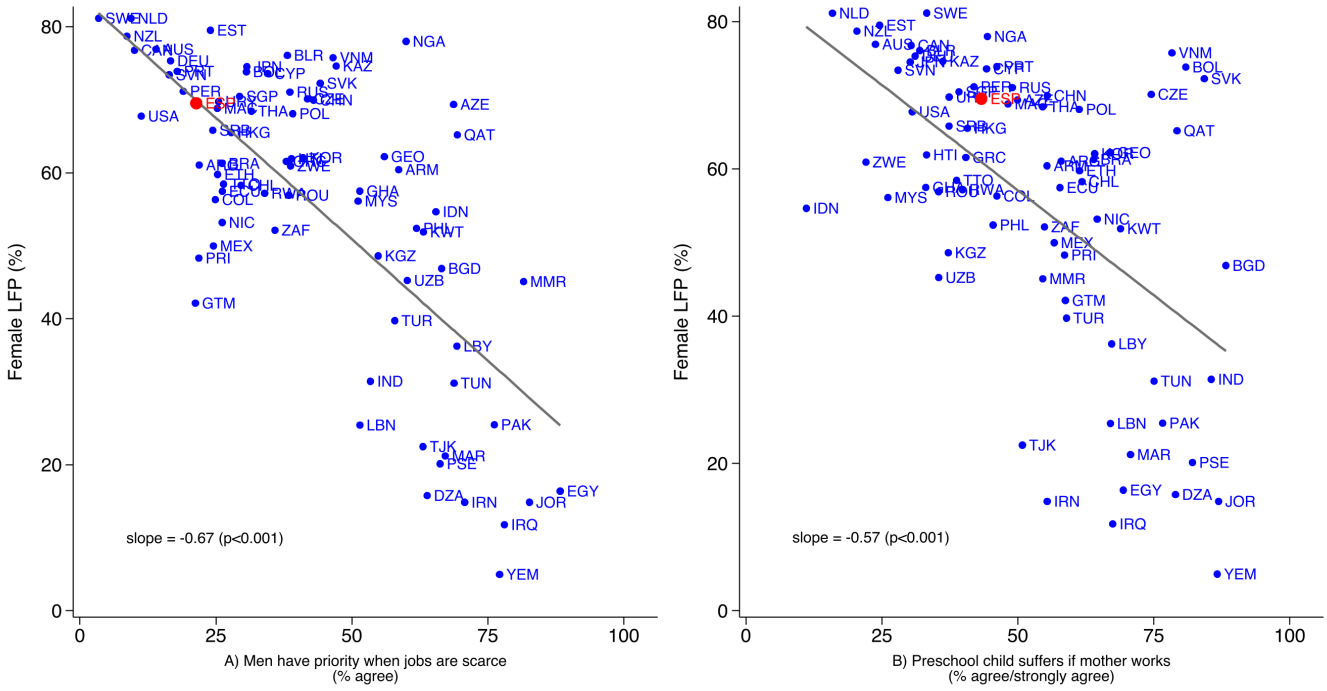
Figure A1.4. Spain: gender norms index by skill, 1994 vs 2012



Notes: Respondent-weighted means of an equally weighted five-item agreement index: preschool children suffer if mothers work; family life suffers if women work full time; women primarily want a home and children; housekeeping is as fulfilling as paid work; and men should earn while women care for the home. All five items must have valid responses. Agreement combines codes 1–2; valid scales are 1–5 in 1994 and the Spanish 1–4 scale in 2012, which omits the neutral option. In 1994, high skill comprises reported professional, technical, administrative, clerical and military occupations explicitly classified in the GESIS ISCO-1968 dictionary; listed sales, service, agricultural, production and labourer occupations are low skill. Unlisted codes are unclassified. In 2012, completed education through lower secondary (DEGREE 0–2) is low skill and levels 3–6 are high skill. These proxies are not interchangeable over time. No cohort restriction, regression, fixed effects or clustering applies. Sources: ISSP 1994 (ZA2620, v5–v8 and v12; v315 weights) and 2012 (ZA5900, Spanish ES_V6–ES_V9 and ES_V11; WEIGHT).

Figure A1.5 documents a cross-country relationship between gender norms and female labor force participation (LFP). Each point is a country; the y-axis is female LFP, and the x-axis is agreement with traditional statements about gender roles. The fitted lines indicate a negative association, and Spain is highlighted within the international distribution. This descriptive relationship motivates measuring norms but does not identify their structural coefficient. Identification instead comes from the within-Spain 2018 Fertility Survey partial first-birth-age gradient; the model does not treat the reform as changing norms or estimate an insecurity–norm interaction.

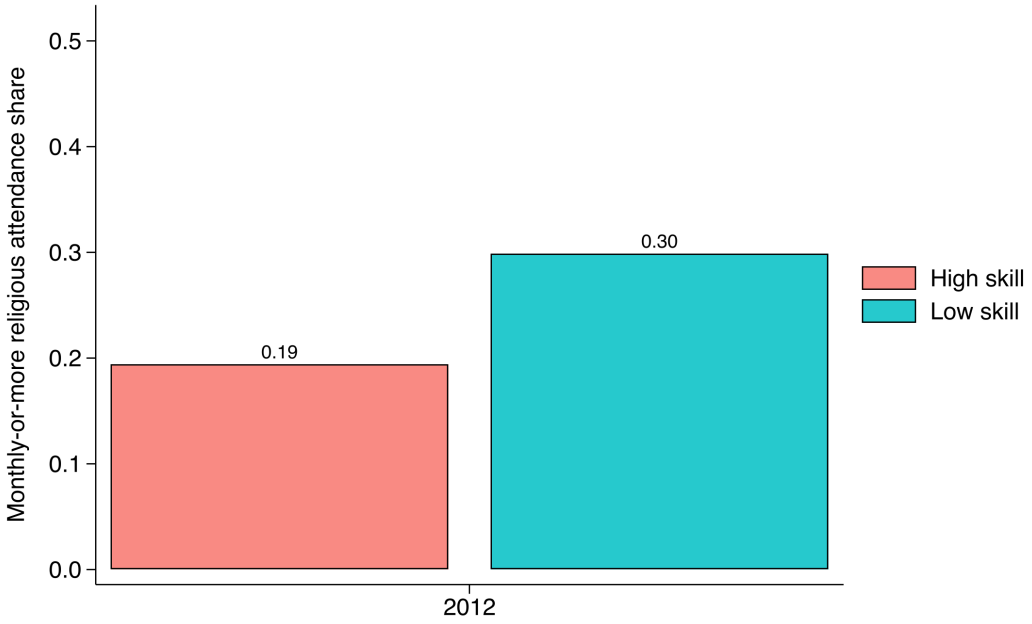
Figure A1.5. Gender norms and female labor force participation across countries



Notes: Each dot is a country with valid observations for both attitude measures and female labor force participation. Panel A reports agreement with "Men have priority when jobs are scarce" (WVS C001, response 1); Panel B reports agreement or strong agreement with "A preschool child suffers if the mother works" (D061, responses 1–2). Country attitude means pool available 1981–2020 waves using respondent weights S017; valid response counts and country-specific survey years are exported with the figure. The y-axis reports female labor force participation at ages 15–64 in 2022 (percent). Solid lines and reported slopes and p-values are from unweighted bivariate country regressions, with conventional OLS standard errors and no fixed effects. Spain is highlighted. *Sources:* World Values Survey time-series file, 1981–2020, version 2.0; World Bank World Development Indicators, modeled ILO estimate SL.TLF.ACTI.FE.ZS, archived January 2026.

Figure A1.6 displays ISSP religiosity in Spain by skill, measured as the share attending religious services monthly or more often. It documents preference heterogeneity not captured by labor-market conditions alone but is not itself targeted in estimation. The model uses a separate standardized 2018 Fertility Survey index combining affiliation and practice intensity and targets its partial completed-fertility gradient. The paper treats the estimated coefficient as a descriptive preference association, not a causal effect of religiosity.

Figure A1.6. Spain: religiosity by skill (monthly or more attendance)



Notes: Shares attending religious services at least monthly (ATTEND 1–4), among Spanish respondents with valid attendance responses 1–8 and completed education DEGREE 0–6. Respondent WEIGHT is applied. Low skill is completed education through lower secondary (0–2); high skill is levels 3–6. Missing or unclassified answers are excluded; both sexes and all respondent ages remain. No reform-cohort restriction, regression, fixed effects or clustering applies. Source: ISSP 2012, ZA5900 version 4.0.0.

A.2. *The Spanish Labor Market in Figures*

Table A2.1 summarizes the menu of contract types in Spain around the 1984 reform and highlights the key institutional feature exploited in the paper: the reform expanded the set of temporary contracts and allowed firms to rely on fixed-term hiring at substantially lower dismissal costs than standard open-ended contracts. The table distinguishes compensation for dismissal from permanent contracts from compensation at the ordinary expiry of temporary contracts, which typically have limited duration and little or no expiry compensation. This institutional gap provides the micro-foundation for interpreting the reform as a large, nationwide shock to job security that is plausibly exogenous to individual fertility preferences, and it motivates the paper's focus on labor-market duality as a constraint on family formation.

Table A2.2 documents the evolution of the share of workers on temporary contracts in Spain over time and shows how temporary employment is concentrated in groups most relevant for fertility decisions. The table highlights three patterns: (i) the released aggregate temporary share rises sharply between 1987 and 1990, documenting the diffusion of fixed-term contracts after the reform; (ii) temporary employment is higher among younger workers (15–24 in the published series), precisely when many individuals enter the labor market and make early family-formation choices; and (iii) there is a clear education gradient, with

Table A2.1. Contract types in 1984 and their terms

Type of Contract	Duration	Severance
Permanent contracts		
Permanent (standard)	Indefinite	20 or 45 days/year
Temporary (with cause)		
Work or service	Until completion	None stipulated
Production needs	Max. 6 months	None stipulated
Interim	Until return	None stipulated
Launching new activity	6–36 months	None stipulated
Employment promotion and training		
Fixed-term (employment promotion)	6–36 months	12 days/year
Training (practicas)	3–36 months	None stipulated
Apprenticeship (formacion)	3–36 months	None stipulated
Other		
Part-time	Permanent or temporary	Based on type
Retirement replacement	Until retirement	None stipulated

Notes: Terms enacted in the 1984 reform. For permanent contracts, 20 days of salary per year applies to justified objective dismissal (maximum 12 months), and 45 days to unfair dismissal when compensation replaces reinstatement (maximum 42 months). Temporary entries describe compensation at ordinary expiry, not early or unfair dismissal. *None stipulated* indicates no specific expiry indemnity in the cited regime; other applicable rules or agreements may provide compensation. Production-needs contracts are limited to six months within twelve months; new-activity contracts cannot extend beyond the three-year launch period. Retirement replacement ends at the replaced worker's ordinary retirement age. This is an institutional summary; no estimator, weights, fixed effects or clustering apply. *Sources:* Original BOE texts of Law 8/1980, Articles 53 and 56, as amended by Law 32/1984; Royal Decrees 1989/1984, Article 3; 1992/1984, Articles 3 and 8; 2104/1984, Articles 2–5 and 9; and 1991/1984, Articles 3 and 8. Article-level provenance accompanies the replication CSV.

higher temporary exposure among those with lower schooling. These stylized facts support the paper's identification strategy and help interpret the heterogeneous fertility effects by skill as operating through differential exposure to insecurity.

Table A2.3 reports one-year transition outcomes for workers who start in temporary employment, providing descriptive evidence on persistent insecurity and limited conversion to permanent contracts during 1987–1996. A large fraction remain temporary one year later, while a non-trivial share moves into unemployment or exits the labor force. These historical aggregates motivate the career-history mechanism but are not inserted into the model. The current transition matrices are estimated by age and skill from the pre-reform cohort MCVL histories.

Table A2.4 summarizes major parental leave reforms in Spain and clarifies how the institutional environment governing family-work trade-offs evolved over the study period. The table shows that maternity

Table A2.2. Total share of temporary employment (%) by educational level, age and gender in Spain

Years	**1983**	**1987**	**1990**	**1998**	**2007**	**2018**
Average (15–64)	—	15.6	29.9	32.9[b]	31.6	26.9
By educational level						
Lower secondary or less	—	—	—	37.7[b]	37.4	32.0
Upper secondary/non-tertiary	—	—	—	29.9[b]	31.3	28.3
Tertiary	—	—	—	25.9[b]	24.8	22.4
By age						
15–24	—	36.1	65.8	73.0[b]	62.7	71.2
25–39	—	13.7	27.9	35.4[b]	34.9	33.8
40–59	—	8.5	15.1	16.8[b]	20.4	19.0
By gender						
Males	—	14.5	27.8	32.0[b]	30.6	26.0
Females	—	18.5	34.3	34.4[b]	32.9	27.8

Notes: Each rate is 100 times the released number of temporary employees divided by all employees in the same country, year and subgroup. The average, education and sex rows cover ages 15–64. Age 25–39 counts are calculated by subtracting 15–24 counts from 15–39 counts before taking the ratio. Education groups are the released ISCED aggregates 0–2, 3–4 and 5–8, not primary-only or university-only categories; historical classification changes limit comparisons over time. An em dash denotes unavailable matching source counts; no later-year composition is used to fill an earlier cell. Superscript *b* retains Eurostat's break-in-series flag. Counts are published in thousands to one decimal; underlying counts and source flags are exported. These are descriptive EU-LFS annual-series ratios, not estimates from the administrative reform sample. No regression, fixed effects, trends or clustering applies; survey weighting is already incorporated in the released counts. The series differs in coverage and historical vintage from the compiled series in Figure 2. *Source:* Eurostat, EU Labour Force Survey, lfsa_etgaed and lfsa_eegaed, archived June 2026 release retrieved on 6 September 2026.

Table A2.3. One-year transitions of temporary workers (1987–1996)

Period	Stay Temp	Perm	Unemp	Out of LF	Other[±]
1987.II–1988.II	48.6	22.0	18.0	5.5	5.9
1988.II–1989.II	56.6	19.1	14.6	3.8	5.9
1989.II–1990.II	57.6	17.8	14.9	5.0	4.6
1990.II–1991.II	63.7	12.8	14.1	4.3	5.1
1991.II–1992.II	57.5	14.0	17.0	6.5	5.1
1992.II–1993.II	56.9	9.4	23.7	5.5	4.5
1993.II–1994.II	63.0	9.2	19.6	4.6	3.5
1994.II–1995.II	64.4	10.8	16.4	4.9	3.6
1995.II–1996.II	62.7	11.0	17.3	5.3	3.7
Average	59.0	14.0	17.3	5.1	4.7

Notes: Published one-year transition percentages for temporary employees in EPA-flujos, using second-quarter pairs. The source's rounded percentages and reported average are retained; the underlying microdata are not re-estimated here. No reform regression is reported. [±]Other includes unclassified workers, self-employed workers and conscripts. *Sources:* Adam and Canziani (1998), Table 11, drawing on Toharia (1997), Table I.2, and EPA-flujos.

leave expanded and was structured around a compulsory post-birth period for mothers, with later reforms allowing partial transferability; fathers' leave remained minimal until the 2007 reform introduced a distinct, non-transferable paid paternity leave entitlement. This timeline is important for interpretation because it indicates that substantial changes in leave policy—especially for fathers—occur relatively late compared with the initial expansion of temporary contracts. The timing distinguishes these leave reforms from the initial expansion of fixed-term hiring in 1984.

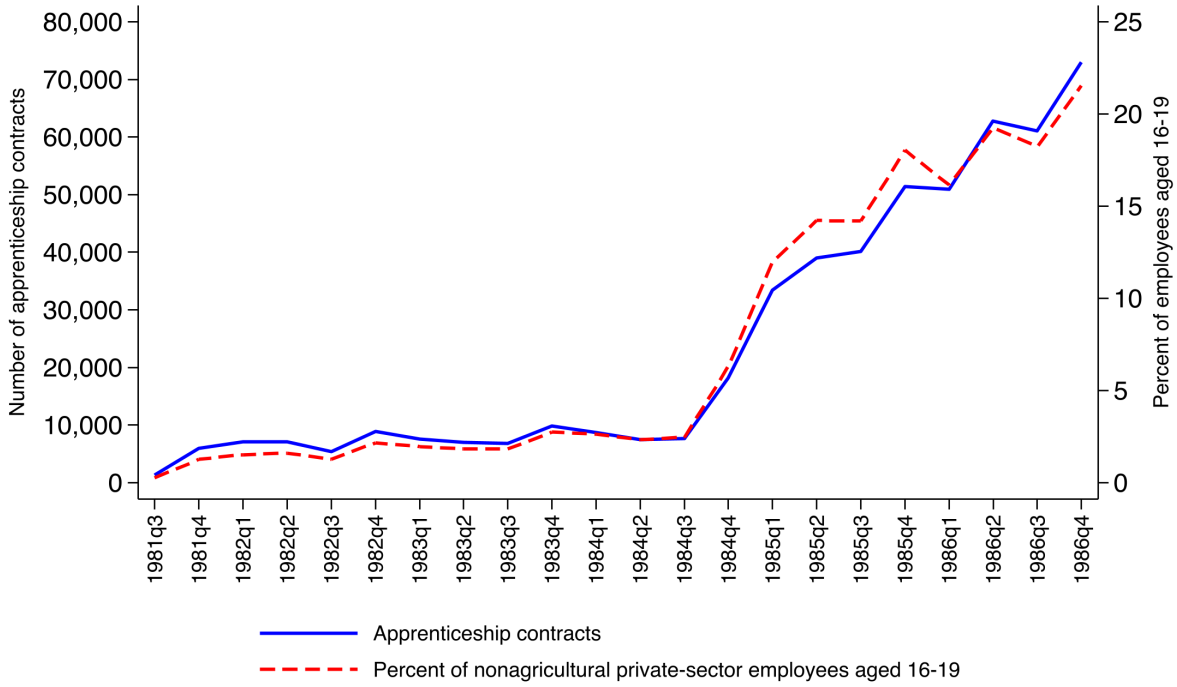
Table A2.4. Parental leave reforms in Spain.

	March 1980[a]	**March 1989**[b]	**November 1999**[c]	**March 2007**[d]
Fathers	2 days of paid job absence after the baby's birth	2 days of paid job absence after the baby's birth	2 days of paid job absence after the baby's birth	2 days of paid job absence after the baby's birth; 13 days of job-protected paid leave (non-transferable to the mother)
Mothers	14 weeks of job-protected paid leave (non-transferable to the father)	16 weeks of job-protected paid leave. The first 6 weeks after birth are compulsory and exclusively reserved to the mother. The last 4 weeks (of the 16) can be transferred to the father	The first 6 weeks after birth are compulsory and exclusively reserved to the mother. The other 10 weeks can be transferred to the father and taken simultaneously or subsequently to the mother's leave	No change

Notes: Selected historical statutory entitlements; this is an institutional timeline, not an estimated policy effect. [a] Workers' Statute (Law 8/1980, *Estatuto de los Trabajadores*). [b] Law 3/1989. [c] Law 39/1999. [d] Law 3/2007. *Source:* Farré and González (2019), Appendix Table 1.

Figure A2.1 documents the short-run diffusion of apprenticeship contracts around the 1984 reform. The blue and red dashed series reproduce approximate digitizations of the published apprenticeship-contract count and its share among non-agricultural private-sector employees aged 16–19. Both paths rise sharply after the reform; the figure provides historical context rather than a new estimate of the reform effect. This evidence complements Table A2.2: the table describes the broader diffusion of temporary employment across groups and years, while the figure shows the timing and concentration of the initial contract response among young entrants.

Figure A2.1. Apprenticeship contracts among young workers around the 1984 reform



Notes: Approximate reconstruction from the vector paths of Figure 3 in García-Pérez, Marinescu and Vall Castelló (IZA Discussion Paper 9777, 2016), retained as a descriptive illustration. The solid blue line gives approximate apprenticeship-contract counts; the dashed red line gives their percentage relative to non-agricultural private-sector employees aged 16–19. The underlying EPA series has not been independently re-estimated. No regression, cohort trend, fixed effects, analytical weights or clustering is applied to these digitized points. *Source:* Authors' extraction from the published figure, which cites INE's EPA.

B. ECONOMIC INTUITION FOR THE LIFE-CYCLE MODEL

This appendix gives the economic intuition for the estimated model in Section 7; it is not a second quantitative model. At each age, a woman compares attempting a birth now with waiting. Let $\Delta_a(x) = v_a^1(x) - v_a^0(x)$ denote the choice-specific value difference at state x. A birth attempt is more attractive when the value of the next child is high, current resources are sufficient, and conception is likely. Waiting is more attractive when employment is unstable or the expected return to reaching a permanent job is high. Years since the last birth are carried as an accounting state for the untargeted spacing validation, but do not receive a separate payoff coefficient. Extreme-value choice shocks turn this comparison into the logit probability used in the estimated model.

Temporary employment affects this comparison through three distinct margins. First, temporary jobs pay less than permanent jobs in the MCVL earnings profiles. Second, current contract status changes the distribution of future employment states. Third, accumulated temporary exposure remains in the state even after the current contract ends. Its direct cost depends on expected employment stability during early

parenting, separately from the resource channel. The 1984 reform changes entry, subsequent temporary exposure, and early-career earnings; it does not directly set fertility choices.

The life-cycle horizon makes postponement economically consequential. At young ages, waiting can preserve the option to have the same child after employment stabilizes. As age rises, declining conception probabilities and the finite horizon reduce that option value. Persistent instability can therefore shift first births to later ages and, for some women, leave insufficient time for the same parity progression. This is why the model jointly uses the age-at-first-birth distribution, parity among older mothers, and reform-induced timing and child-count changes.

Desired family size, religiosity, and traditional gender norms shift the baseline value of children but are held fixed under reform exposure. Their coefficients are disciplined by joint partial gradients in the 2018 Fertility Survey, conditional on the other preference measures, skill, age, and region. This treatment allows measured preference heterogeneity without interpreting culture as a response to the labor-market reform. The reduced-form Fertility Survey results likewise show no systematic reform effect on the available religiosity and attitude measures.

Partner resources could affect the same comparison, and the father-side results provide suggestive household-level mechanism evidence. However, the birth records observe fathers only at realized births and do not follow couples before fertility decisions. Adding endogenous partnership or bargaining would therefore introduce transition and preference parameters that the maintained data cannot identify. The estimated model instead keeps the woman's labor-risk channel transparent and treats the partner evidence as external validation of the broader household-resource interpretation.

C. DATA DETAILS

C.1. *Spanish Birth Certificate Micro-data (1980–2017)*

The data used in this study come from the Birth Statistics (Estadística de Nacimientos) compiled by the Instituto Nacional de Estadística (INE) as part of the Movimiento Natural de la Población (MNP). These records originate from birth registrations in Civil Registries across Spain. The complete harmonized birth-record window used in this paper is 1980–2017; this is the window of the available project inputs, rather than the beginning of INE's historical microdata series.[45]

Birth registrations supply the information collected in the statistical childbirth bulletin (*Boletín Estadístico de Parto*) and transmitted to INE. Electronic communication of births from hospitals began on 15 October 2015, alongside registration through Civil Registries.[46] The collection channel matters for field availability:

[45] INE provides historical birth microdata from 1975; see its MNP methodology, Section 4.4.

[46] See the instruction of 9 October 2015 on electronic hospital birth registration.

INE documents that cohabitation information was not collected through electronic hospital registrations in the final years of our window.[47]

The annual source files contain birthweight, gestational age, plurality and maternal live-child information throughout the harmonized 1980–2017 window, although variable names and the meaning of child-count fields change. The replication verifies the year-specific fields and uses the corresponding first-delivery definitions described in the data section. C-section and cohabitation information are available from 2007; unavailable fields and unanswered questions are not coded as zero. Birthweight is retained between 450 and 6,200 grams and gestational age between 22 and 45 completed weeks. These outcome-specific validity rules affect regression counts without changing the underlying maternal-cohort eligibility. The microdata are publicly available through INE's online portal.

C.2. *Births per woman (cohort-level): constructing the denominator*

For maternal cohort-quarter c, province p and skill group s, births per woman (cohort-level) is $Y_{cps} = B_{cps}/D_{cps}$, where B_{cps} counts eligible infant records registered in 1980–2017, including all nationalities, and $D_{cps} = H_{cp} f_c \widehat{\pi}_s$. Here H_{cp} is the historical number of registered births in the province and quarter, f_c is the national female share of births in that cohort year, and $\widehat{\pi}_s$ is a common skill share estimated with original respondent weights from the 1999 Fertility Survey. The reference sample contains 1,337 eligible women born in 1965–1969, the overlap of the two skill groups' cohort windows. Childless women remain in this sample. The estimated shares are 54.46% low-skilled and 45.54% high-skilled; the same shares are applied to all provinces and cohort quarters.

The survey uses the paper's education-based skill classification and includes women of all nationalities. Education proxies the occupational classification in the register, and the shares are observed after the reform. Historical registered births and a national female share approximate the size of female birth cohorts; they do not measure the resident female population throughout the childbearing years or adjust for migration and mortality. Thus births per woman (cohort-level) is a denominator-scaled cohort-level birth measure, not completed fertility for unique women. Its levels are not directly comparable to individual survey child counts, which have different observation ages and sampling units. Ceuta and Melilla lack the required historical denominator and are excluded from quantity specifications, while their eligible infant records remain in timing and health samples.

Table 1 and Figure 3 weight the same cell outcome by D_{cps}, so each skill-specific descriptive mean equals total eligible births divided by the total estimated female cohort denominator. The descriptive standard deviation measures dispersion across cells, not across women. The DD and annual event-study regressions

[47] See the MNP methodology, Sections 4.2–4.3.

retain analytical weights equal to B_{cps}. Their percentage effects use the treated pre-reform mean from the same regression sample and with the same regression weights. Reported clustered regression standard errors condition on the estimated skill shares. Sources are the INE historical provincial birth series, the female complements of the birth-sex percentages in Table 2.1.7 of the 1968 and 1977 *Anuario Estadístico de España*, the 1999 Fertility Survey, and the 1980–2017 birth register.

C.3. *Birth-record coverage and cohort support, 1980–2017*

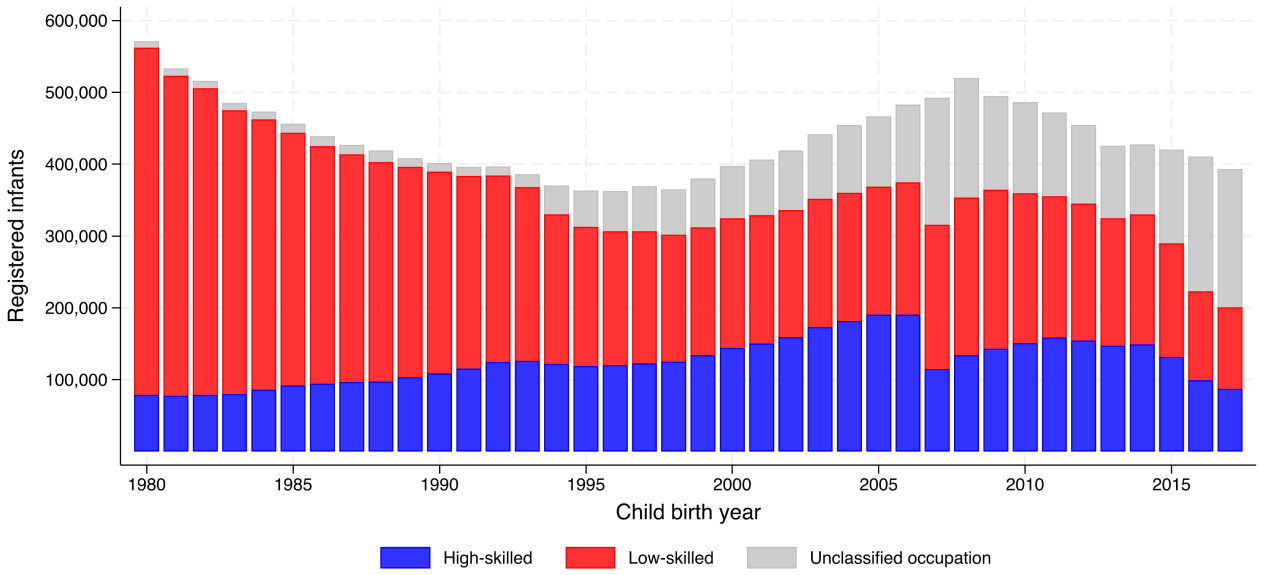
Annual registered births and occupation coverage—Figure C3.1 counts every infant record in the harmonized birth records by child birth year, before imposing the maternal cohort windows. The blue and red segments apply the harmonized high- and low-skilled occupation definitions from Table D1. The gray segment separately retains missing or unclassified maternal occupations. This distinction makes the relationship between the full register and the classified analysis sample explicit: a change in the classified count can reflect both the number of registered births and the availability of a classified occupation. These source counts document coverage within the available file; agreement with a separately published annual INE series has not been established by this exercise.

Table C3.1 compares the full classified register with the main maternal-cohort sample. The first three columns retain classified mothers from every maternal cohort observed in 1980–2017. The last three apply the high-skilled 1962Q1–1969Q4 and low-skilled 1965Q1–1972Q4 windows. All columns include every nationality. Unclassified maternal occupations remain outside these skill comparisons and their record count is reported separately in the notes.

The table reports both eligible infant counts and the numbers with valid timing outcomes. The birth-record eligibility rules are held fixed when age information is unavailable or invalid. Maternal-age and first-delivery-age moments in the main-sample columns use exactly the same unweighted infant observations as Panel A of Table 1, with the skill columns displayed in the opposite order. Full-register age moments apply the same occupation, age and delivery definitions while allowing all maternal cohorts. Thus the difference between the full and main samples comes from the explicit maternal-cohort restriction, rather than a different timing-outcome construction.

Analysis cohorts in the full birth-record window—Figure C3.2 counts eligible infants by the mother's birth-cohort quarter and skill. Both groups are observed in the same 1980–2017 birth-record window, but their skill-specific maternal-cohort windows differ because the relevant labor-market entry ages are 16 and 18. Consequently, there are quarters at either edge of the combined maternal-cohort support that contain only

Figure C3.1. Annual registered births by maternal skill, 1980–2017



Notes: Stacked bars count every source infant record by child birth year before restricting maternal cohorts. Maternal occupations are high-skilled (blue), low-skilled (red), and unclassified (gray). No nationality restriction is imposed. The unclassified category remains separate and is never assigned treatment status. These source counts document administrative coverage; they do not establish equality with a separately published INE series. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

Table C3.1. Full register and main sample by maternal skill, 1980–2017

	Full classified register			Main cohort sample		
	High-skilled	Low-skilled	Total	High-skilled	Low-skilled	Total
Infant records	4,766,560	9,144,080	13,910,640	1,325,611	2,058,507	3,384,118
Share of classified records (%)	34.3	65.7	100.0	39.2	60.8	100.0
Valid maternal-age records	4,766,306	9,140,413	13,906,719	1,325,546	2,056,645	3,382,191
Maternal age: mean	31.363	28.584	29.536	31.576	28.068	29.443
Maternal age: SD	4.673	5.715	5.540	4.829	6.151	5.923
Valid first-delivery-age records	2,773,024	4,213,769	6,986,793	775,401	1,104,814	1,880,215
First-delivery age: mean	30.177	26.442	27.924	30.232	26.031	27.763
First-delivery age: SD	4.628	5.511	5.491	4.796	5.880	5.838

Notes: Counts are infant records, including all nationalities. Full-register columns retain every maternal cohort with classified maternal occupation; the raw register additionally contains 2,674,003 records with unclassified maternal occupation, out of 16,584,643 total records. These unclassified records are shown separately in Figure C3.1 and never enter a skill comparison. Main columns apply the skill-specific maternal cohort windows in the paper. Age statistics are unweighted and use each outcome's valid values under the same canonical definitions as Table 1. First deliveries retain all infants from multiple deliveries. Field availability and invalid ages affect age-row counts, not infant eligibility. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). No agreement with an external INE annual series is asserted by this descriptive support table.

one skill group. Raw infant counts combine realized fertility, cohort size and the ages at which each cohort contributes births. They do not directly measure the family size of uniquely identified women.

Figure C3.3 presents the same eligible infants by child birth year. The figure documents the 1980–2017 calendar support behind the cohort-based analysis and the concentration of realized births in the years when these maternal cohorts were of childbearing age. Neither this calendar profile nor the preceding maternal-cohort counts are causal estimates. The quantity regressions scale the corresponding province–cohort–skill infant totals by the estimated female cohort denominator described in Appendix C.2.

Figure C3.4 presents the numerator-only outcomes in the cohort-profile layout. Panels A)–B) show the cell-size-weighted mean of province-cell infant counts, and C)–D) show the corresponding mean of log counts. These outcomes require no population denominator, so they retain all provinces with eligible infant counts, including Ceuta and Melilla. The panels share a common vertical scale within each row. The fitted lines summarize cohort means using a level shift and slope change at the relevant cutoff; they are distinct from DD level coefficients and annual event-study interactions.

Figure C3.5 uses first-delivery outcomes from birth records. Before 2007, the classification combines available live-child and plurality information to identify deliveries with no preceding live-born children; from 2007 onward, it uses the official previous-live-birth count. Every infant from a twin or other multiple first delivery is retained throughout. Maternal age is the verified delivery-age field, with the same valid-age support as the principal timing regression.

The figure reports the share of first-delivery infant records occurring at maternal age 20 or younger and at age 35 or older, by skill and cohort quarter relative to the reform. It describes how observed first deliveries are distributed across age thresholds. Because the denominator is the set of observed first-delivery infant records with valid age, these shares differ conceptually from both the births per woman at the cohort level and a woman's probability of becoming a mother. The survey contributes no maternal-age observations to these profiles.

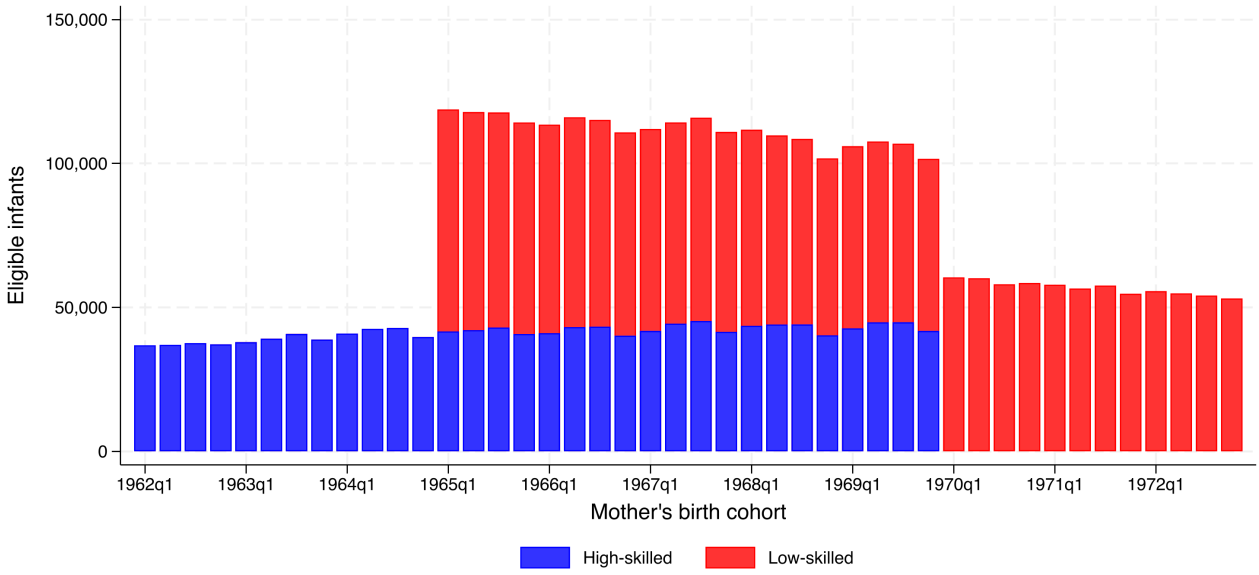
References—

1. Instituto Nacional de Estadística (INE). (n.d.). *Movimiento Natural de la Población: Nacimientos. Nacimientos (cifras mensuales). Por lugar de inscripción (serie histórica 1941–1974): Total nacional y provincias* [Conjunto de datos]. Madrid: INE.

2. Instituto Nacional de Estadística (INE). (1975). *Anuario estadístico de España*. Madrid: INE.

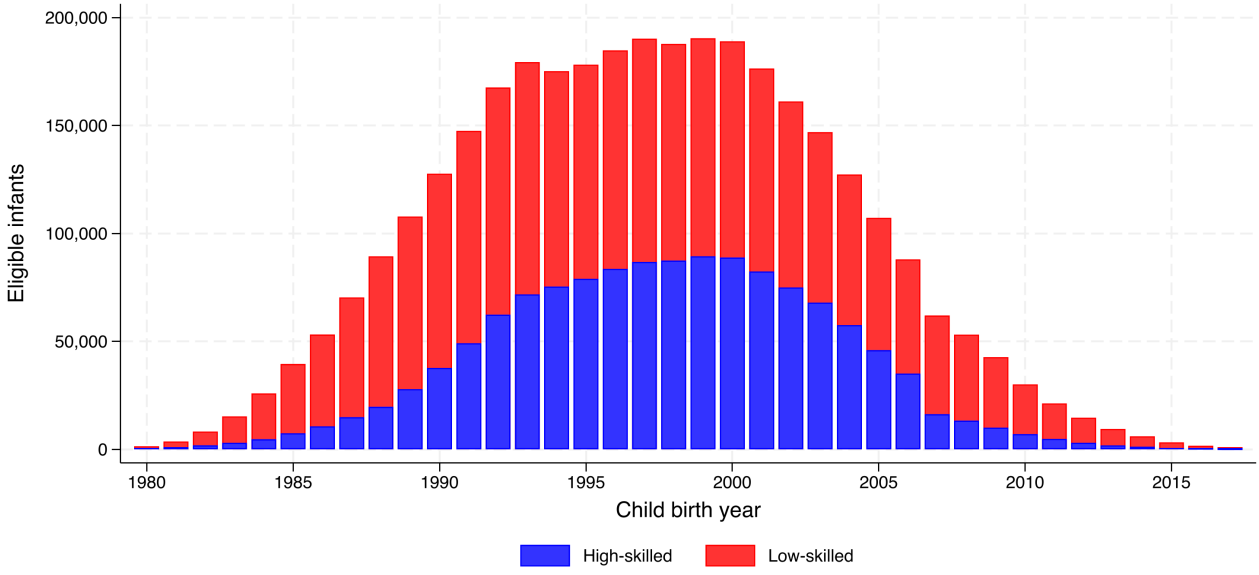
3. Instituto Nacional de Estadística (INE). (n.d.). *Encuesta de Fecundidad 1999* [Microdatos]. Madrid: INE.

Figure C3.2. Registered births by analysis maternal cohort and skill



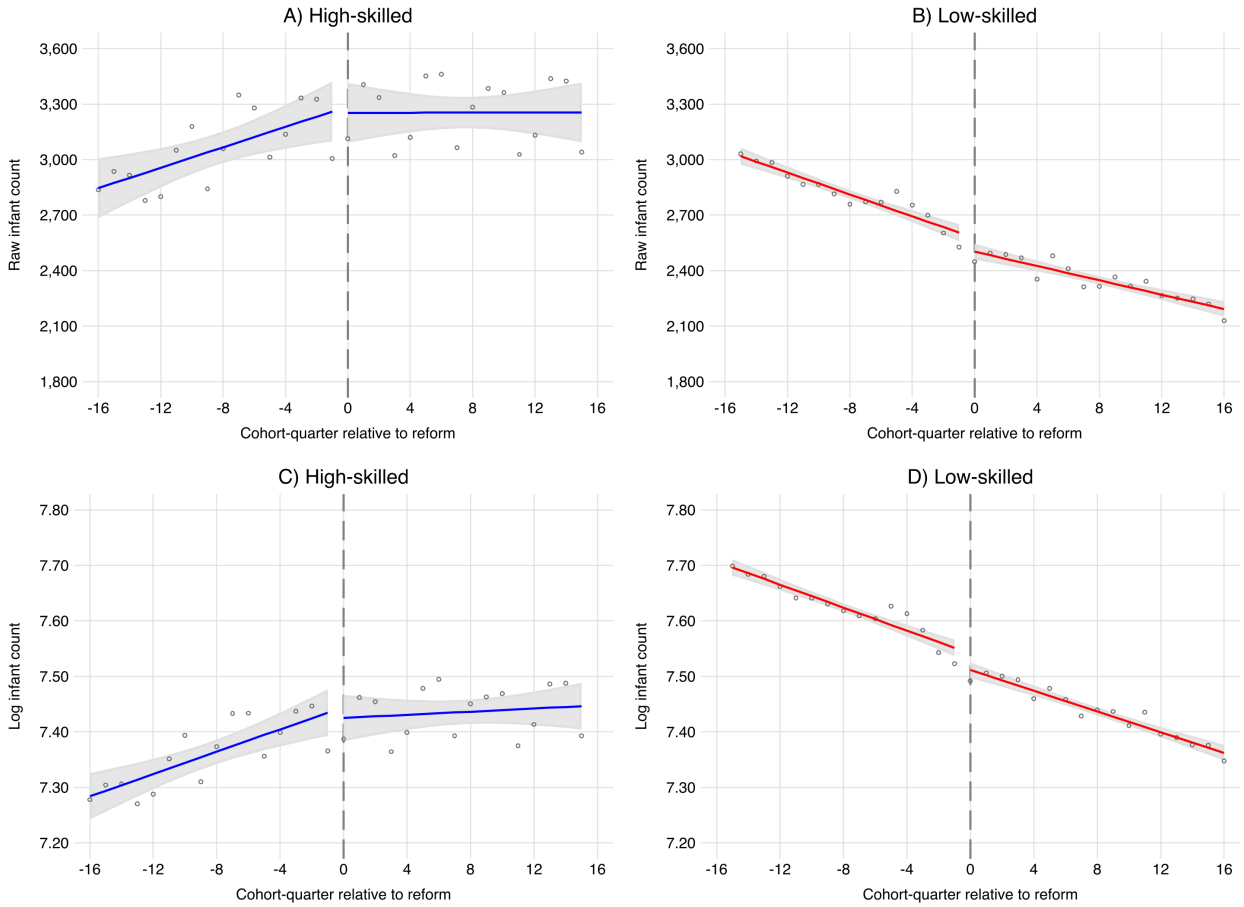
Notes: Stacked bars count eligible infant records by maternal birth-cohort quarter. Blue denotes high-skilled mothers and red low-skilled mothers. Skill groups outside their designated cohort windows contribute no analysis observations. Counts include every eligible infant regardless of maternal-age outcome validity; they are not survey-weighted or divided by the fertility denominator. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

Figure C3.3. Registered births by child birth year in the analysis cohorts



Notes: Stacked bars count eligible infant records by child birth year; blue denotes high-skilled mothers and red low-skilled mothers. Counts are unweighted, include every infant, and use the same skill and maternal-cohort eligibility as the main administrative sample. The figure documents calendar-year coverage rather than a causal contrast. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

Figure C3.4. Raw and log birth-count outcomes by maternal cohort and skill

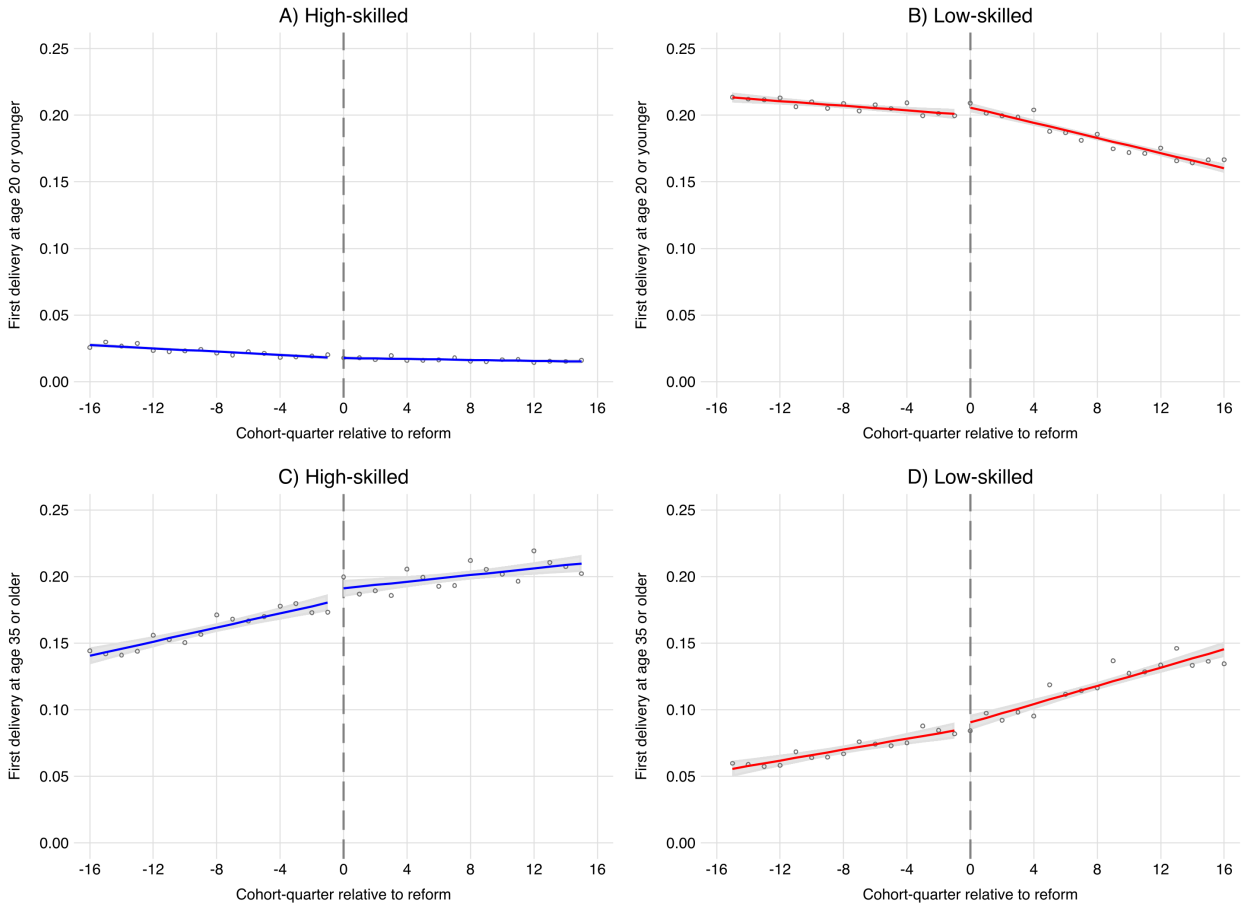


Notes: A)–B) plot cohort-quarter means of province-cell infant counts; C)–D) plot means of their natural logarithms. Cell-size analytical weights are used, including provinces without the estimated female cohort denominator because numerator-only outcomes do not require it. No denominator or rescaling enters these outcomes. Lines are descriptive piecewise-linear cohort-mean fits, allowing a level shift and slope change at the skill-specific cutoff, with 95 percent confidence bands for fitted means. Axes are common across the two skills within each row. These fits are distinct from the DD coefficients and annual event-study estimates. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

4. Instituto Nacional de Estadística (INE). (n.d.). *Encuesta de Fecundidad 2018* [Microdatos]. Madrid: INE.

5. Reher, D. S., & Iriso-Napal, J. (1989). Marital fertility and its determinants in rural and urban Spain, 1887–1930. *Population Studies*, 43(3), 405–427. https://doi.org/10.1080/0032472031000144176

Figure C3.5. First-delivery age thresholds by maternal cohort and skill



Notes: A)–B) plot the share of first-delivery infant records with maternal age at most 20; C)–D) plot the share at least 35. All infants from multiple first deliveries are retained. Outcomes use the same valid administrative ages as the first-delivery-age regression and unweighted infant records. Lines are descriptive piecewise-linear fits to cohort means, allowing a level and slope change at the skill-specific cutoff, with 95 percent confidence bands for fitted means. Axes are common across skills within each row. No survey birth-age observations enter these outcomes. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

C.4. *Spanish Fertility Survey (1999 – 2018)*

This study also draws on the Spanish Fertility Survey (Encuesta de Fecundidad), conducted by the Instituto Nacional de Estadística (INE) in 1999 and 2018. Both surveys are nationally representative, cross-sectional datasets designed to capture the determinants, timing, and outcomes of fertility and reproductive behavior in Spain. In this paper, those survey histories are not the main reduced-form fertility-outcome source; the birth records carry that role. The survey fertility history enters the reduced-form evidence only through the appendix number-of-children robustness check, while the main use of the surveys is to measure mechanisms and preference-related variables. Although both surveys share a common structure and comparable core

indicators, the 2018 wave introduced important innovations in questionnaire design, population coverage, and attitudinal content, reflecting the profound social and demographic transformations that occurred in Spain over the preceding two decades.

The 1999 survey was carried out at the end of the fertility decline that began in the late 1970s, when Spain reached one of the lowest fertility rates in Europe. It collected retrospective reproductive histories for all women aged 15–49, along with a limited set of socioeconomic and attitudinal variables. Respondents provided detailed information on the timing and number of live births, contraceptive use, and union history, as well as basic background characteristics such as education, occupation, and region. Although the 1999 wave included a few questions on gender roles and family values, its main focus was on fertility timing and completed family size rather than on attitudes or household division of labor.

Nearly two decades later, the 2018 Encuesta de Fecundidad represented a major redesign of the survey, both conceptually and methodologically. It included men for the first time (aged 18–55), moved to a digital data collection system, and significantly broadened the scope of attitudinal and behavioral variables. The new modules captured not only detailed reproductive histories but also:

1. Fertility ideals and intentions, including desired number of children, perceived ideal age for motherhood and fatherhood, and reasons for delaying or foregoing childbearing.

2. Work–family reconciliation, covering perceptions of institutional support, maternity and paternity leave experiences, and difficulties balancing employment and family life.

3. Gender-role attitudes, such as agreement with statements on the appropriate division of household tasks, parental responsibilities, and the role of women in the labor market.

4. Partnership dynamics, including cohabitation history, partner's characteristics, and shared decision-making about fertility and domestic work.

These additions make the 2018 survey uniquely suited to examine how attitudes toward fertility, gender norms, and the sharing of household tasks interact with demographic behavior. The 1999 wave, by contrast, serves as a benchmark for fertility behavior during the earlier phase of Spain's demographic transition, but with much less information on underlying beliefs and family norms. Together, the two surveys allow for a rare intertemporal comparison of reproductive intentions and gender attitudes over nearly twenty years of social change.

The Spanish Fertility Surveys are cross-sectional, not panel datasets; hence, there is no attrition across waves. However, both surveys collect retrospective fertility and union histories, which may be affected by

recall bias, particularly for older cohorts. To mitigate this issue, INE incorporated detailed event-history modules that anchor births and partnerships to calendar years and key life events (education completion, first job, etc.), improving temporal accuracy.

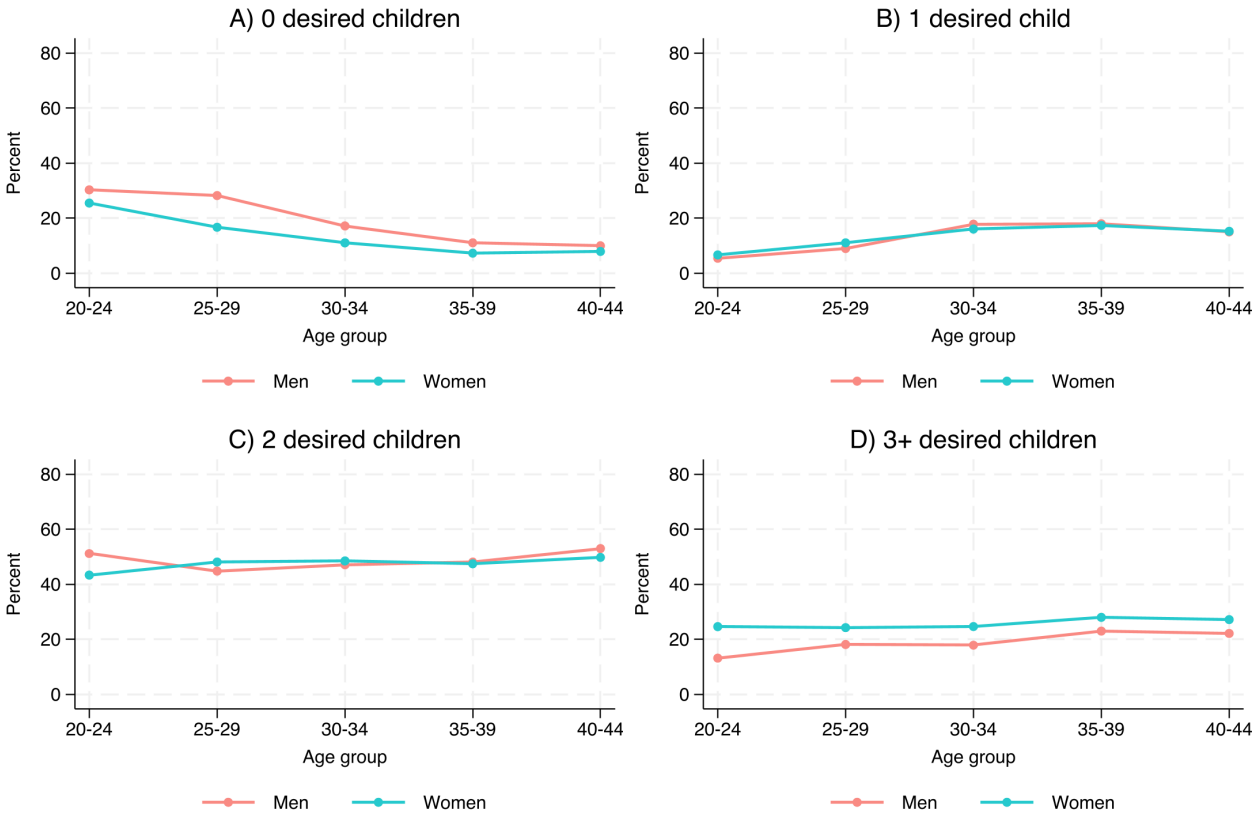
Both surveys exhibit generally low levels of item non-response due to interviewer training and automated consistency checks (in 2018). However, some variables—especially attitudinal questions on fertility ideals, work–family reconciliation, or division of household tasks—show higher rates of missing values.

Despite differences in coverage and questionnaire structure, both surveys maintain comparable fertility-history modules, enabling consistent measures of parity, age at first birth, and timing of subsequent births. The inclusion of attitudinal and gender-norm variables in 2018 offers the opportunity to link actual fertility behaviour with stated beliefs about family and work roles—a dimension not systematically observed in the 1999 data. The microdata include identifiers for regions (NUTS-2 level), allowing researchers to explore spatial differences in fertility behaviour and gender attitudes across Spain. Both datasets are publicly available for research through the INE microdata portal.

Figure C4.1 describes cross-sectional differences in stated desired fertility across age groups for men and women in Spain. It provides a descriptive benchmark for separating what respondents say they want from what is realized under biological and economic constraints. The canonical model uses the all-women desired-family-size measure to identify a preference gradient, while realized parity, childlessness, and first-birth timing remain separate outcomes determined under labor-market and biological constraints.

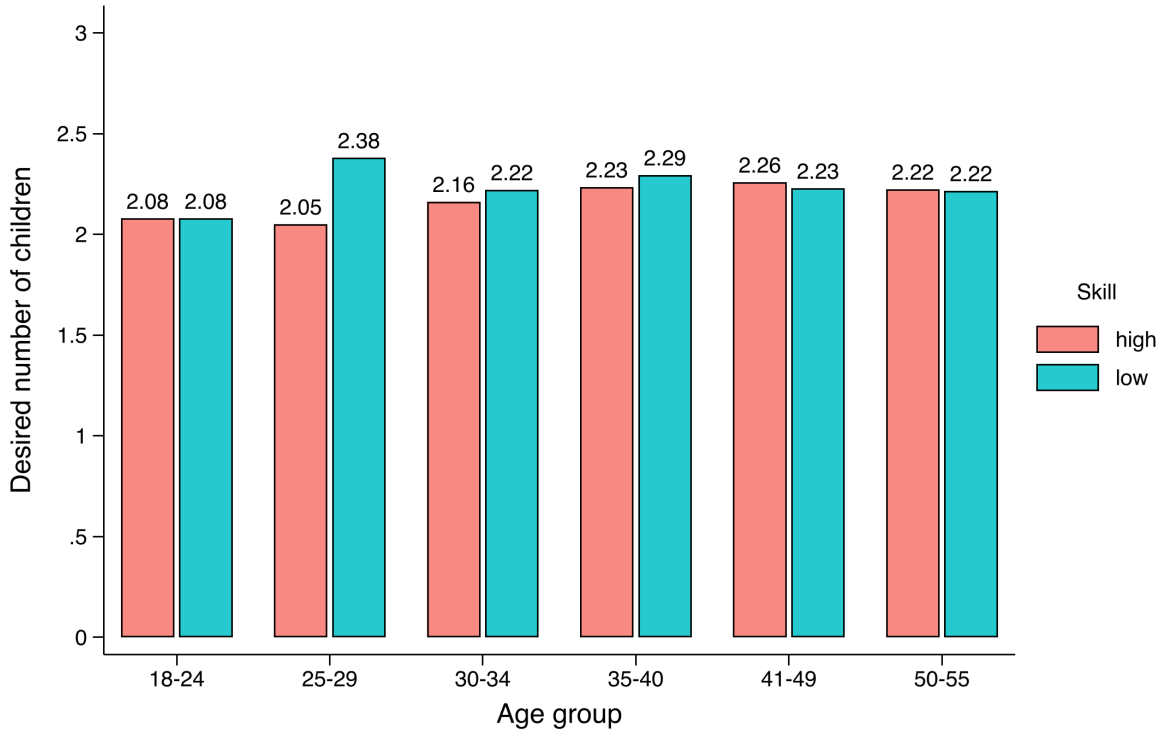
Figure C4.2 reports desired-fertility differences across skill groups among biological mothers and highlights preference heterogeneity that may coexist with differential temporary-employment exposure. The figure remains descriptive. Structural identification instead uses the respondent-level partial gradient of realized on desired fertility, conditioning jointly on religiosity, gender norms, skill, age, and region; it does not target the plotted unconditional skill means.

Figure C4.1. Desired Fertility by gender and age in Spain



Notes: Respondent-weighted shares within each sex and five-year age group, ages 20–44, including parents and childless respondents of all nationalities. The outcome is INE's combined number of desired children (NDESEOHIJO), grouped as 0, 1, 2 and 3 or more; every observed nonnegative integer answer is retained. Each person enters once with the original respondent FACTOR. Panels share a 0–80 percent scale. These are cross-sectional descriptive age profiles, with no regression, fixed effects or reform interpretation. Source: INE, 2018 Fertility Survey, original male and female respondent files.

Figure C4.2. Desired Fertility by age and skill in Spain



Notes: Source: 2018 Fertility Survey, INE, original women respondent file. Bars are respondent-weighted means of desired children (valid values 0–6), with each mother counted once. This contextual sample contains biological mothers aged 18–55 of all nationalities; no child-nationality, child birth-year or reform-cohort restriction is imposed. Low skill comprises highest attained education codes 1–3, and high skill codes 4–9, as in the survey analysis. Age bins show their actual support. These descriptive means do not estimate a reform effect and are not structural targets.

D. VARIABLES AND TREATMENT DEFINITION ROBUSTNESS

Table D1 documents how the treatment groups (high- vs. low-skilled workers) are defined in the birth-record analysis and in the Fertility Survey robustness and mechanism specifications. Because the birth records do not contain a consistent education measure prior to the mid-1990s, the paper uses occupation/status at delivery to classify skill in the birth records, while the Fertility Survey relies on educational attainment given missingness and measurement limitations in occupational variables. By presenting both mappings side-by-side, the table clarifies how the exposure design is implemented across datasets despite differences in available covariates. This transparency is important for interpreting the reduced-form results as capturing differential exposure to the reform across skill groups, rather than artifacts of dataset-specific coding choices.

Table D1. Harmonized treatment definitions by data source

Administrative occupation/status	Fertility Survey education
High-skilled	
(1) Professional, technical and related workers	Upper-secondary education and above
(2) Public-administration and business executives and directors	1999: observed completed-study codes outside 1–7; 2018: completed-study codes 4–9
(3) Clerical and related workers	
(4) Traders and salesworkers	
Low-skilled	
(5) Service workers	Less than upper-secondary education
(6) Agricultural, livestock, forestry, fishing and hunting workers	1999: completed-study codes 1–7; 2018: completed-study codes 1–3
(7) Production workers, transport-equipment operators and nonagricultural laborers	
(8) Professional armed-forces personnel	
(10) People engaged in household work	
Excluded administrative categories	
(9) Students	
(11) Retired persons, pensioners and rentiers	
(12) Persons who cannot be classified	

Notes: Administrative skill uses the harmonized merged occupation/status fields at delivery, applying the same mapping to mothers and fathers throughout 1980–2017. High-skilled is codes 1–4; low-skilled is codes 5–8 and 10. Maternal codes 9, 11, 12 and missing classification are excluded. Principal father interactions require classified father skill, with missing/unclassified fathers considered separately in the appendix. The survey uses source-specific completed education. The pooled 1999/2018 survey child-count sample includes women of all nationalities, including childless women. The 2018 mechanism sample includes women of all nationalities, including childless women, without child-year or child-nationality filters. These source rules do not restrict the all-nationality administrative sample. *Sources:* INE Childbirth Statistics Bulletin occupation dictionaries and the 1999 and 2018 Fertility Survey codebooks.

Table D2 describes the construction of the Egalitarian Views Index used in the reduced-form mechanisms analysis. The table lists each survey item and its exact contribution to the final index. The original response scale runs from agreement to disagreement; the formulas orient every item so that higher values correspond to more egalitarian views. The reduced-form index averages the available recoded items on a 0–1 scale. The structural traditional-norm index uses a parsimonious seven-item subset of the same 2018 Fertility Survey,

oriented so that higher values are more traditional and requiring at least four valid responses. Appendix G.3 documents the exact structural coding and keeps the two indices distinct.

Table D2. Composition of the Egalitarian Views Index

Variable	Variable information	Index score (0–1)
v_juntos	It is acceptable for a couple to live together even if they do not intend to marry.	$(3 - x)/2$
v_divorcio	It is acceptable for an unhappy married couple to divorce even if they have children.	$(3 - x)/2$
v_hijosm	A woman must have children to feel fulfilled.	$(x - 1)/2$
v_hijosh	A man must have children to feel fulfilled.	$(x - 1)/2$
v_crecerfeliz	A child needs a home with both father and mother to grow up happily.	$(x - 1)/2$
v_criarsola	A woman can raise a child on her own if she does not wish to maintain a stable relationship with a man.	$(3 - x)/2$
v_homosexual	It is acceptable for same-sex couples to have the same rights as different-sex couples.	$(3 - x)/2$
v_prioridadm	For a woman, family should take priority over a professional career.	$(x - 1)/2$
v_princuidador	The father or mother should be the main caregiver during ages 0 to 3.	$(x - 1)/2$
v_relcercamad	A working mother can have a relationship with her child as close as a non-working mother.	$(3 - x)/2$
v_mganamas	If the woman earns more than her partner, that is not good for the relationship.	$(x - 1)/2$
v_casafam	Taking care of the home and family is as satisfying as paid work.	$(x - 1)/2$
v_tardom	Men should participate in household tasks to the same extent as women.	$(3 - x)/2$
v_quedarconm	If parents divorce, it is better for the child to stay with the mother than with the father.	$(x - 1)/2$
v_trabescaso	When jobs are scarce, men should have greater right to a job than women.	$(x - 1)/2$
v_relcercapad	A working father can have a relationship with his child as close as a non-working father.	$(3 - x)/2$

Continued on next page

Table D2 (continued)

Variable	Variable information	Index score (0–1)
`v_prioridadh`	For a man, family should take priority over a professional career.	$(3 - x)/2$

Notes: The original response scale is 1 = Agree, 2 = Neither agree nor disagree, 3 = Disagree. The formulas show each item's contribution to the final egalitarian index, where larger values are more egalitarian. The index averages available valid items on the 0–1 scale; it is not a z-score. Missing items are excluded from the respondent's item average. The seventeen-item reduced-form index remains distinct from the structural model's seven-item index. *Source:* INE 2018 Fertility Survey, original T_OPINA coding.

Table D3 documents an alternative occupation-based Fertility Survey classification. The table lists the detailed categories assigned to low-skilled and high-skilled groups. The mapping uses the current occupation reported in the 2018 survey; missing or unclassified occupation remains unclassified. The table documents this alternative classification; the reported survey estimates use education-based skill groups. The birth-register sample-definition check in Table F8 is separate and uses the occupation/status measure in birth records recorded at delivery.

Table D3. Alternative treatment definition: Fertility Survey occupations

Fertility Survey Dataset	
Low-skilled workers	**High-skilled workers**
Employees of travel agencies, receptionists, and telephone operators; counter clerks and related workers (excluding ticket sellers).	Members of the executive branch and legislative bodies; leaders of public administration and social interest organizations; executive directors
Administrative employees with public-facing tasks not classified under other headings.	Directors of administrative and commercial departments
Waiters and cooks who own their businesses.	Directors of production and operations
Salaried workers in the restaurant services.	Directors and managers of accommodation, restaurant, and retail businesses
Salesclerks in stores and warehouses.	Directors and managers of other service enterprises not classified under other headings
Store owners.	Healthcare professionals
Salespeople (except in stores and warehouses).	Professionals in early childhood, primary, secondary, and post-secondary education

Continued on next page

Table D3 (continued)

Fertility Survey Dataset	
Low-skilled workers	**High-skilled workers**
Cashiers and ticket sellers (excluding banks).	Other education professionals
Workers in healthcare services for personal care.	Professionals in physical, chemical, mathematical, and engineering sciences
Other workers in personal care.	Legal professionals
Workers in personal services.	Specialists in public administration and business organization and marketing
Workers in protection and security services.	Information technology professionals
Skilled workers in agricultural activities.	Professionals in social sciences
Skilled workers in livestock activities (including poultry, beekeeping, and similar).	Professionals in culture and entertainment
Skilled workers in mixed agricultural and livestock activities.	Technicians in physical and engineering sciences
Skilled workers in forestry, fishing, and hunting activities.	Supervisors in mining engineering, manufacturing industries, and construction
Workers in structural construction and related occupations.	Health technicians and alternative therapy professionals
Finishers in construction and installation (except electricians), painters, and related workers.	Financial and mathematical support professionals
Welders, sheet metal workers, metal structure assemblers, blacksmiths, toolmakers, and related workers.	Representatives, commercial agents, and related professionals
Machinery mechanics and adjusters.	Administrative support professionals; technicians in law enforcement
Skilled workers in electricity and electrotechnology.	Support professionals in legal, social, cultural, sports, and related services
Precision metalworkers, ceramists, glassworkers, craftsmen, and workers in the graphic arts.	Information and communication technology (ICT) technicians
Workers in the food, beverage, and tobacco industry.	Employees in accounting, financial, production support, and transportation services
Workers in wood, textiles, clothing, leather, footwear, and other craft-related occupations.	Library, postal, and related service employees

Continued on next page

Table D3 (continued)

Fertility Survey Dataset	
Low-skilled workers	**High-skilled workers**
Operators of fixed installations and machinery.	Other administrative employees without public-facing tasks
Assemblers and installers in factories.	
Locomotive engineers, agricultural machinery operators, heavy mobile equipment operators, and sailors.	
Drivers of vehicles for urban or road transport.	
Domestic employees.	
Other cleaning staff.	
Food preparation assistants.	
Urban waste collectors, street vendors, and other elementary occupations in services.	
Agricultural, forestry and fishing labourers.	
Construction and mining labourers.	
Manufacturing labourers.	
Transport labourers, freight handlers and shelf fillers.	
Military occupations.	

Notes: This table documents the alternative current-occupation classification for the 2018 Fertility Survey: recoded OCUPA categories 01–25 are high-skilled and 26–62 are low-skilled. Missing or unclassified occupation remains unclassified. The reported survey regressions define skill using education. The entries are category definitions; no regression, analytical weights, fixed effects or clustering apply. *Source:* INE, 2018 Fertility Survey individual-file dictionary, T_OCUP.

E. LABOR-MARKET FIRST-STAGE RESULTS

This appendix provides supporting administrative evidence for the labor-market first stage. Low-skilled women are assigned the 1968Q4 cutoff, when they turn 16 after the reform, and high-skilled women are assigned the 1966Q1 cutoff, when they turn 18 after the reform. The entry-age evidence uses a broad low-skilled sample because imposing a subsequent attachment requirement would condition the entry outcome on future employment. The objective is not to estimate fertility effects in the MCVL, which does not contain the full fertility histories used in the main analysis, but to assess recorded affiliation at the labor-market ages directly exposed to the liberalization. Main-text Figure 4 reports the formal early-career job-stability, employment-intensity, and contribution-based income estimates; Appendix Figure E2 reports the corresponding descriptive cohort paths.

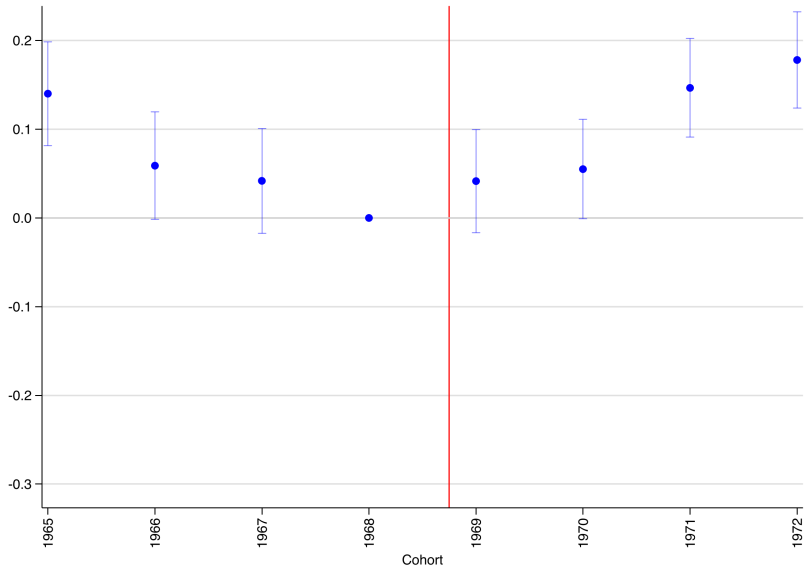
Figure E1 shows birth-year coefficients for first recorded affiliation by age 18 in this low-skilled entry sample.

The career outcomes use the ten calendar-age years and affiliation-support restrictions specified in the data section. The source panel selects an annual main affiliation; we retain this selection rather than substituting a secondary job when the main record is not employee work. A year counts toward years of non-permanent employment when the main affiliation is employee work classified as temporary or covered by the agricultural, domestic-service, maritime, or mining employee regimes. Special-regime status does not establish a fixed-term contract. Where contract status is missing, the source rule classifies an affiliation as stable if at least two of three conditions hold: one spell in the year, at least 180 main-job days in that year, and at least 365 main-job days across the observed history; otherwise it classifies it as temporary. This duration-based imputation is imperfect and uses subsequent history. The replication reports its classification performance when observed contract codes are masked, separately from the broad special-regime assignment.

Days worked are reconstructed from valid employee spell dates. Concurrent spells are combined as a union of calendar days, then each year's total is capped at 365 and the decade total divided by ten. Benefit, own-account, and other nonemployment affiliations contribute no employee days. An invalid potentially relevant employee spell makes this outcome unavailable for that woman.

For contribution-based income, we link monthly bases to employee and benefit-related contribution accounts. Repeated records for the same woman, account, year and month contribute the largest nonzero recorded base once; concurrently observed eligible accounts are then summed. Accounts also linked to other affiliation categories are treated as unavailable. We deflate the bases to 2011 euros and retain an annual sum only when all eligible account-months are observed and its sum is nonnegative. The person-level outcome averages these complete years within the ten-year window, requiring at least one. Years without eligible

Figure E1. Probability of first recorded affiliation by age 18.



Notes: Points are birth-year coefficients for the probability of having a first recorded Social Security affiliation by age 18, including age 18. The regression includes birth-year dummies and birth-quarter fixed effects, with 1968 omitted, in the low-skilled MCVL entry sample without the subsequent career-attachment restriction. Whiskers are 95% heteroskedasticity-robust confidence intervals. This figure has no high-skilled comparison group and does not plot the Table 2 post-cutoff slope coefficient. The vertical line marks 1968Q4, the first fully exposed low-skilled cutoff. *Source:* MCVL 2006.

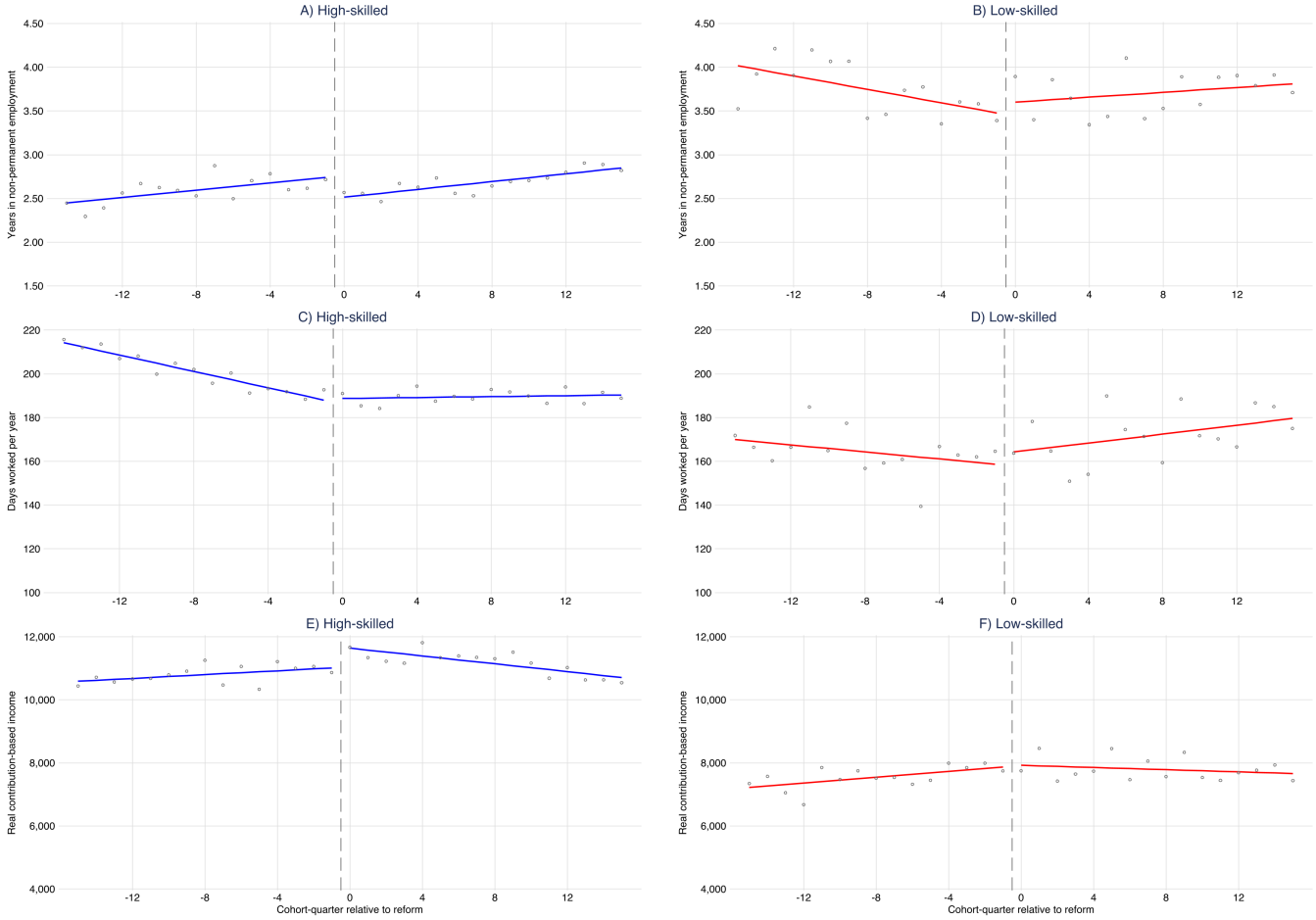
affiliations, missing bases, and ambiguous histories are not imputed as zero. This conditioning differs from the fixed ten-year denominator for days worked. The MCVL documentation reports missing agricultural employer contribution bases in the 2006 edition; these are treated as unavailable. Contribution bases can be capped or statutory and do not measure total cash wages, benefits, or disposable income. The sample-size differences in Figure 4 reflect these support rules.

For the lower-schooling entry-age evidence, we estimate:

$$Y_{ic} = \alpha + \beta Post_c + \gamma Trend_c + \delta\left(Post_c \times Trend_c\right) + \pi_q + \varepsilon_{ic}, \tag{E1}$$

where i indexes women, c indexes birth cohorts, q indexes birth quarter, $Trend_c$ is measured relative to the 1968Q4 cutoff, and β captures the discontinuous change for lower-schooling women at the reform threshold. The transition specifications additionally include indicators for 1968Q3 and 1968Q4. Main-text Table 2 reports a positive post-cutoff change in the cohort slope of early recorded affiliation. Figure E1 presents the corresponding birth-year coefficients. These entry-age results complement the three early-career outcomes reported in Figure 4.

Figure E2. Labor-market outcomes in the first 10 years of potential labor-market experience, by skill group



Notes: High-skilled cohorts appear on the left and low-skilled cohorts on the right. Dots are cohort-quarter means for the outcome-specific Figure 4 samples. Years of non-permanent employment includes years in the specified employee special regimes; income is average annual real contribution-based income (2011 euros per year). Blue (red) lines fit high-skilled (low-skilled) means separately before and after the dashed exposure cutoff. Within each outcome row, both skill groups use identical y-axis limits and tick intervals. Construction and sample details are in the text above. *Source:* MCVL 2006.

Figure E2 plots cohort-quarter means separately by skill using the Figure 4 outcome definitions and their respective available observations. Split linear fits describe the cohort paths; formal inference comes from the DD estimates in Figure 4.

F. REDUCED-FORM EVIDENCE AND ROBUSTNESS

This appendix consolidates the reduced-form material that supports the empirical narrative in the manuscript. It first reports birth-record count robustness checks and, in the same table, the Fertility Survey number-of-children estimate as an appendix child-count robustness check. It then reports the outcome-fitted synthetic-control diagnostics for the birth-record outcomes and the father-side mechanism event-study diagnostics before turning to fertility timing, age-threshold estimates, and survey-calibrated selection sensitivities. The final parts report inference checks, multiple-testing corrections, and the robustness figures referenced in the main text.

Unless otherwise noted, event time −1 is the omitted reference period and event time 0 is the first fully treated cohort under the baseline coding. Circles report annual regression coefficients and bars report 95 percent confidence intervals.

F.1. *Birth-record fertility checks and event-study diagnostics*

The raw- and log-birth-count estimates in Table F1 check whether the fertility pattern also appears without the estimated cohort denominator, while the respondent-weighted Fertility Survey estimate provides a comparison based on individual child counts, including childless women.

Birth-record synthetic-control checks.—The synthetic-control checks compare the low-skilled outcome path with a fitted combination of high-skilled province paths. Following (4), weights sum to one, may be negative, and are chosen using annual pre-period bins −4 through −1 only. The same weights are carried into the post period, and the gap is normalized at bin −1. This makes the comparison easy to interpret: a negative post-period gap indicates a decline relative to the path fitted before the reform. Births per woman uses the estimated historical female cohort denominator; Figures F1 and F2 check whether the raw and log birth-count numerators also decline.

Father-side mechanism event studies.—Table F2 and Figure F3 extend Table 8 to annual event bins. A) is low-skilled mothers with low-skilled fathers; B) is low-skilled mothers with high-skilled fathers. Both interactions are estimated jointly, comparing each group with high-skilled mothers pooled across classified father skill. Missing or unclassified father occupation is excluded. Exposure uses the mother's cohort and cutoff. Births per woman at the cohort level is split by realized-birth father shares, preserving its denominator and weights.

The annual lead tests reject joint zero pre-period coefficients in both panels (Table F2); these pre-period movements limit a causal reading of the dynamic paths. The synthetic overlay uses donor weights that combine high-skilled province paths to match each low-skilled pre-reform outcome path. Weights sum to

Table F1. Birth-number robustness and weighted Fertility Survey child counts

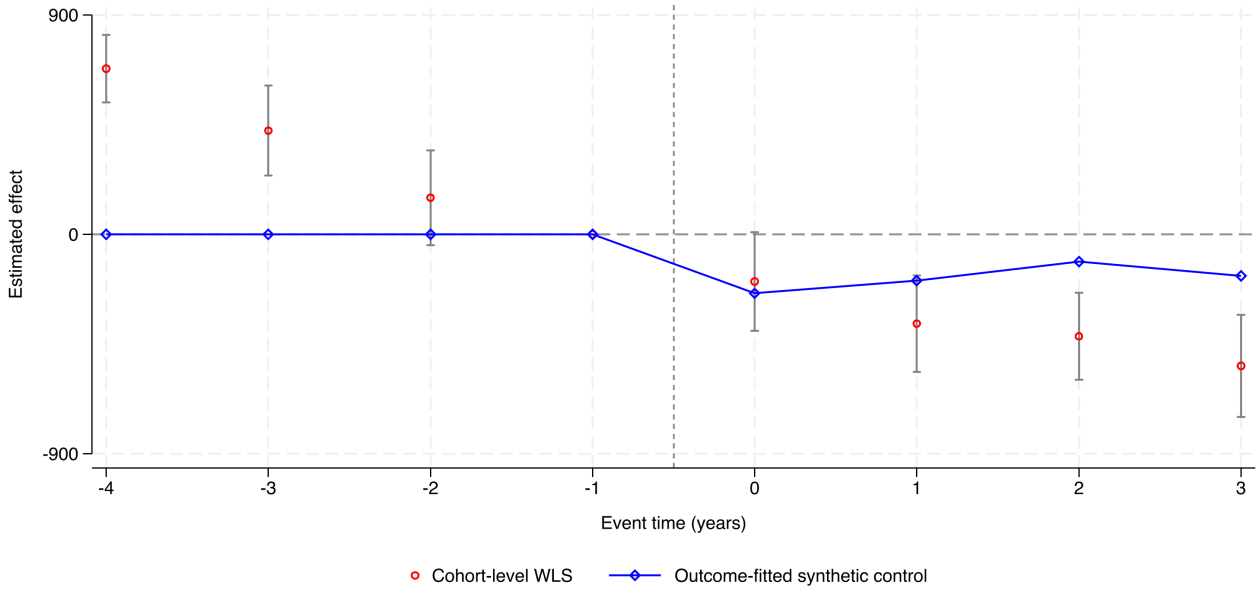
	Log births per cohort cell	Raw birth count	Biological children per survey respondent
Post × low-skilled	-0.265***	-686.097***	-0.217***
	(0.019)	(60.419)	(0.061)
Mean outcome	7.483	2817.731	1.228
Treated pre-reform mean	7.626	2818.347	1.520
Coefficient (% of pre mean)	-3.5	-24.3	-14.3
Observations	3,328	3,328	5,621
R-squared	0.944	0.950	0.079

Notes: Columns 1–2 use administrative cohort-quarter by province by skill cells, with eligible birth-count analytical weights and province fixed effects. Log births use positive cell counts. Raw/log counts include Ceuta and Melilla because they do not require the historical denominator. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. Column 3 uses one respondent per eligible woman in the pooled 1999/2018 Fertility Surveys, including childless women with zero biological children, source-specific education groups and respondent probability weights; it includes region and survey-wave fixed effects and the same DD cohort trends. Women of all nationalities are included in both survey waves. Survey counts need not measure completed fertility, particularly in 1999. The percent row scales each coefficient by its own treated pre-reform outcome mean; the log-outcome ratio is not a percent change in the number of births. Stars denote p below .10, .05, .01. *Sources:* INE Childbirth Statistics Bulletin, 1980–2017, and 1999/2018 Fertility Surveys.

one, may be negative, and remain fixed after fitting bins −4 through −1. The gap is normalized at event time −1, and its post-reform path is a design-based magnitude diagnostic. Exact pre-period fit is by construction and does not resolve the lead-test evidence. The final low-skilled cohort has no high-skilled support in annual bin 4; it remains in the DD sample but is excluded from these annual paths.

The missing-father diagnostics in Table F2a assess how the availability of recorded father characteristics varies with exposure, complementing the estimates for classified fathers.

Figure F1. Annual raw birth count with outcome-fitted synthetic control



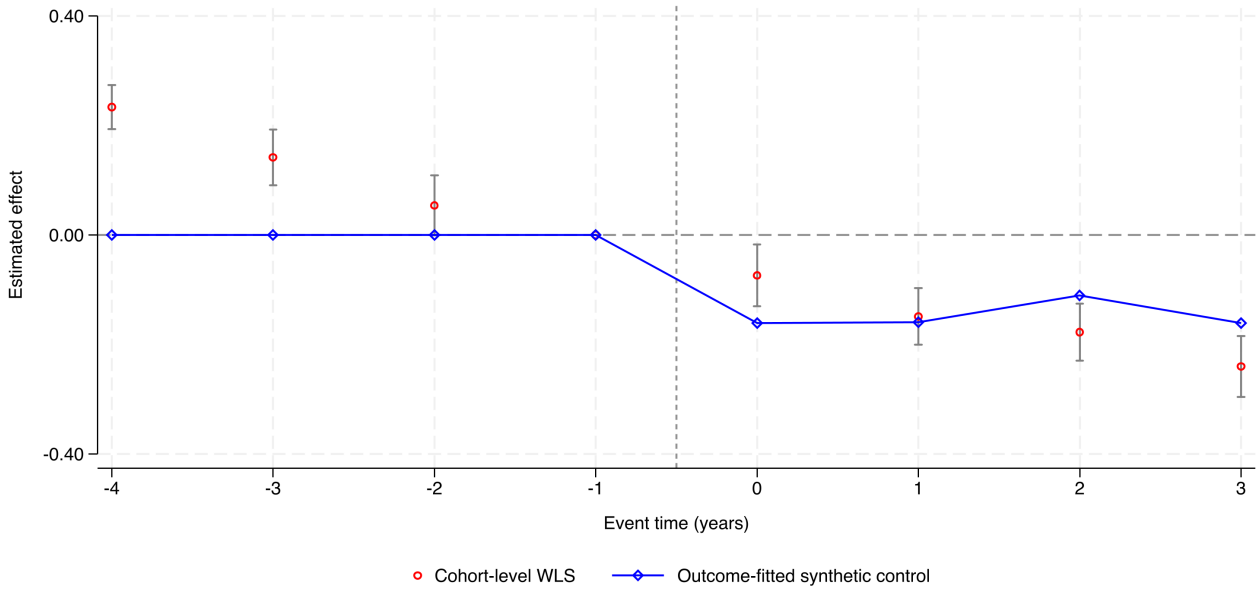
Notes: The outcome is the raw number of registered births in each cohort-quarter by province by skill cell. One observation is a maternal cohort-quarter by province by skill cell. Cohort-level WLS denotes the annual event-study regression with eligible birth-count analytical weights. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Annual skill interactions and province fixed effects are included. Intervals are 95% confidence intervals clustered by maternal cohort-quarter; bin -1 is omitted and the unsupported final-bin contrast is not plotted. Symmetric axes retain all intervals, with common limits within same-unit panel pairs. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). Weights combine high-skilled province paths to match the low-skilled pre-reform path in bins -4 through -1; they sum to one and may be negative. The gap is normalized at -1. Its post-reform path is a design-based magnitude diagnostic; pre-period fit is imposed.

Table F2. Father-skill annual event-study diagnostics

Panel	Pre joint p	Post joint p	Mean post	SE	Observations
A	.000646	1.59e-07	-0.114	0.023	4,700
B	.0104	.301	-0.017	0.012	4,700

Notes: The outcome is births per woman (cohort-level), split by recorded father skill. Samples and estimators match Figure F3. A and B are the low-mother/low-father and low-mother/high-father interactions from one joint regression. Each comparison pools high-skilled mothers with classified fathers; missing/unclassified fathers are excluded. Pre tests cover bins -4 through -2; bin -1 is omitted. Post averages use bins 0–3 and their full clustered covariance. Birth-count weights, province fixed effects and maternal cohort-quarter clustering are used. The unsupported final-bin contrast is not reported. Synthetic paths are magnitude diagnostics, not confidence intervals. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. *Sources:* INE birth register, historical births and sex shares, and 1999 Fertility Survey.

Figure F2. Annual log birth count with outcome-fitted synthetic control



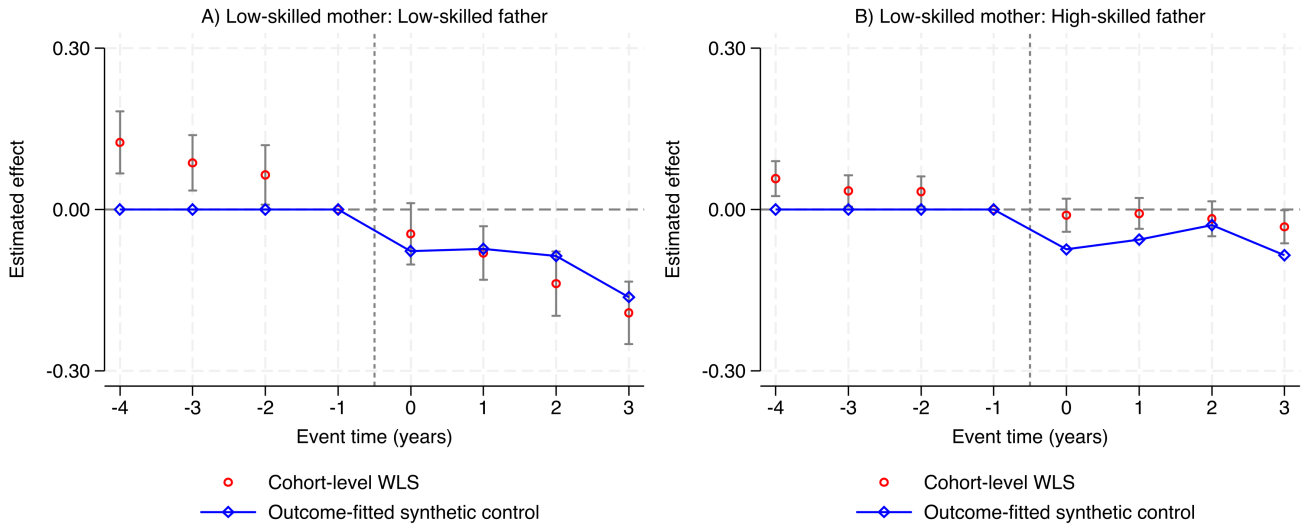
Notes: The outcome is the log registered birth count in positive cohort-quarter by province by skill cells. One observation is a maternal cohort-quarter by province by skill cell. Cohort-level WLS denotes the annual event-study regression with eligible birth-count analytical weights. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Annual skill interactions and province fixed effects are included. Intervals are 95% confidence intervals clustered by maternal cohort-quarter; bin -1 is omitted and the unsupported final-bin contrast is not plotted. Symmetric axes retain all intervals, with common limits within same-unit panel pairs. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). Weights combine high-skilled province paths to match the low-skilled pre-reform path in bins -4 through -1; they sum to one and may be negative. The gap is normalized at -1. Its post-reform path is a design-based magnitude diagnostic; pre-period fit is imposed.

Table F2a. Missing father occupation: separate sample diagnostics

Outcome and sample	DD	SE	Observations
Unclassified father share: all births	0.005	0.001	3,384,118
Maternal age: unclassified fathers only	0.231	0.091	233,775
First-delivery age: unclassified fathers only	0.161	0.099	143,039

Notes: These are separate diagnostics, not additional categories in Table 8. The first row uses every eligible infant and an indicator for unclassified father occupation. Timing rows restrict to that category and compare low- with high-skilled mothers under the mother's exposure definition. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. First deliveries retain every infant from multiple deliveries. The appendix discussion following Table F3 reports sensitivity to conflicting parity information. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE).

Figure F3. Annual father-skill mechanism event studies



Notes: Red circles give annual weighted event-study estimates with 95% clustered intervals; blue diamonds give synthetic gaps. A and B are the two interactions from one joint regression: low-skilled mothers with low-skilled and high-skilled fathers, respectively. All comparisons pool high-skilled mothers across classified father skills; missing/unclassified fathers are excluded. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. Cohort-province couple-group cells use the main mother-cohort denominator and birth-count weights, annual interactions and province fixed effects. Inference clusters by maternal cohort-quarter; bin -1 is omitted and the unsupported final-bin contrast is not plotted. Signed donor weights sum to one and fit the low-skilled pre-period path; the gap is normalized at -1 and is a design-based magnitude diagnostic. Axes share symmetric limits containing every interval. *Sources:* INE birth register, historical births and sex shares, and 1999 Fertility Survey.

F.2. *Fertility Timing, Age Thresholds, and Selection Sensitivity*

This subsection collects the fertility evidence that is useful for interpretation but secondary to the birth-record results emphasized in the main text. The ordering follows Section 5.2: first the birth-record timing-estimator table, then the age-threshold tables that complement Figures 7 and 8, and finally the selection sensitivities. The common message is that the birth records provide the most precise evidence on timing: exposed mothers in the low-skill occupation/status group have fewer very young births and more births, including first births, at older maternal ages.

The comparison in Table F3 distinguishes the timing change at the cohort cutoff, measured by DD, from the average change across supported post-reform event years.

First-delivery measurement. The direct live-child count and plurality determine first deliveries throughout. The previous-child date is an auxiliary consistency check: before 1996 it refers to a previous delivery and need not indicate a live birth. From 1996 onward, some count-based first deliveries report an earlier previous-live-child date. The principal classification retains these observations because the conflict does not establish which field is incorrect. Excluding the 5,577 timing observations with such a date (valid month and year

Table F3. Maternal-age outcomes: regression estimators

	DD	Event-study
A) Maternal age at birth		
Estimated coefficient	0.526***	0.440***
	(0.067)	(0.088)
Level effect (% of pre mean)	1.9	1.6
Observations	3,382,191	3,382,191
R-squared	0.107	0.108
B) First-delivery age		
Estimated coefficient	0.441***	0.350***
	(0.060)	(0.082)
Level effect (% of pre mean)	1.7	1.4
Observations	1,880,215	1,880,215
R-squared	0.154	0.154

Notes: Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. The event-study column averages annual interactions 0–3 relative to omitted bin -1, using their full covariance. Both specifications retain all eligible cohorts; the unsupported final-bin contrast is not reported. Percentages use the corresponding treated pre-reform mean. First deliveries retain every infant from multiple deliveries. The appendix discussion following Table F3 reports sensitivity to conflicting parity information. One observation is an infant record; timing each infant receives equal weight. Maternal ages require at least 15 years; first-delivery ages also require at most 55 years. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). Stars denote p below .10, .05, .01.

preceding the current delivery, including earlier months within the same year) gives a first-delivery-age DD estimate of 0.442 years (standard error 0.060; 1,874,638 observations), or 1.7% of the corresponding treated pre-reform mean of 25.349 years. This check uses the same eligibility, province fixed effects, cohort trends and cohort-quarter clustering as Table F3; regressions are unweighted. It evaluates sensitivity to inconsistent reports rather than validating either response. *Source:* INE, birth records, 1980–2017.

The age-threshold estimates in Table F4 describe whether the increase in maternal age reflects fewer realized births at younger ages, more at older ages, or both, with separate estimates for all births and first deliveries.

Selection sensitivities.—Tables 4 and 5 report conditional empirical-trimming and joint-normal selection sensitivities. Both use the 2018 Fertility Survey, when the eligible women are near the end of their repro-

Table F4. Age-threshold DD estimates for any birth and first delivery

	Pr(age ≤20)	Pr(age ≥35)	Pr(age ≥40)
A) Any birth			
Post × low-skilled mother	-0.013***	0.015***	0.008***
	(0.001)	(0.004)	(0.002)
Mean outcome	0.080	0.208	0.039
Treated pre-reform mean	0.131	0.134	0.028
Effect (% of pre mean)	-9.9	11.6	29.5
Observations	3,382,191	3,382,191	3,382,191
R-squared	0.048	0.028	0.003
B) First delivery			
Post × low-skilled mother	-0.016***	0.007**	0.005***
	(0.002)	(0.003)	(0.001)
Mean outcome	0.123	0.129	0.025
Treated pre-reform mean	0.207	0.070	0.015
Effect (% of pre mean)	-7.8	10.0	32.3
Observations	1,880,215	1,880,215	1,880,215
R-squared	0.082	0.026	0.003

Notes: One observation is an infant record. Outcomes are probability indicators constructed from valid administrative maternal age. Panel A uses all births with valid age; Panel B uses first deliveries with valid age. Regressions are unweighted. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. First deliveries retain every infant from multiple deliveries. The appendix discussion following Table F3 reports sensitivity to conflicting parity information. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. All three thresholds in a panel use the same regression sample. Maternal ages below 15 are missing; first-delivery age additionally requires at most 55. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). Stars denote p below .10, .05, .01.

ductive life cycle. The sample includes women of all nationalities, including those without children, and uses official respondent weights and the maintained cohort design.

We estimate two selection probabilities. For maternal age across all births, selection means having at least one biological child. For first-delivery age and first-delivery perinatal health, the reference selection event is a first biological birth during 1980–2017 at maternal ages 15–55. We identify the earliest biological birth from the complete child history before applying these calendar and age limits. Childless women and women whose first birth falls outside these limits remain in the denominator. Neither definition restricts the woman's nationality or the child's nationality or country of birth.

The motherhood specification gives an exposed selection probability of 78.801% and an estimated no-reform probability of 83.127%, implying a proportional loss of 5.204%. For the first-delivery reference event, the corresponding probabilities are 78.705% and 82.684%, implying a loss of 4.813%. Each loss is calculated as $q = 1 - s_T/s_C$ from the weighted survey DD. The first-delivery calibration is shared by the first-delivery outcomes; each health regression retains its original sample and outcome availability, including C-section records from 2007–2017.

Both calculations assume weakly decreasing individual selection into the relevant event. They apply its estimated proportional loss to high-skilled post-reform comparison records within maternal cohort-quarter–province cells and refit the maintained administrative DD. This links the survey and register through common proportional selection effects across skill groups, source populations and infant-record contributions. Maternal age across all births also reflects changes in the composition of births by parity. The results are therefore survey-calibrated selection sensitivities rather than conventional sharp Lee bounds.

The conditional calculation estimates the relevant selection probability separately by maternal birth quarter, interacting the DD level terms with birth-quarter indicators while retaining the common cohort trends and region effects. Selection is assumed to weakly decrease within each birth quarter; an estimated increase is assigned zero loss under this restriction. For each quarter, it retains a fraction $1 - q_j$ of the administrative comparison-cell distribution. The lowest and highest attainable means come from its lower and upper tails, with fractional retention at tied values. The DD endpoints combine these means according to the signs of the regression weights. Maternal birth quarter is determined before reform exposure. No outcome distribution is imposed.

The joint-normal calculation uses the comparison-group selection probability for the same reference event: 72.133% for motherhood and 71.812% for first deliveries. Write this probability as s_H, define $s_A = s_H(1-q)$, $c(s) = \Phi^{-1}(1-s)$, $\lambda(s) = \phi(c(s))/s$, and $a_H = \lambda(s_H)[\lambda(s_H) - c(s_H)]$. For observed comparison-cell mean μ_H and standard deviation σ_H, the normal moment restriction gives

$$\mu_A = \mu_H + \frac{\rho\sigma_H}{\sqrt{1-\rho^2 a_H}}\big[\lambda(s_A) - \lambda(s_H)\big].$$

We refit the DD across the full common correlation range $\rho \in [-1, 1]$, including its limiting endpoints. This is a continuous normal moment approximation for ages, birthweight and gestation. Binary outcomes use latent normal thresholds with unrestricted cell correlations, giving feasible means $[\max\{0, (\mu_H - q)/(1-q)\}, \min\{1, \mu_H/(1-q)\}]$; at the same selection rate, these coincide with the feasible empirical retained-mean envelope.

The tables report point endpoints. Separate endpoint uncertainty calculations use 999 paired survey–register cohort-cluster resamples. Both overall and birth-quarter-specific selection probabilities are reesti-

mated in every draw, with probability and monotonicity constraints applied consistently. The smaller-cluster bootstrap draws 20 of 44 cohorts within birth-season and cutoff-regime strata, restores stratum mass and uses basic intervals scaled by $\sqrt{20/44}$, with seed 20260905. These are pointwise asymptotic intervals; unidentified conditional draws are retained and flagged, and conditional intervals are left unavailable if any draw is unidentified. Positive point endpoints alone do not establish statistical significance.

F.3. *Inference, Multiple Testing, and Robustness Checks*

This subsection consolidates the inference and robustness checks referenced in the main text. Appendix Figures F1–F3 report the annual numerator and father-side diagnostics.

The coefficients, clustered standard errors, p-values, and sample sizes in Table F5 allow the precision of the headline estimates to be compared on the same basis.

To account for testing several related mechanisms, Table F6 pairs raw p-values with Benjamini–Hochberg adjustments, calculated separately within the five economic outcomes in Table 6 and the five preference and support outcomes in Table 9.

Table F5. Inference for headline reduced-form estimates

Outcome	Estimate	SE	p-value	N
Births per woman (cohort-level)	-0.257	0.018	< 0.000001	3,200
Maternal age at birth	0.526	0.067	< 0.000001	3,382,191
First-delivery age	0.441	0.060	< 0.000001	1,880,215
Survey biological children	-0.217	0.061	0.000978	5,621

Notes: These p-values use the unrounded code-generated coefficients, clustered covariance and regression degrees of freedom. Administrative estimates reproduce Table 3/Table 4: Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. Quantity uses cell birth-count analytical weights; timing uses unweighted infant records and outcome-specific age support. The pooled survey includes childless respondents with zero biological children, original 1999/2018 respondent weights, region and wave fixed effects and the shared DD cohort trends. Women of all nationalities are included in both survey waves. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). The survey source is the INE 1999/2018 Fertility Surveys.

Table F6. Multiple-testing corrections for mechanism outcomes

Outcome	Raw p-value	BH within table
Married	0.8610	0.8610
Homeownership	0.1896	0.4739
Difficulty making ends meet	0.5037	0.6297
Temporary contract among employees	0.0331	0.1656
Monthly net income $\geq$1000 EUR	0.2901	0.4835
Age at economic independence	0.5380	0.6724
No childcare used (children under 14)	0.3811	0.6352
Unmet desired fertility	0.0116	0.0579
Egalitarian attitudes (0-1)	0.3444	0.6352
No religious affiliation	0.7858	0.7858

Notes: Benjamini–Hochberg (BH) adjustments are calculated separately for the five economic outcomes in Table 6 and the five preference/support outcomes in Table 9. All outcomes in each table are retained. Raw two-sided t-test p-values use the same respondent-weighted DD and maternal cohort-quarter clustered standard errors as those tables, with their outcome-specific regression samples and degrees of freedom. BH adjusts false-discovery rates under independence or suitable positive dependence. *Source:* 2018 Fertility Survey, INE.

F.4. *Additional specification robustness*

The following tables report the treatment-definition, cohort-position, functional-form, and bandwidth checks summarized in the main text.

The survey checks in Table F7 examine employment, schooling, disability, and the stated need for flexible childcare to consider having a child, providing evidence on other possible changes accompanying reform exposure.

Adding students to the birth-record comparison group in Table F8 checks whether the results depend on excluding mothers who are still in education.

Tables F9–F12 examine how the fertility results vary across post-reform cohorts and alternative choices of cohort trends and sample windows. Table F9 shows larger reductions in births per woman at the cohort level and greater delays in maternal and first-delivery age among later post-cutoff cohorts; these gradients describe cohort position rather than measured individual exposure to temporary employment. Adding quadratic and cubic cohort trends and their post-cutoff interactions leaves the baseline estimates largely unchanged (Table F10). Restricting the window to sixteen quarters per skill group yields smaller estimated effects (Table F11), whereas adding four quarters at each end of the main windows yields larger effects (Table F12). Across these specifications, the estimates retain the pattern of fewer births and later childbearing, although their magnitudes depend on the cohort window.

Table F7. Alternative explanations: employment, schooling, disability, and childcare (DD)

	(a) Paid work	*(b)* Long-term unemployment	*(c)* Age at first job	*(d)* Age completing studies	*(e)* Disability	*(f)* Flexible childcare needed for fertility
Exposed cohort × low-skilled	0.067	−0.015	−0.276	−1.921	0.033	−0.023
	(0.112)	(0.099)	(1.452)	(1.873)	(0.052)	(0.097)
Mean dep. var.	0.648	0.593	23.765	22.819	0.066	0.483
Treated pre mean	0.462	0.654	24.042	20.433	0.085	0.527
Effect (% of pre mean)	14.5	-2.4	-1.1	-9.4	38.3	-4.3
R^2	0.085	0.020	0.028	0.060	0.012	0.026
Observations	3,611	3,611	3,318	2,630	3,611	3,566

Notes: One observation per woman, with valid responses for each outcome. Women of all nationalities are included, whether or not they have children. General outcomes impose no restrictions on children's nationality or birth years. Low-skilled women (education below high-school completion) are born in 1965Q1–1972Q4, with cutoff 1968Q4; high-skilled women (high-school completion or above) are born in 1962Q1–1969Q4, with cutoff 1966Q1. All regressions include Post, Treat, a common linear cohort-quarter trend and its post-cutoff interaction, region and maternal birth-year fixed effects, original 2018 respondent weights, and standard errors clustered by maternal calendar birth-cohort quarter. Means and percentage effects use each regression's sample and respondent weights. Long-term unemployment means at least six months. School-completion age retains source winsorization. The flexible-childcare indicator records whether access to flexible childcare would be needed to consider having a child in the next three years, among respondents who do not intend to have a child in that period (question 7.5.14); it differs from actual childcare use in Table 9. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. *Source:* 2018 Fertility Survey, INE.

Table F8. Student-inclusive administrative comparison group (DD)

	(1) Births per woman (cohort-level)	(2) Maternal age at birth	(3) First-delivery age
Exposed cohort × low-skilled	−0.244***	0.487***	0.418***
	(0.018)	(0.062)	(0.055)
Mean dependent variable	1.486	29.323	27.607
Comparison group mean, pre	1.372	30.777	29.239
Treatment group mean, pre	1.693	27.428	25.368
Effect (% treated pre mean)	-14.4	1.8	1.6
R^2	0.440	0.091	0.127
Observations	3,200	3,438,228	1,929,373

Notes: Birth records from Spain, 1980–2017, all nationalities. Low-skilled mothers born 1965Q1–1972Q4 are compared with high-skilled mothers and students born 1962Q1–1969Q4. Students (occupation 9) enter only this appendix comparison; retired and unknown occupations remain excluded. Exposure uses the mother's cohort and the same skill-specific cutoffs as the main design. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. Quantity is measured in cohort-quarter by province by skill cells, with analytical weights equal to eligible infant counts; timing regressions use unweighted infant records. Maternal age requires an observed delivery age of at least 15; first-delivery age additionally requires age at most 55. All infants from multiple first deliveries are retained. Quantity uses the unchanged main-sample denominator; no extra population denominator is attributed to students. This check assesses sensitivity to comparison-group composition, rather than treating student status as an observed skilled occupation. Percentage effects divide each coefficient by its own regression sample's treated pre-reform mean. Significance: $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). Historical cohort totals: INE.

Table F9. Post-cutoff cohort-position gradients

Inverse position bin	Births per woman (cohort-level)	Maternal age	First-delivery age
Index 1–4 × low-skilled	-0.345***	0.919***	0.820***
	(0.026)	(0.049)	(0.047)
Index 5–8 × low-skilled	-0.290***	0.830***	0.726***
	(0.020)	(0.053)	(0.045)
Index 9–12 × low-skilled	-0.209***	0.571***	0.494***
	(0.021)	(0.057)	(0.055)
Index 13–16 × low-skilled	-0.163***	0.162*	0.080
	(0.017)	(0.087)	(0.079)

Notes: This is a post-cutoff cohort-position gradient, not measured contract duration or observed individual exposure intensity. For post cohorts with event quarter t, the retained index is min(16,17-t) for low-skilled mothers and 16-t for high-skilled mothers; pre cohorts have index 0. Thus the first two low-skilled post quarters both receive 16. The categories compare interactions of low skill with index bins 1–4,5–8,9–12,13–16 against the pooled pre-cutoff category. The regression includes skill, index-bin indicators and interactions, a common linear cohort-quarter trend and its post interaction, province fixed effects, and cohort-quarter clustered SEs. Larger index means earlier post-cutoff cohorts. Registered births in Spain, 1980–2017, all nationalities. Low-skilled cohorts: 1965Q1–1972Q4; high-skilled cohorts: 1962Q1–1969Q4. Respective exposure cutoffs: 1968Q4 and 1966Q1. The cohort measure uses a historical female-cohort denominator allocated with weighted 1999 survey skill shares, including childless women; it is not completed fertility. Descriptive means use denominator weights; regressions use birth-count weights. Percentages use the regression-weighted treated pre-reform mean. Standard errors condition on estimated shares. Ceuta/Melilla lack denominator support. Sources: INE birth register, historical births and sex shares, and 1999 Fertility Survey. Timing columns use unweighted valid infant ages and the maintained first-delivery classification. First deliveries retain every infant from multiple deliveries. The appendix discussion following Table F3 reports sensitivity to conflicting parity information. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). Stars denote p below .10, .05, .01.

Table F10. Cubic cohort-trend robustness

	Births per woman (cohort-level)	Maternal age	First-delivery age
Post × low-skilled	-0.255***	0.529***	0.442***
	(0.018)	(0.069)	(0.062)
Mean outcome	1.470	29.443	27.763
Treated pre-reform mean	1.693	27.428	25.368
Effect (% of pre mean)	-15.1	1.9	1.7
Observations	3,200	3,382,191	1,880,215
R-squared	0.456	0.107	0.154

Notes: The polynomial specification adds quadratic and cubic powers of the cohort-quarter trend and their post interactions to the shared DD equation; all lower-order shared DD terms remain included. Otherwise, samples, denominators, weights and outcome definitions follow Tables 3–4: 1980–2017 births, all nationalities, with every infant from multiple first deliveries retained. DD includes skill and exposure indicators and their interaction, cohort and post-cutoff trends, and province fixed effects. Standard errors are clustered by maternal birth-cohort quarter. Percentages use each regression sample's treated pre-reform mean. *Sources:* INE birth register, historical cohort totals and 1999 Fertility Survey. Stars denote p below .10, .05, .01.

Table F11. Shorter cohort bandwidth: sixteen quarters per skill group

	Births per woman (cohort-level)	Maternal age	First-delivery age
Post × low-skilled	-0.136***	0.262**	0.193*
	(0.022)	(0.111)	(0.103)
Mean outcome	1.471	29.395	27.690
Treated pre-reform mean	1.636	27.657	25.591
Effect (% of pre mean)	-8.3	0.9	0.8
Observations	1,600	1,709,826	952,238
R-squared	0.442	0.104	0.151

Notes: This explicitly labeled bandwidth check restricts event quarter to -8 through 7: low-skilled birth cohorts 1966Q4–1970Q3 and high-skilled cohorts 1964Q1–1967Q4. The canonical main cohort definition is unchanged. Otherwise, samples, denominators, weights and outcome definitions follow Tables 3–4: 1980–2017 births, all nationalities, with every infant from multiple first deliveries retained. DD includes skill and exposure indicators and their interaction, cohort and post-cutoff trends, and province fixed effects. Standard errors are clustered by maternal birth-cohort quarter. Percentages use each regression sample's treated pre-reform mean. *Sources:* INE birth register, historical cohort totals and 1999 Fertility Survey. Stars denote p below .10, .05, .01.

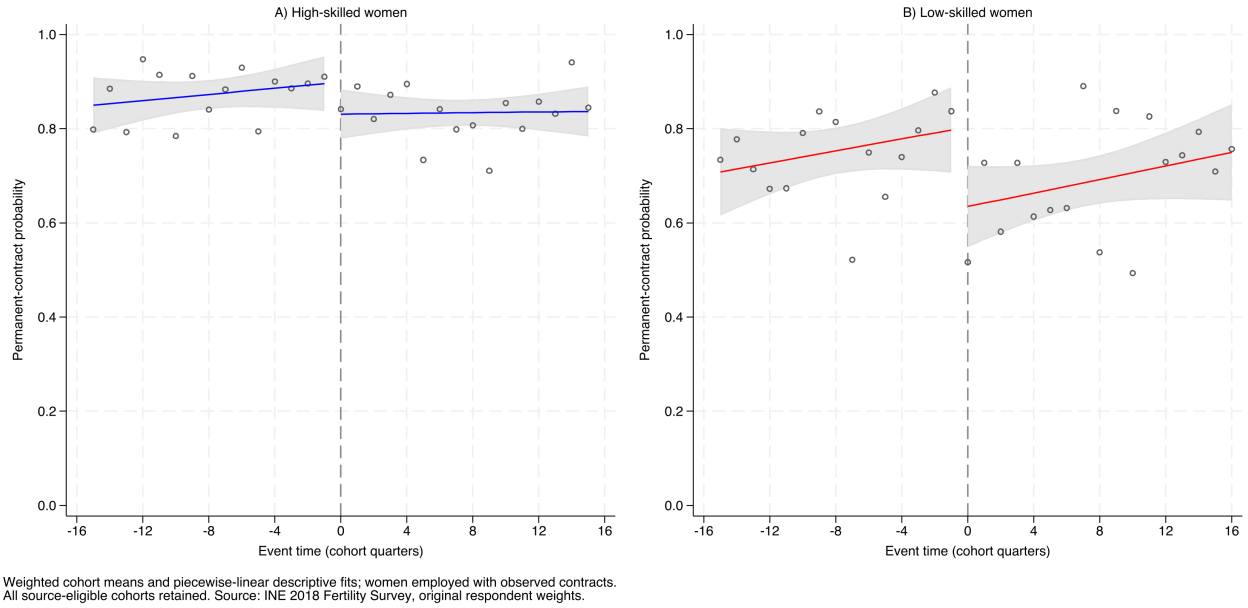
Table F12. Expanded cohort bandwidth, 1961–1973 (DD)

	(1) Births per woman (cohort-level)	(2) Maternal age at birth	(3) First-delivery age
Exposed cohort × low-skilled	−0.327***	0.602***	0.506***
	(0.023)	(0.063)	(0.059)
Mean dependent variable	1.478	29.417	27.753
Comparison group mean, pre	1.311	31.066	29.639
Treatment group mean, pre	1.729	27.299	25.245
Effect (% treated pre mean)	-18.9	2.2	2.0
R^2	0.474	0.114	0.162
Observations	4,000	4,237,197	2,351,300

Notes: Birth records from Spain, 1980–2017, all nationalities. This appendix check expands the low-skilled maternal cohorts to 1964Q1–1973Q4 and high-skilled cohorts to 1961Q1–1970Q4, adding four quarters at each end of each main window. The respective cutoffs remain 1968Q4 and 1966Q1; occupations, first deliveries and outcome validity are unchanged. DD includes skill and exposure indicators, their interaction, common linear cohort and post-cutoff trends, province fixed effects, and maternal cohort-quarter clustered standard errors. Quantity is measured in cohort-quarter by province by skill cells, with analytical weights equal to eligible infant counts; timing regressions use unweighted infant records. Maternal age requires an observed delivery age of at least 15; first-delivery age additionally requires age at most 55. All infants from multiple first deliveries are retained. Every existing cohort denominator is retained exactly. Added cohorts use the same historical-birth normalization; 1961 and 1973 province-quarter birth totals are linearly extrapolated from 1962–1972 totals separately by province and calendar quarter. Extrapolation coefficients and denominators are exported. This sensitivity check treats those projected denominators as fixed. Percentage effects divide each coefficient by its own regression sample's treated pre-reform mean. Significance: $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$. Source: Childbirth Statistics Bulletin, Spanish National Statistics Institute (INE). Historical cohort totals: INE.

Additional robustness figures.—Figure F4 describes permanent-contract holding among employed women with observed contract type in the 2018 Fertility Survey. It uses all source-eligible cohorts, the same education-based skill and exposure definitions, and the original respondent weights. The separate pre- and post-cutoff fits summarize cohort profiles; they are descriptive and do not replace the maintained survey DD equation or the administrative labor-market first stage.

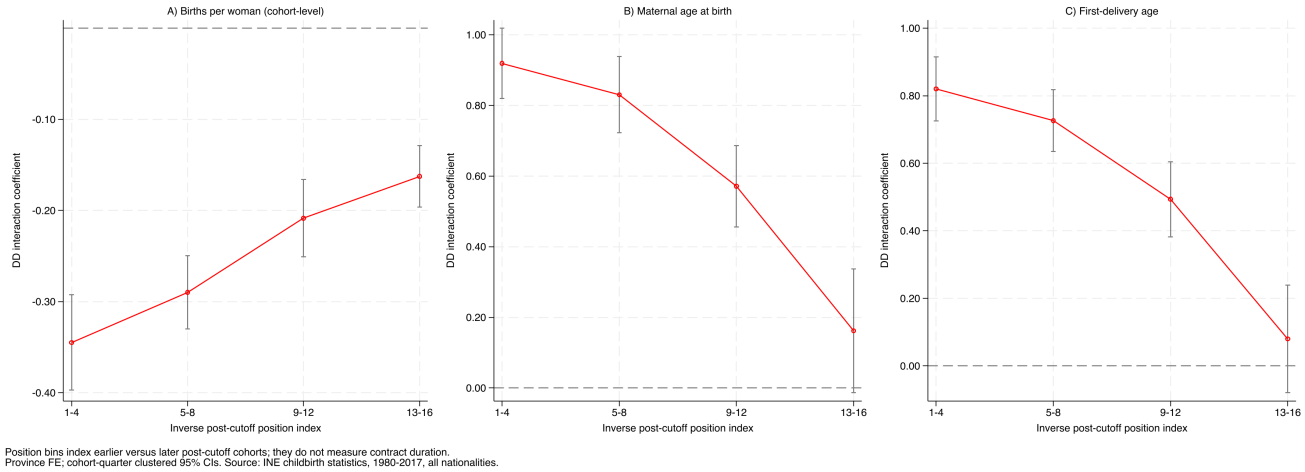
Figure F4. Permanent-contract holding by cohort and skill



Notes: A) high-skilled women; B) low-skilled women. The outcome equals one for a permanent contract among currently employed women with observed contract type. Dots are cohort-quarter means using the original 2018 respondent weights; fitted lines are separate linear profiles before and after each skill-specific cutoff, weighted by the cohort total of respondent weights. All source-eligible cohorts are retained, with no trimming of cohort means. The sample uses education-based skill and includes women of all nationalities, including childless women, without restrictions on children's nationality or birth years. This descriptive figure is separate from the region- and maternal-birth-year-fixed-effect mechanism regressions. *Source:* 2018 Spanish Fertility Survey, INE.

Figure F5 plots the cohort-position interactions from Table F9 for the birth-record quantity and timing outcomes. The inverse index is largest for earlier post-cutoff cohorts and must not be read as observed years of fixed-term employment.

Figure F5. Post-cutoff cohort-position gradients



Notes: A) births per woman (cohort-level); B) maternal age at delivery; C) first-delivery age. Points are maternal-low-skill interactions with inverse cohort-position bins; bars show 95 percent confidence intervals clustered by maternal birth-cohort quarter. For post-cutoff event quarter t, the index is $\min(16, 17-t)$ for low-skilled mothers and $16-t$ for high-skilled mothers; pre-cutoff cohorts have index zero. Larger values indicate earlier post-cutoff cohorts, not measured contract duration. The comparison is the pooled pre-cutoff category, conditional on skill and index-bin indicators, a common linear cohort trend and its post interaction, and province fixed effects. Quantity uses cohort–province–skill cells, the estimated female cohort denominator and eligible-cell-size analytical weights; timing uses unweighted valid birth records, retaining every infant from first multiple deliveries in the first-delivery outcome. The 1980–2017 all-nationality sample and skill-specific cohort windows are unchanged. *Source:* Childbirth Statistics Bulletin, INE; estimated female cohort population denominator for quantity.

G. MODEL ENVIRONMENT AND SOLUTION

G.1. *State space and timing*

The state in Equation (8) is

$$x_a = (s, e_a, n_a, g_a, h_a),$$

where a denotes age, $s \in \{L, H\}$ is skill, $e_a \in \{O, T, P\}$ is current labor-market status, $n_a \in \{0, 1, 2, 3\}$ is parity, $g_a \in \{0, 1, 2, 3\}$ is years since the last birth, and $h_a \in \{0, \ldots, 25\}$ records accumulated temporary-employment exposure. Skill s is fixed over the life cycle. Labor status e_a evolves according to the age- and skill-specific transition probabilities estimated from the MCVL. Parity n_a increases following a successful birth and is capped at three, and no further birth is feasible at $n_a = 3$. The terminal model parity is compared with the empirical three-or-more category. The spacing state g_a resets following a birth and otherwise increases up to its cap of three. Accumulated temporary employment exposure h_a counts every year of temporary work over the reproductive horizon and does not fall when a woman subsequently moves to another labor state. This last feature allows earlier employment insecurity to affect later fertility decisions separately from her current contract.

The successful-birth payoff includes the prospective cost $c_T \mathcal{I}^r_{s,a}(e,h)$, where r denotes the employment regime and

$$\mathcal{I}^r_{s,a}(e,h) = \frac{\mathbf{1}\{e \neq P\}\log(1+h) + \beta \sum_{e'} P^r_{s,a}(e,e')\mathbf{1}\{e' \neq P\}\log(1+h+\mathbf{1}\{e'=T\})}{1+\beta},$$
$$h_{16} = \mathbf{1}\{e_{16} = T\}, \qquad h_{a+1} = h_a + \mathbf{1}\{e_{a+1} = T\}, \quad a = 16, \ldots, 39. \tag{G1}$$

The expectation covers the birth year and the following infant-care year. The 2-year parenting window follows the infant-stage duration in Guner, Kaya and Sánchez-Marcos (2024); it is a calibration, not an additional estimated parameter. The state grid retains all 25 possible temporary years from ages 16 through 40, with no two-year cap or erasure of past exposure. Logarithmic curvature limits the marginal cost of additional history. Permanent employment removes the current-year component, while the risk of subsequently leaving permanence can still make past insecurity relevant. The same cost applies at every parity below three, and c_T is its only estimated coefficient.

For a birth at age 40, the second parenting year is valued using the age-40 labor-transition law and the corresponding update of history. This is a one-period cost commitment attached to the birth payoff, not an extension of fertility choices or the value-function horizon. It permits a coherent benefit from waiting for stability without adding a recovery parameter or a child-age state. The literature motivates the timing of the commitment; it does not independently identify the log-history cost or its causal interpretation.

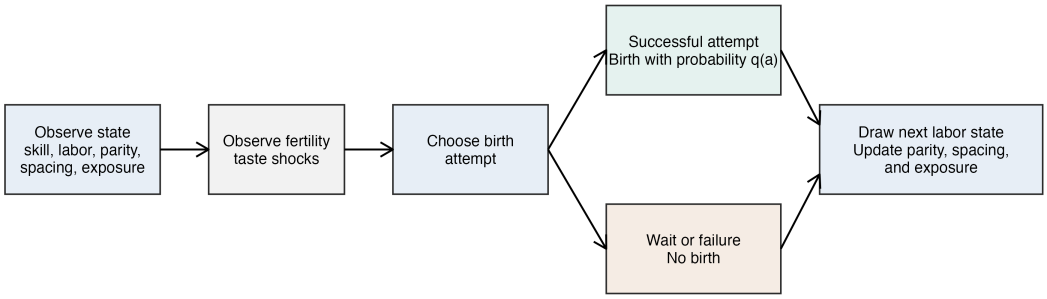
At each age, the woman observes her current state x_a and the choice-specific fertility shocks ε_{ad} for $d \in \{0,1\}$, where $d = 1$ denotes attempting a birth and $d = 0$ denotes waiting. The feasible action set is $\mathcal{D}(n) = \{0,1\}$ below parity three and $\mathcal{D}(3) = \{0\}$; she chooses between the available actions. Conditional on an attempt, birth success is realized according to $C_a \sim \text{Bernoulli}\{q(a)\}$, where $q(a)$ is the age-specific probability that an attempt results in a live birth. The realized birth indicator is therefore $B_a = d_a C_a$. Employment then evolves according to $P_{s,a}(e_a, \cdot)$, the distribution of next period's labor state conditional on age, skill, and current employment. The resulting birth and employment realizations determine next period's parity, birth spacing, accumulated temporary-employment exposure, and labor status.

When deciding whether to attempt a birth, a woman conditions on her current employment state and the probabilities of future labor-market transitions, but not on the next period's realized employment state. Future employment conditions affect fertility through expected early-parenting insecurity and the continuation value, rather than through knowledge of the next contract realization. Desired family size (D_i), religiosity (R_i), and traditional gender norms (G_i) enter the birth payoff as fixed observed characteristics and do not evolve as state variables over the life cycle.[48]

[48] The model follows annual fertility decisions from ages 16 through 40 and includes births realized at age 40 in final parity. The Fertility Survey sample contains women aged 45–55, but dated biological-child histories restrict the structural outcomes

Figure G1 shows the resulting within-period sequence, in which fertility taste shocks are observed before the attempt decision, while birth success and the next labor-market state are realized afterward. A successful birth increases parity and resets birth spacing, whereas the employment realization updates both current labor status and accumulated temporary-employment exposure before the woman makes the following year's fertility decision.

Figure G1. Within-period timing and state transitions



Notes: The sequence repeats annually from ages 16 through 40. Employment transitions depend on age, skill, and current labor status. Birth success depends on age through $q(a)$. Following a realized birth, parity increases and the birth-spacing state resets; otherwise spacing advances toward its cap. Temporary-employment exposure is accumulated in h_a and can therefore affect subsequent fertility choices after current employment status changes.

G.2. *Payoffs and Bellman recursion*

Current utility depends on the resources available after supporting existing children. If the woman waits, available resources at age a are

$$c_a^0(x) = \max\{\underline{c}, y_{s,a,e} - \kappa n\}, \tag{G2}$$

where $y_{s,a,e}$ is contribution-based income for a woman of skill s, age a, and employment state e, κ is the annual resource cost per child, and $\underline{c}$ is a lower bound that keeps consumption positive. The value of waiting is therefore

$$v_a^0(x) = \log c_a^0(x) + \beta E[V_{a+1}(x') \mid x, d = 0], \tag{G3}$$

where $v_a^0(x)$ is the choice-specific value of waiting, β is the discount factor, and $V_{a+1}(x')$ is the continuation value associated with next period's state x'.

If the woman attempts a birth and the attempt succeeds, available resources become

$$c_a^1(x) = \max\{\underline{c}, y_{s,a,e} - \kappa(n+1)\}. \tag{G4}$$

to births realized through age 40. The model therefore measures fertility by 40 rather than eventual lifetime fertility. Mothers whose first birth preceded age 16 account for 0.55 percent of low-skilled and 0.17 percent of high-skilled mothers in these horizon-matched profiles; the parity-zero initial condition approximates this small tail without deleting those women from the data. The construction and weighting of D_i, R_i, and G_i are described in Appendix H.

Let $b_a(x) \equiv \Delta u_a(x; D_i, R_i, G_i)$ denote the payoff from a successful birth defined in Equation (10). Because an attempt succeeds only with probability $q(a)$, the value of attempting a birth averages over the successful- and unsuccessful-birth branches,

$$v_a^1(x) = \log c_a^0(x) + q(a)b_a(x) + \beta \left\{ q(a)E[V_{a+1}(x'_B)] + (1 - q(a))E[V_{a+1}(x'_0)] \right\}, \tag{G5}$$

where $v_a^1(x)$ is the value of attempting a birth, x'_B is the next-period state following a successful birth, and x'_0 is the corresponding state when the attempt does not produce a birth. The difference $v_a^1(x) - v_a^0(x)$ therefore compares the current and future value of attempting a birth with the value of waiting, taking into account both birth-success risk and future employment opportunities.

Under the Type-I extreme-value fertility shocks introduced in the main text, the probability of attempting a birth below parity three is

$$\Pr(d_a = 1 \mid x_a) = \Lambda\big(v_a^1(x_a) - v_a^0(x_a)\big),$$

where $\Lambda(\cdot)$ is the logistic cumulative distribution function. At parity three this probability is zero. For $n < 3$, the expected value before the fertility shocks are realized is

$$V_a(x) = \log\big[\exp\{v_a^0(x)\} + \exp\{v_a^1(x)\}\big].$$

At parity three, $V_a(x) = v_a^0(x)$ under the maintained mean-zero shock normalization. Solving these equations backward from the end of the reproductive horizon gives the fertility policy at each age and state, while iterating the state distribution forward produces the corresponding population fertility outcomes.[49]

The age component of the successful-birth payoff is

$$c_A(a) = c_E w_E(a) + c_L w_L(a) + w_V(a), \qquad \begin{aligned} w_E(a) &= \min\{1, \max\{0, (29 - a)/7\}\}, \\ w_L(a) &= \min\{1, \max\{0, (a - 29)/6\}\}, \\ w_V(a) &= \max\{0, (a - 35)/5\}. \end{aligned} \tag{G6}$$

The coefficients c_E and c_L are estimated jointly with the other preferences; the age breakpoints and the coefficient of w_V are fixed normalizations. The final component is an age-related utility cost, distinct from biological success $q(a)$. Appendix I reports the re-estimated check that removes this post-35 normalization.

The opportunity cost of childbearing for high-skilled women is tied to the age profile of contribution-based income in permanent employment. Let $\ell_H(a) = \log y_{H,a,P}$ denote the log permanent-employment income

[49] We impose $V_{41}(x) = 0$ for all states, so age 40 is the final period in which a birth can occur. This also truncates modeled child-related resource costs after the reproductive horizon and is therefore a finite-horizon approximation. Population moments are obtained by forward iteration of the state distribution rather than by simulation. With the maintained annual cost and discount factor, the monetary present value of the modeled child-cost stream is €9594 for a birth at 16 and €600 for a birth at 40, before the consumption floor. This quantifies the truncation incentive; it is not an equivalent-utility calculation or evidence that post-40 obligations disappear.

profile for high-skilled women. We summarize the discounted income growth that remains after age a as

$$K(a) = \sum_{t=a+1}^{40} \beta^{t-a} \left[\ell_H(t) - \ell_H(t-1)\right], \qquad \omega_H(a) = \frac{K(a)}{K(16)}. \tag{G7}$$

The function $\omega_H(a)$ therefore measures how much career progression remains at each age relative to age 16. It enters the successful-birth payoff through $c_H 1\{s = H\}\omega_H(a)$, where c_H is estimated and determines the strength of the high-skill career opportunity cost. The shape of $\omega_H(a)$ comes from the pre-cutoff-cohort MCVL income profile, while its importance for fertility is determined by the estimated coefficient c_H.[50]

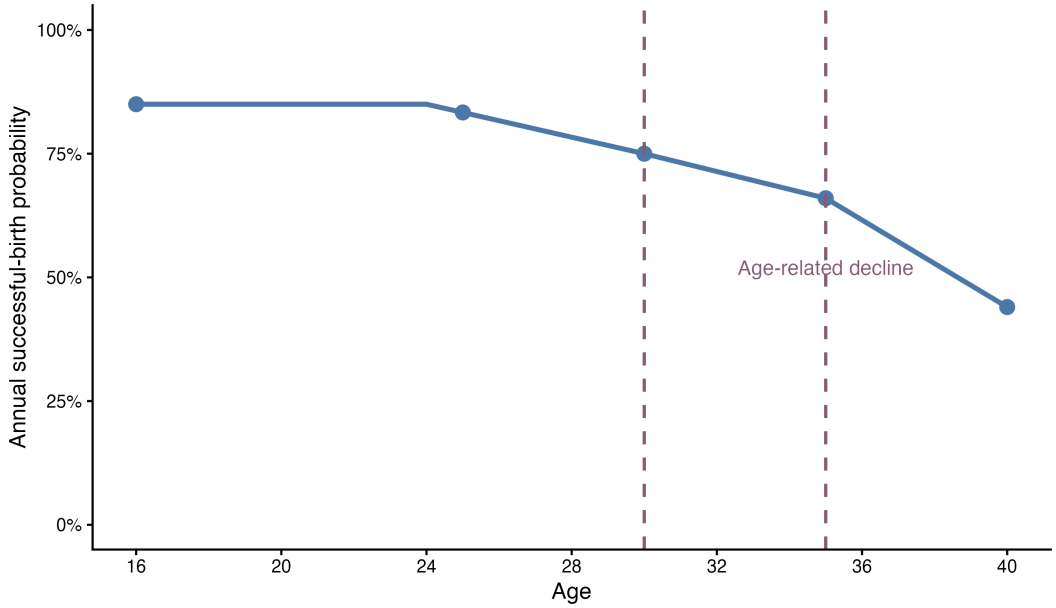
The remaining resource parameters are calibrated rather than estimated with the fertility moments. The discount factor is $\beta = 0.96$, and contribution-based income is expressed in units of €10,000. The child-resource parameter $\kappa = 0.06$ therefore corresponds to 600 euros per child and year, while the lower bound $\underline{c} = 0.05$ corresponds to €500 in annual resources.[51]

Birth-success risk varies with age through $q(a)$. As shown in Figure G2, the model uses a success probability of 0.85 through age 24 and then lets it decline with age, reaching 0.75 at age 30, 0.66 at age 35, and 0.44 at age 40. The age-30, age-35, and age-40 values are anchored to Leridon (2004), with linear interpolation between the calibration points. This schedule is common across skill and employment states and is fixed before the fertility parameters are estimated. It uses population one-year probabilities as annual conditional success rates, with a common schedule across women of the same age. Under continuous attempts starting at 30, the implemented schedule implies 99.4 percent success over four years, compared with 91 percent in Leridon (2004); starting at 35 gives 97.4 versus 84 percent. These cumulative probabilities describe continuous attempts; actual motherhood also depends on the endogenous decision to attempt a birth. The source follows conceptions eventually ending in a live birth, while the annual model records success in the decision period. The difference in cumulative success reflects the approximation involved in applying a common annual schedule without persistent fecundity heterogeneity. The biological accounting is conditional on this schedule.

[50] The permanent-employment income profile is interpolated from the observed MCVL age-bin medians. Its use determines the age pattern of the opportunity cost but does not interpret the MCVL profile as a causal wage penalty from childbearing. The mechanism is motivated by Adda, Dustmann and Stevens (2017).

[51] For women in the outside-employment state, resources are normalized to the larger of €2,000 and 40 percent of median permanent-employment contribution-based income. The child-resource deduction should be interpreted as a parsimonious subtraction from the resources available to the woman rather than as a measure of total household expenditure on children. Appendix Table I3 reports sensitivity to this calibration.

Figure G2. Age-dependent successful-birth probability



Notes: Annual probability $q(a)$ that a birth attempt results in a live birth. The model uses a probability of 0.85 through age 24 and linear interpolation between the age-specific calibration points thereafter. The values at ages 30, 35, and 40 are anchored to Leridon (2004). The schedule is common across skill and employment states and is not estimated from the fertility moments. Appendix Table I3 varies the post-age-35 segment by plus or minus 20 percent.

G.3. *Desired fertility, religiosity, and gender norms*

The 2018 Spanish Fertility Survey measures realized fertility together with the three preference characteristics that enter Equation (10). Desired family size, D_i, is observed for both mothers and childless women and captures differences in the number of children women would like to have. We cap both desired and realized fertility at three to match the model's parity state. Religiosity, R_i, combines religious affiliation with the intensity of religious practice, while traditional gender norms, G_i, summarize views about women's roles in work, family, and household production. Higher values of G_i correspond to more traditional views. These measures allow women with otherwise similar labor-market histories to differ in the value they attach to having a child.

The preference parameters α_D, α_R, and α_G are disciplined by the corresponding relationships between these characteristics and fertility in the survey. Desired family size and religiosity are related to completed fertility, while traditional gender norms are related to age at first birth among mothers. The empirical relationships are estimated jointly, so each coefficient captures the association of its preference characteristic with fertility while holding the other preference measures and observable demographic characteristics fixed.[52]

[52] The survey regressions additionally control for skill, age, and region and use the official survey weights. Desired family size is centered at the population mean, while religiosity and traditional gender norms are standardized and population-centered before entering utility. Missing religiosity is replaced by the corresponding skill-group weighted mean when constructing level profiles, while the gradient regressions use respondents with complete information. These survey gradients are used as conditional associations rather than interpreted as causal effects of changing preferences or beliefs.

Because the model is designed to study the fertility consequences of the labor-market reform, D_i, R_i, and G_i are held fixed when reform exposure changes. Fertility can therefore respond to the reform through employment conditions, contribution-based income, accumulated temporary-employment exposure, birth success, and the remaining reproductive horizon, but not through an imposed change in preferences. This is a maintained restriction. Desired fertility and attitudes are reported retrospectively and may reflect subsequent experience. Appendix I examines how the estimates vary with the age sample used to measure these preference gradients. Table G1 summarizes how each observed preference characteristic enters utility and which empirical relationship disciplines its coefficient.

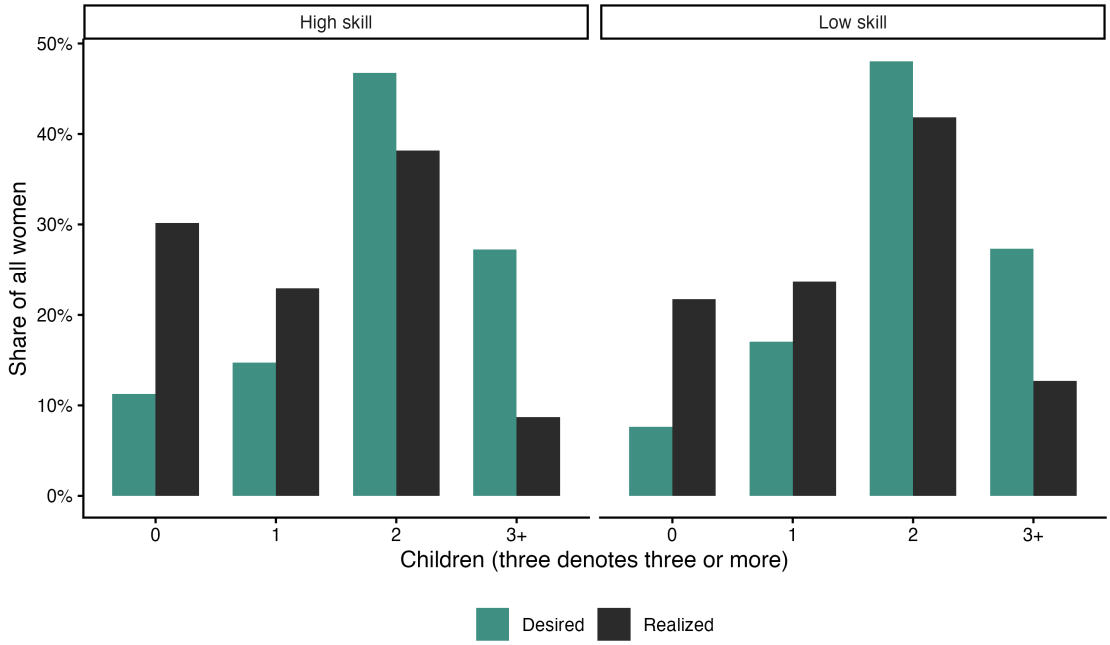
Table G1. Preference measures and structural roles

Measure	Construction	Structural role	Identifying moment
Desired family size	Desired children (0–3), population-centered	Birth value	Completed-children gradient
Religiosity	Affiliation/practice index, standardized	Birth value	Completed-children gradient
Traditional gender norms	Seven-item traditional-norm index, standardized	Birth value and timing	First-birth-age gradient

Notes: The table links each preference characteristic to its role in Equation (10) and to the survey relationship used in estimation. Desired and realized fertility are capped at three, while religiosity and traditional gender norms are standardized. *Source:* 2018 Spanish Fertility Survey and authors' calculations.

Desired family size remains conceptually distinct from realized fertility in the model. Women who report the same desired number of children can end the reproductive life cycle with different numbers of births because their employment histories, contribution-based income, birth success, and timing differ. Figure G3 compares the marginal distributions of desired and realized fertility within each skill group. These marginal distributions describe the difference between reported family-size preferences and realized parity; they do not display realized fertility conditional on a particular desired family size.

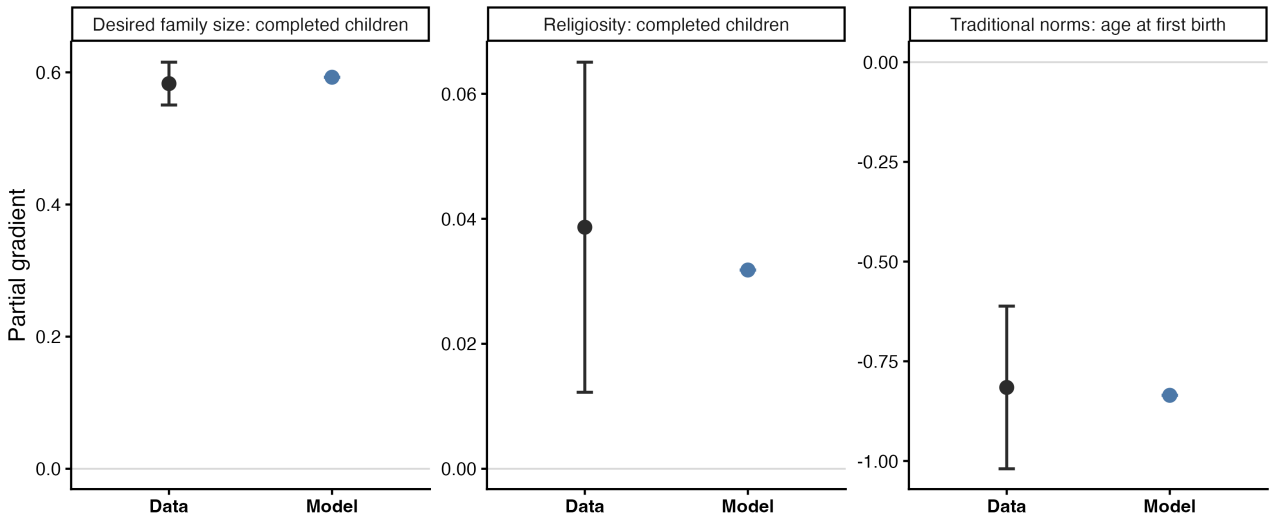
Figure G3. Desired and realized completed fertility



Notes: Survey-weighted marginal distributions of desired children and biological children realized through age 40, by skill for women surveyed at ages 45–55, including childless women. Both desired and realized fertility are capped at three, with three denoting three or more children. *Source:* 2018 Spanish Fertility Survey and authors' calculations.

The model must also reproduce the empirical relationships used to discipline α_D, α_R, and α_G. Figure G4 compares the survey gradients with their model counterparts. Women who desire more children have higher completed fertility, greater religiosity is associated with somewhat higher completed fertility, and more traditional gender norms are associated with earlier first motherhood. The model reproduces the direction and broad magnitude of these relationships while keeping all three preference characteristics fixed across the reform counterfactuals. Appendix Table H4 reports the corresponding numerical moments and model-data differences.

Figure G4. Preference gradients: data and model



Notes: Points compare the partial relationships observed in the 2018 Spanish Fertility Survey with their model counterparts. Desired family size is measured as a one-child increase, while religiosity and traditional gender norms are measured as one-standard-deviation increases. Whiskers show 95 percent confidence intervals for the survey estimates; panel scales differ. *Source:* 2018 Spanish Fertility Survey and authors' model calculations.

G.4. *Resources and employment progression*

Employment conditions affect fertility through both the resources available today and the employment opportunities women expect in the future. We construct these two components from the pre-cutoff-cohort MCVL histories before estimating fertility preferences.

For current resources, let $\widetilde{y}_{sbe}$ denote median annual contribution-based income for skill group s, age group b, and employment state e, measured in units of €10,000. For temporary (T) and permanent (P) employment, these medians are calculated directly from positive annual contribution-based income in the MCVL. Because the MCVL does not provide a comparable income measure when a woman is outside covered dependent employment, we normalize income in the outside state (O) to

$$y_{sbe} = \begin{cases} \max\{0.20,\, 0.4\widetilde{y}_{sbP}\}, & e = O, \\ \widetilde{y}_{sbT}, & e = T, \\ \widetilde{y}_{sbP}, & e = P. \end{cases} \tag{G8}$$

Thus, y_{sbe} is the contribution-based income assigned to a woman with skill s, in age group b, and labor state e. The outside-employment value is the larger of €2,000 per year and 40 percent of the corresponding permanent-employment median. We observe the employment-state medians in four age groups and interpo-

late log income between their midpoints, holding the endpoints constant, so that resources evolve smoothly with age rather than changing discretely at the boundaries of the age groups.[53]

The resources entering current utility depend on this income profile and on the number of children already born,

$$c_a(x) = \max\{\underline{c},\ y^R_{s,a,e} - \kappa n\}, \tag{G9}$$

where $y^R_{s,a,e}$ is annual contribution-based income interpolated across age-group midpoints under the relevant policy regime, n is current parity, κ is the annual resource deduction per child, and $\underline{c}$ is the minimum resource level defined in Appendix G.2. Each existing child therefore reduces available resources in every subsequent period in which the woman remains in the model. Under reform exposure, the parameter Δ_y in Equation (11) changes contribution-based income through age 25 for women in temporary or permanent employment. Its value is chosen to reproduce the observed first-stage change in contribution-based income after accounting for the reform-induced change in the distribution of women across employment states.

Table G2 reports the contribution-based income profiles that enter this calculation. The temporary-to-permanent ratio in the final column shows how the resources associated with temporary employment compare with those associated with permanent employment at different ages and for each skill group. These differences are also the basis for the counterfactual that equalizes the income profiles of temporary and permanent workers.

[53] Contribution-based income is the annual contribution measure available in the MCVL and should not be interpreted as household disposable income. The construction uses complete observed contribution years. The reduced-form analysis of days worked is retained as separate evidence; the annual structural employment state does not introduce an additional days-worked margin.

Table G2. Median annual contribution-based income by employment state, age, and skill

Skill	Age	Temporary	Permanent	Temp./perm.
High	16–22	2,211	10,804	0.20
High	23–29	7,703	17,227	0.45
High	30–34	10,155	21,789	0.47
High	35–40	14,065	23,391	0.60
Low	16–22	2,810	8,547	0.33
Low	23–29	6,363	11,168	0.57
Low	30–34	6,807	11,724	0.58
Low	35–40	7,802	12,894	0.61

Notes: Median annual contribution-based income among MCVL person-years with positive observed income. The final column reports the ratio of the temporary-employment median to the permanent-employment median. Income profiles are first-step inputs to the structural model rather than fertility moments used to estimate preferences. *Source:* MCVL 2006 and authors' calculations.

Future employment opportunities are described by a first-order Markov process estimated separately by skill and age group,

$$\Pr(e_{a+1} = j \mid e_a = i, s, a) = P_{s,b(a)}(i, j), \qquad \sum_{j \in \{O,T,P\}} P_{s,b(a)}(i, j) = 1. \tag{G10}$$

The element $P_{s,b(a)}(i, j)$ is the probability that a woman currently in labor state i moves to state j in the following year, conditional on her skill group and age. The lookup $b(a)$ is annual at ages 16–22 and pooled over ages 23–29, 30–34 and 35–40. The age-bin table summarizes these inputs; it does not impose a constant young-age transition matrix. Each transition row is estimated from adjacent MCVL person-years. We add one-half observation to each possible destination before normalizing the row, so that sparsely populated age-by-skill cells do not generate mechanically zero transition probabilities.[54]

The reform changes these transition probabilities through two distinct margins. The parameter ρ in Equation (11) changes the probability of reaching temporary employment during the early career, through age 25. The parameter δ instead changes the persistence of temporary employment after a woman is already in state T. To implement the latter, define

$$u_a = P_{L,a}(T, O)$$

[54] This is the Jeffreys correction used throughout the labor first step. It has its largest effect in sparse cells and preserves the requirement that the three destination probabilities sum to one.

as the baseline probability that a low-skilled temporary worker moves outside covered employment in the following year, and

$$v_a = \frac{P_{L,a}(T,T)}{1-u_a}$$

as the probability of remaining temporary conditional on remaining in covered employment. The persistent reform component is then

$$\begin{aligned}
\widetilde{P}^{\delta}_{L,a}(T,O) &= u_a, \\
\widetilde{P}^{\delta}_{L,a}(T,T) &= (1-u_a)\Lambda\{\text{logit}(v_a)+\delta\}, \\
\widetilde{P}^{\delta}_{L,a}(T,P) &= 1-u_a-\widetilde{P}^{\delta}_{L,a}(T,T),
\end{aligned} \tag{G11}$$

where $\Lambda(z) = (1+\exp(-z))^{-1}$ is the logistic function. A positive δ raises the conditional probability of remaining in temporary employment rather than moving into permanent employment, while leaving the probability of moving outside covered employment unchanged. The other origin states are not changed by δ. After this persistent component is applied, the early-career parameter ρ adjusts destination probabilities through age 25 and the row is renormalized so that all transition probabilities continue to sum to one.[55]

The model maps the reform into the full labor histories observed in the MCVL, rather than treating each annual observation independently. Let

$$\mathcal{H}_i = (e_{i,16}, \ldots, e_{i,25})$$

denote woman i's observed sequence of labor states during the first ten potential career years. For policy regime r, the relative likelihood of observing this history is

$$L_i^r = \frac{p_{16}^r(e_{i,16})}{p_{16}^0(e_{i,16})} \prod_{a=16}^{24} \frac{P_{L,a}^r(e_{i,a}, e_{i,a+1})}{P_{L,a}^0(e_{i,a}, e_{i,a+1})}. \tag{G12}$$

The first ratio captures the reform-induced change in the distribution of labor states at entry, while the product captures the change in the probability of each subsequent transition. A history that becomes more likely under the reform receives a larger value of L_i^r, whereas a history that becomes less likely receives a smaller value.

Because the first-decade employment and contribution-based income regressions use different empirical samples, these likelihood ratios are normalized separately within each sample,

$$w_{i,k}^r = \frac{L_i^r}{\sum_{j \in \mathcal{S}_k} L_j^r}, \qquad i \in \mathcal{S}_k, \tag{G13}$$

[55] This decomposition is motivated by the distinction between access to permanent employment, job destruction, and resources in Guner, Kaya and Sánchez-Marcos (2024). We use our own annual MCVL transitions to estimate these objects rather than importing the quarterly transition probabilities or household structure from that study. A $T \to P$ transition in our annual data may involve a change of employer and should therefore be interpreted as access to permanent employment, not as a within-firm promotion.

where $\mathcal{S}_C$ denotes eligible low-skilled histories from pre-cutoff entry cohorts in the career sample, and $\mathcal{S}_Y$ denotes the corresponding pre-cutoff contribution-income sample. Histories of exposed cohorts do not enter either baseline set. Each woman therefore receives one policy weight based on her full labor history.

For the career outcome, let z_{ia} equal one when year a counts toward the empirical measure of non-permanent employment and zero otherwise. The model counterpart under regime r is

$$\sum_{i \in \mathcal{S}_C} w^r_{i,C} \sum_{a=16}^{25} z_{ia},$$

which gives expected years of non-permanent employment during the first decade. For contribution-based income, let $\mathcal{A}_i$ denote the complete contribution years observed for woman i. The corresponding model moment is

$$\sum_{i \in \mathcal{S}_Y} w^r_{i,Y} \left[\frac{1}{|\mathcal{A}_i|} \sum_{a \in \mathcal{A}_i} y^r_{L,a,e_{ia}} \right].$$

Income years are therefore averaged within each woman before applying her policy weight, preserving the dependence between employment history and the years in which income is observed. The benchmark maps the empirical proportional income change into a resource index built from state medians and calibrated outside resources. These resources summarize the budget entering fertility decisions, whereas observed contribution bases also reflect benefit receipt and the number of observed contribution years. Outside-state years account for 15.7 percent of income-observation weight, with mean observed contributions of €6861 and calibrated model resources of €4021. Appendix Table I4 therefore reports re-estimations using observed contribution marks for outside years and, separately, for all labor states, while retaining the same observation-year weights and labor-history transport rule. The comparison measures the sensitivity of the labor shock to the distinction between contribution income and resources in utility. Contribution income includes eligible benefit-related bases and uses complete observed contribution years; its contrast with days worked, averaged over the fixed decade, does not identify a cash-wage-per-day effect.[56]

The annual T/P classification also retains the final duration-based proxy when recorded contract status is missing. In particular, early-career state probabilities and transitions should be interpreted conditional on that measurement rule, which uses observed employment duration, including subsequent history. They are not estimates based entirely on directly recorded historical contract codes. The model-state entry regressions improve the alignment of empirical and simulated statistics without eliminating this measurement restriction.

[56] The reweighting changes the relative frequency of complete $O/T/P$ labor histories but keeps the empirical observation and sample eligibility rules conditional on those histories fixed. Years used to establish sample eligibility through benefit-related affiliation are not reclassified as employee years. In addition, some jobs in employee special regimes count toward the empirical measure of non-permanent employment even when their observed contract places them in structural state P. Such years contribute to z_{ia} but do not increase accumulated temporary-employment exposure h_a, which rises only when the structural state is T.

The labor first step also uses information after the first decade to discipline whether temporary employment remains persistent later in the career. We construct a common eighteen-year potential-career window, corresponding to ages 16–33 for low-skilled women and ages 18–35 for high-skilled women. Within potential-career years 11–18, the temporary-employment share is calculated among employed person-years, while the transition into permanent employment is calculated among years that begin in temporary employment. These later-career moments use the same education groups as the structural model and allow cohort trends to differ by skill and on either side of the reform cutoff.[57]

For each later-career outcome, we estimate

$$\begin{aligned}\overline{Y}_i = \alpha + \tau L_i \,\mathrm{Post}_i + \gamma L_i + \eta \,\mathrm{Post}_i + b r_i + c \,\mathrm{Post}_i \, r_i \\ + d L_i r_i + f L_i \,\mathrm{Post}_i \, r_i + \mu_{p(i)} + \nu_{q(i)} + u_i,\end{aligned} \tag{G14}$$

where $\overline{Y}_i$ is woman i's later-career temporary-employment share or temporary-to-permanent transition rate, L_i is an indicator for low skill, Post_i identifies cohorts on the exposed side of the skill-specific reform cutoff, and r_i measures birth-cohort quarter relative to that cutoff. The coefficient τ is the DD effect at the cutoff. The coefficients γ and η allow level differences by skill and reform side, while b, c, d, and f allow linear cohort trends to differ by skill and before and after the cutoff. The terms $\mu_{p(i)}$ and $\nu_{q(i)}$ are birth-province and quarter-of-year fixed effects, and u_i is the regression residual. Temporary-employment regressions are weighted by the number of employed years, temporary-to-permanent regressions by the number of temporary-origin years, and standard errors are clustered by birth-cohort quarter.

The temporary-share outcome therefore measures the fraction of employed years spent in temporary work, while the mobility outcome measures the fraction of temporary-origin transitions followed by permanent employment. The latter is conditional on being temporary and consequently reflects both the transition process and the composition of women who remain in temporary employment. We use the stock and transition moments jointly to discipline δ. The original first-decade specification is left unchanged for its own target; the unemployment rate at first affiliation used there is not added to the broader later-career sample because doing so would condition the sample on early observed entry.

The 2018 Fertility Survey contributes one additional later-career labor moment, the reform effect on temporary employment among women observed in temporary or permanent jobs. The empirical regression uses the survey weights and the same cohort design used elsewhere in the paper, while the model applies the same DD projection to its age-specific temporary-employment shares. All women in these cohorts are older than 40 when observed in 2018, so this employment moment uses labor distributions projected beyond age 40.

[57] The structural education crosswalk aligns the MCVL categories as closely as possible with the Fertility Survey distinction between less than completed upper secondary education and completed upper secondary education or more. One aggregated MCVL education category cannot be mapped perfectly, so the reduced-form proxy groups and the structural skill groups are not identical.

The final baseline transition matrix is held fixed, and the reform-specific persistent conversion adjustment δ continues to apply; the early destination and entry adjustments have expired. This extension is used only to construct the employment moment; fertility choices still end at age 40 and the Bellman problem retains the terminal condition $V_{41} = 0$.

The labor-market parameters are estimated before the fertility preferences. Initial T/P effects and covered entry by age 19 use the same annual states and education groups as the model, with a common cohort trend, a post-cutoff slope change, quarter controls and cohort-clustered uncertainty. The initial-stock effects are -0.038 and +2.024 percentage points. Covered entry by 19 changes by +13.244 points; its structural counterpart is one minus the probability of remaining in O from potential entry through age 19. We match this change through an O-to-employee log-odds adjustment $\eta = +0.3868$ at ages 16–18, rather than moving the full conditional affiliation DD into the age-16 stock. These additional structural first steps leave the published Stata estimates unchanged. Sampling uncertainty and covariance with the other labor inputs enter the joint inference.

Conditional on the initial-stock effects, the parameters η, ρ, δ, and Δ_y are chosen jointly to match the remaining labor-market evidence. The first-decade target is an increase of 0.415 years of non-permanent employment. The contribution-based income target is a decline of €535.82 from a corresponding pre-reform mean of €7573.38. After allowing for the change in employment-state composition, the fitted early-career destination adjustment is $\rho = 0.1940$ and the fitted contribution-income shift is $\Delta_y = -0.0106$. The entry-flow and temporary-destination adjustments jointly match covered entry by 19 and the frozen first-decade exposure target; the persistent component is disciplined by the later labor moments.

The timing of this first-decade response is not itself targeted. As a diagnostic, the model implies an average annual increase in non-permanent employment of +4.42 percentage points during the first five potential career years, compared with +3.21 percentage points in the data, and a change of +3.88 percentage points during years six through ten, compared with +5.09 percentage points in the data. Matching the ten-year total is therefore assessed separately from the distribution of exposure within the decade. These within-decade profiles remain untargeted checks rather than additional restrictions on the fertility model. Likewise, agreement with marginal labor-state shares does not independently validate the joint distribution of accumulated histories. Pre-cutoff refers to assigned potential entry, not to every observed job occurring before 1984; the reference labor process includes subsequent histories of those cohorts. Employment transitions remain conditional inputs to fertility decisions, so these checks do not identify fertility-induced exits or selection out of the 2006 MCVL frame.

The fitted persistent parameter is $\delta = 0.0575$ with a standard error of 0.0774, implying a modest increase in the tendency to remain temporary rather than move into permanent employment. In potential-career years 11–18, the MCVL estimate for the temporary-employment share is +0.37 percentage points (standard error 1.14), compared with +1.64 percentage points in the model. The estimated change in temporary-to-permanent mobility is −0.52 percentage points (standard error 0.88), compared with −0.66 percentage points in the model. The 2018 Fertility Survey estimate is larger, at +15.72 percentage points (standard error 7.13), whereas the model predicts +0.69 percentage points. The latter comparison therefore contains the largest remaining discrepancy among the later-career labor moments, while the MCVL stock and transition moments support a comparatively modest persistent component.

The labor estimates and fitted reform adjustments are collected in Table G3. Panel A reports the empirical first-stage moments, Panel B reports the fitted labor-process parameters, and Panel C compares the matched moments with their model counterparts and standardized residuals. These objects are held fixed when the fertility preference parameters are estimated, although their sampling uncertainty is propagated through the structural estimates and counterfactuals.[58]

[58] The covariance matrix for the MCVL labor moments is constructed jointly by birth cohort so that correlated first-stage estimates remain correlated in structural inference. The 2018 survey employment moment forms a separate data-source block. Estimation of the labor first step therefore precedes preference estimation, but its uncertainty is not treated as zero.

Table G3. Labor-market first-step estimates and fitted reform adjustments

A. Empirical labor regressions				
Labor outcome	Estimate	SE	Women	Cohorts
Conditional affiliation by 19 (pp; diagnostic)	14.30	1.53	34,680	44
Broad non-permanent exposure (years/decade)	0.415	0.142	27,191	42
Contribution resources (euros/year)	-535.82	249.86	26,382	42
Employee days (days/year; check)	4.26	6.24	27,026	42
Temporary share, career years 11–18 (pp)	0.37	1.14	50,657	44
T-to-P mobility, career years 11–18 (pp)	-0.52	0.88	29,949	44
Temporary share, 2018 employees (pp)	15.72	7.13	1,812	43
Temporary stock at potential entry (pp)	-0.04	0.40	60,125	44
Permanent stock at potential entry (pp)	2.02	0.43	60,125	44
Covered employee entry by 19 (pp)	13.24	0.86	60,125	44

B. Jointly fitted labor-process adjustments				
Labor-process parameter	Estimate	SE	MCVL SE	Survey SE
O-to-employee log-odds shift, ages 16–18	0.3868	0.0701	0.0701	0.0022
Early T-destination log-odds adjustment	0.1940	0.0812	0.0811	0.0037
Persistent T/P retention log-odds shift	0.0575	0.0774	0.0768	0.0097
Employed-resource log multiplier	-0.0106	0.0327	0.0327	0.0001

C. Matched labor moments and fitted counterparts				
Moment	Data	Model	SE	Residual/SE
Covered entry by 19 (pp)	13.24	13.24	0.86	0.00
Broad exposure (years/decade)	0.415	0.415	0.142	0.00
Resource-index change (euros equivalent)	-535.82	-535.82	249.86	-0.00
Temporary share, career years 11–18 (pp)	0.37	1.64	1.14	1.11
T-to-P mobility, career years 11–18 (pp)	-0.52	-0.66	0.88	-0.15
Temporary share, 2018 employees (pp)	15.72	0.69	7.13	-2.11

Notes: Panel A reports the labor-market moments used to discipline the structural employment process together with clustered standard errors and sample sizes. The first-decade estimates retain the reduced-form samples and specifications used in the main analysis; days worked is reported as an additional check rather than an estimation target. Panel C uses the appropriate observation equation for each matched target; it does not equate conditional first affiliation with covered entry. Later-career MCVL estimates use Equation (G14). Temporary-to-permanent transitions condition on a temporary origin and may involve a change of employer. The 2018 employment estimate uses official Fertility Survey weights. Panel B reports the fitted structural labor parameters and their uncertainty. These parameters are estimated in the labor first step and are not additional fertility preference parameters. *Source:* MCVL 2006, 2018 Spanish Fertility Survey, and authors' calculations.

Table G4 reports the baseline one-year transition probabilities from temporary employment that enter Equation (G10). For each age and skill group, the three columns give the probability that a woman currently in temporary employment moves outside covered employment, remains temporary, or moves into permanent employment in the following year. These probabilities determine both the future employment states and the contribution-based income paths that women anticipate when making fertility decisions.

Table G4. One-year transitions from temporary employment

Skill	Age	To outside	Remain temporary	To permanent
High	16–22	29.7%	51.0%	19.3%
High	23–29	25.8%	50.1%	24.0%
High	30–34	18.7%	66.3%	14.9%
High	35–40	13.4%	72.0%	14.5%
Low	16–22	30.8%	52.4%	16.9%
Low	23–29	31.9%	53.6%	14.5%
Low	30–34	23.2%	62.2%	14.6%
Low	35–40	19.9%	65.6%	14.5%

Notes: One-year transition probabilities estimated from adjacent MCVL person-years. Outside employment includes schooling, nonparticipation, and years without covered dependent wage employment. Each row sums to one. A transition from temporary to permanent employment may involve a change of employer. *Source:* MCVL 2006 and authors' calculations.

H. ESTIMATION, IDENTIFICATION, AND VALIDATION

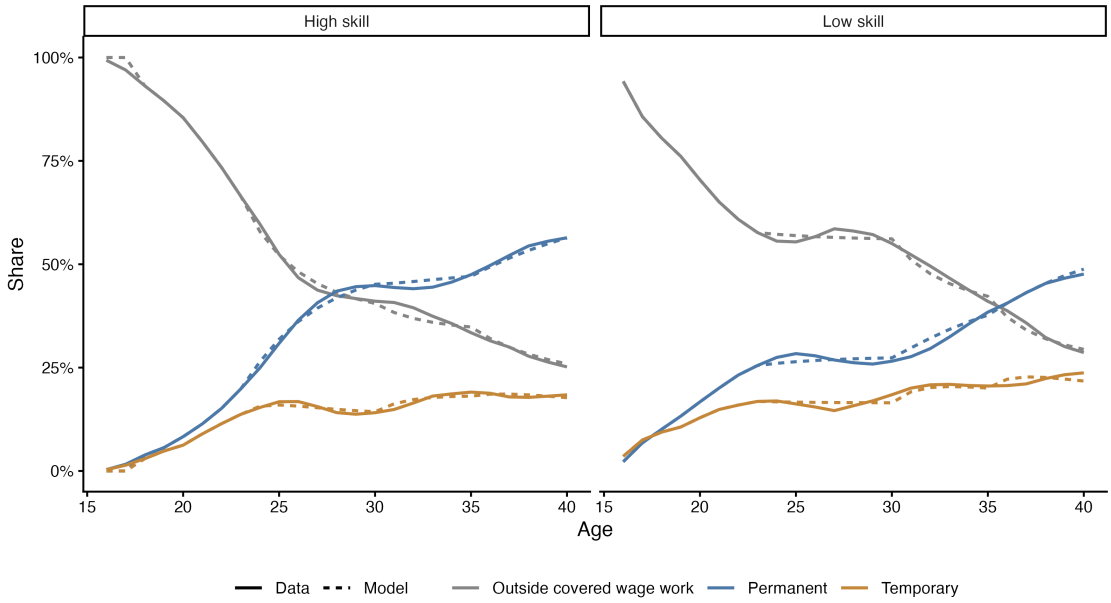
H.1. *First-step labor data and reform mapping*

The labor-market inputs are estimated before the fertility preference parameters. Pre-cutoff-cohort MCVL histories determine the distribution of women across the outside (O), temporary (T), and permanent (P) states at potential labor-market entry, the probabilities of moving across these states thereafter, and the contribution-based income associated with temporary and permanent employment. Outside-state resources are calibrated using the floor and permanent-income fraction in Appendix G; they are not estimated outside-state medians. Transitions are estimated annually at younger ages and pooled into broader age groups later in the career, where the data become thinner. The construction of these transition probabilities, including the one-half Jeffreys correction applied to each transition row, is described in Appendix G.

Figure H1 compares the resulting model-implied labor-state profiles with the corresponding pre-cutoff-cohort MCVL profiles. Low-skilled women are initialized using the observed distribution of labor states at age 16, while high-skilled women are initialized at age 18. Under the potential-entry convention, high-skilled women are assigned to the outside state at ages 16–17. The close correspondence between the empirical and model profiles shows how well the estimated transition process reproduces the labor-market histories that serve as inputs to the fertility model.[59]

[59] Because these same pre-cutoff-cohort histories are used to estimate the transition process, Figure H1 should be interpreted as a first-step fit diagnostic rather than as an independent out-of-sample test. The outside state includes schooling and nonparticipation and is classified by the main annual affiliation; an outside-state year may still contain some employee days.

Figure H1. Labor-state profiles: MCVL and model



Notes: The figure compares pre-cutoff-cohort MCVL labor-state shares with the profiles generated by the estimated transition process. Low-skilled women enter from the observed age-16 distribution and high-skilled women from the observed age-18 distribution. High-skilled women are assigned to the outside state at ages 16–17 under the model's potential-entry convention. The outside state is classified by main annual affiliation and includes schooling and nonparticipation; some outside-state years contain employee days. *Source:* MCVL 2006 and authors' calculations.

The reform changes this baseline labor process through the three margins described in Equation (11). The entry shift is disciplined by the estimated increase in the first annual temporary or permanent main affiliation between potential entry and age 19, early-career exposure is disciplined by the increase in years of non-permanent employment during the first ten potential career years, and the resource channel is disciplined by the estimated change in contribution-based income. Later MCVL employment shares and temporary-to-permanent transitions, together with the 2018 temporary-employment moment, discipline the persistence of the labor-market shock. Appendix Table G3 reports these empirical estimates and their fitted structural counterparts, while Appendix Table G4 reports the baseline transition probabilities.

The survey's first paid job and the MCVL's first annual employee main affiliation are distinct entry measures. For currently employed respondents with no previous job, the survey routes past retrospective first-job questions; their current-job start dates complete the first-entry observation. Respondents reporting no employment history are distinguished from missing dates. Recorded education codes reproduce the maintained skill mapping: MCVL lower-secondary code 31 is assigned to low skill and vocational code 32 to high skill. This differs from the reduced-form proxy partition. Generic education categories and the absence of historical attainment dates remain transport assumptions. Administrative skill is based on occupation

at delivery, which cannot be reconstructed from the survey's current occupation; matching the age support and plurality weights therefore does not remove every cross-source population difference.

These labor-market objects are fixed before the fertility preference parameters are estimated. Their joint sampling uncertainty is nevertheless carried into the structural estimation, including the covariance among the MCVL first-stage estimates. The Fertility Survey effects on completed children and motherhood and the birth-record effect on first-birth age enter separately as fertility moments and do not determine the labor-market mapping.

H.2. *Moment conditions and parameter identification*

We estimate the ten preference parameters using 18 empirical moments. Twelve moments describe fertility levels and first-birth timing by skill, three describe the relationships of desired family size, religiosity, and traditional gender norms with fertility, and three measure the fertility response to reform exposure. The Fertility Survey moments and the birth-record first-birth-age effect both include women of all nationalities. The profile sample contains 5,367 women aged 45–55; the structural reform sample contains 3,611 women in the 2018 cohort window. Official respondent weights are used, and childless women are retained. Complete biological-child birth months and years are matched to maternal birth dates to construct parity and motherhood through age 40; missing histories are not recoded as zero children. First-birth timing conditions on motherhood by that age, and the preference-gradient regressions use these same horizon-specific outcomes. Simulated profiles reproduce the survey's exposure mixture: each low-skilled respondent on the exposed side of the cutoff receives the reform labor regime, while earlier low-skilled respondents and high-skilled respondents receive the reference regime. Population moments integrate these predictions using official respondent weights; they are not predictions from the reference regime alone.[60]

Table H1 clarifies which objects are estimated jointly with fertility preferences and which enter the model from the first step or from external calibration. In particular, the labor-market process and the reform-induced changes in employment and contribution-based income are determined before preference estimation, while fertility choices and completed fertility are model outcomes.

The administrative structural target restricts first-delivery age to 16–40 while retaining the source DD design, controls, cohort clustering and infant weighting. It is +0.412 years, compared with the published broader-support estimate of +0.441 years. The restriction defines a conditional age-support counterpart; it is not a selection correction or a revision of the reduced-form estimate. For this target, model first-birth

[60] The structural estimation assumes that the timing effect from birth records can be mapped to the population represented by these survey moments. The survey targets are −0.108 capped children (SE 0.064) and −0.041 motherhood probability (SE 0.030); their estimands differ from the pooled uncapped child-count DD. The resident-at-entry diagnostic restricts the survey sample, but does not reconstruct an equivalently restricted administrative population. The reduced-form general household and preference mechanism estimates are not among the 18 fertility targets; the temporary-contract estimate enters separately through the labor-market first step.

probabilities are weighted by the pre-reform age-specific ratio of infant records to deliveries and the administrative DD cohort weights. This retains every infant from multiple first deliveries in the empirical counterpart. Survey preference distributions are transported within treated low-skill cohort quarters, and plurality conditional on age is held fixed across regimes. The corresponding sampling uncertainty is propagated jointly with the administrative target. Counterfactual tables separately report woman-level first-birth ages conditional on motherhood. The exact timing decomposition gives 0.096 years of within-type delay and -0.020 from the changing composition of mothers. Composition accounts for part of the conditional response, but not the remaining shortfall. Administrative types use occupation/activity at delivery, whereas survey types use attained education; their transport remains a maintained assumption. A same-design, education-group survey timing check is imprecise and is retained as a diagnostic rather than substituted for the administrative target.

Equation (12) chooses the parameter vector that brings the model moments as close as possible to their empirical counterparts. Because some moments are estimated from the same respondents and cohorts, we account for their sampling covariance rather than treating the 18 targets as independent. Let $\widehat{V}$ denote the covariance matrix of the empirical moments and A the matrix that combines the detailed first-birth-age cells into the broader age groups used in estimation. We construct

$$\widehat{V} = A \text{ blockdiag}\left(\widehat{V}_{FS,\text{joint}}, \widehat{\sigma}^2_{RF,BR}\right) A', \qquad \widehat{W} = (\widehat{V} + \lambda I)^{-1}, \qquad \lambda = 10^{-8}\,\text{mean}\{\text{diag}(\widehat{V})\}. \tag{H1}$$

Here $\widehat{V}_{FS,\text{joint}}$ contains the joint sampling covariance of the Fertility Survey moments, $\widehat{\sigma}^2_{RF,BR}$ is the variance of the birth-record timing effect, and $\widehat{W}$ is the weighting matrix used in the minimum-distance objective. The small regularization term λI stabilizes the numerical inversion without materially changing the relative weighting of the moments.[61]

[61] Respondents who contribute to several Fertility Survey moments remain linked when the covariance matrix is constructed, and scores are aggregated by birth-cohort quarter. Sampling errors are assumed independent across the Fertility Survey, MCVL, and birth records. We also propagate uncertainty from the MCVL labor first step, including the estimated initial shares, transition probabilities, contribution-based income profiles, and reform adjustments. A 500-draw stratified respondent bootstrap and an alternative covariance structure are retained as sensitivity checks.

Table H1. Model objects, empirical discipline, and structural role

Status	Objects	Source or discipline	Structural role
Estimated	Child and parity values; age and skill costs; temporary-exposure cost	Fertility levels, timing, parity, and reform-response moments	Baseline fertility choices and insecurity channel
Estimated heterogeneity	Desired-family-size, religiosity, and gender-norm values	2018 Fertility Survey conditional gradients	Observed preference heterogeneity; fixed across policies
First step	Employment transitions and T/P median contribution resources; calibrated O resources	MCVL pre-reform cohort annual histories	Expectations and current resources
Reduced-form mapping	Entry, early transition adjustment, persistent conversion and resource shifts	MCVL reform histories and 2018 employee-contract DD	Historical labor-market experiment; separate labor first step
Observation first step	Infant-to-delivery ratios and administrative DD projection	Frozen administrative first-delivery sufficient statistics	Map woman-level birth probabilities to infant-weighted timing
Calibrated	Discounting, birth-success schedule, child resources, outside-state resources	External evidence and normalizations	Dynamic horizon and resource accounting
Endogenous	Attempt policies, births, labor distributions under policy, completed fertility	Solved exactly from the model	Fit, mechanisms, and counterfactual outcomes

Notes: The table distinguishes preference parameters estimated with the 18 fertility moments from labor-market inputs estimated beforehand, externally calibrated objects, and endogenous model outcomes.

Identification comes from the fact that different parameters affect different features of fertility behavior. Parity moments discipline the value of additional children, first-birth timing disciplines age and career costs, differences by skill help identify the high-skill opportunity cost, and the reform moments help identify the role of accumulated temporary-employment exposure. The survey gradients separately discipline the coefficients on desired family size, religiosity, and traditional gender norms. Table H1 and the preference definitions in Table G1 provide the qualitative mapping. Table H2 quantifies local sensitivity to the three reform targets; the replication results report all 18 target columns. All ten parameters are estimated jointly and therefore generally affect more than one moment.

Table H2. Local parameter sensitivity to reform targets

Parameter	Child-count target	Motherhood target	Timing target
Birth value	+0.041	-0.070	+0.061
Second-birth shift	-0.045	+0.095	-0.011
Third-birth shift	-0.005	+0.109	-0.059
Early-age shift	+0.042	-0.036	+0.045
Late-age cost	+0.002	+0.010	-0.006
High-skill early cost	-0.032	-0.009	+0.011
Temporary-exposure cost	+0.058	-0.008	+0.090
Desired-family-size value	-0.012	-0.019	+0.016
Religiosity value	+0.006	-0.003	-0.001
Gender-norm value	-0.019	+0.007	+0.006

Notes: Each cell is the local parameter change from increasing the indicated empirical target by one of its standard errors, holding other targets, the weighting matrix and first-step primitives fixed. All ten preferences adjust jointly. The derivative includes residual curvature; the complete matrix and counterfactual sensitivities are supplied with the R results. These are scaled local derivatives: a finite one-standard-error re-estimation can encounter parameter bounds and need not equal the linear approximation.

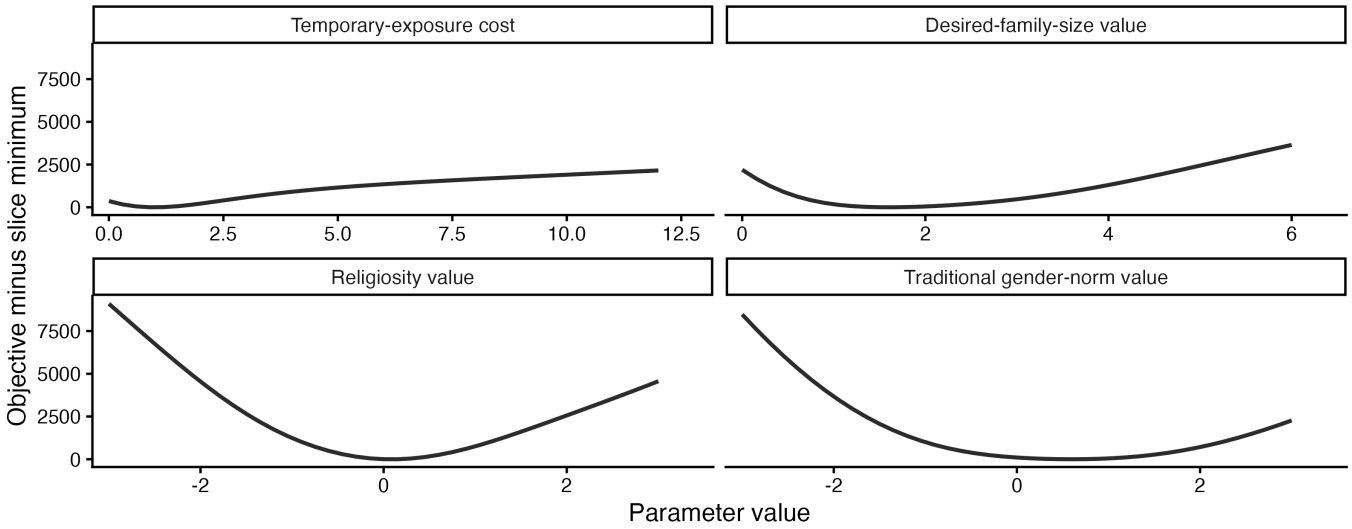
To show more directly how the empirical moments discipline the estimates, let J denote the Jacobian of the model moments with respect to the parameter vector θ, evaluated at the estimates. The local sensitivity matrix is

$$S = \left[J'\widehat{W}J + \sum_k (\widehat{W}r)_k \nabla^2 m_k \right]^{-1} J'\widehat{W}, \quad r = m(\widehat{\theta}) - \widehat{m}, \qquad J = \left. \frac{\partial m(\theta)}{\partial \theta'} \right|_{\widehat{\theta}}. \tag{H2}$$

The usual Gauss–Newton approximation omits the residual-curvature term; the code reports both and checks finite-difference step stability. An element of S measures how an estimated parameter would move locally following a small change in one empirical moment, holding the other moments fixed. Following Andrews, Gentzkow and Shapiro (2017), we use this calculation to make the link between parameters and identifying moments transparent. Larger changes in calibrated objects require solving and re-estimating the model and are therefore examined separately in the sensitivity analysis.

Figure H2 provides an additional check by varying selected parameters around their estimates while holding the remaining parameters fixed. These slices display local objective curvature, holding the other coefficients fixed. Their slopes depend on parameter units and are not a ranking of sampling precision or substitutes for the reoptimized profiles in Appendix I.

Figure H2. Objective-function slices



Notes: Each panel varies the indicated parameter around its estimate while holding the remaining nine parameters fixed. The curves are diagnostics of local curvature and are not confidence intervals.

We also verify that the numerical solution is stable across starting values and that the moments contain enough local variation to distinguish all ten parameters. Table H3 reports the objective value, Jacobian rank and conditioning, convergence diagnostics, and the multistart comparison. Estimation uses eight dispersed starting values and selects the converged solution with the lowest objective. At the reported solution, the covariance-weighted Jacobian has full rank 10, all parameters lie in the interior of their bounds, and the largest projected Gauss–Newton correction is 0.0000162.[62]

[62] The minimized objective is 80.36. We report it as a measure of overall fit alongside a local Gaussian reference that propagates joint survey, administrative-observer and labor-input influences through the same weighting matrix. The reference 95th percentile is 16.26 and the simulated reference probability is < 0.001. This is a first-order null approximation, not a full nonlinear bootstrap; numerical convergence is separate from statistical fit.

Table H3. Numerical estimation diagnostics

Diagnostic	Value
Minimum objective	80.36
Overidentifying restrictions	8
Targeted moments	18
Estimated parameters	10
Jacobian rank	10
Smallest singular value	2
Jacobian condition number	26.7
Dispersed starting values	8
Converged joint starts	8
Joint-polish convergence code	0
Maximum absolute projected score	0.00036
Maximum Gauss–Newton correction	1.6e-05
Fertility Survey bootstrap draws	500
Joint-input Gaussian cone reference p	1e-05
Joint-input Gaussian cone 95 percent critical value	16.26

Notes: The table reports numerical diagnostics for the selected minimum-distance solution. Rank and conditioning are calculated from the covariance-weighted numerical moment Jacobian.

With 18 moments and ten parameters, the model is not free to match every empirical target exactly. Table H4 therefore reports all moments used in estimation together with their model counterparts and standardized differences. The model reproduces the main fertility-level and timing patterns, while the largest remaining differences are concentrated in high-skilled first-birth timing and the magnitude of the reform response in first motherhood. The table also reports finer first-birth-age groups as additional checks; these detailed cells are not separately targeted in estimation.

Table H4. Targeted moments and detailed timing checks

Moment	Data	Model	Std. residual
Motherhood share: high skill	0.698	0.703	0.60
Mean first-birth age: high skill	28.985	28.985	-0.00
One child: high skill	0.230	0.247	2.00
Two children: high skill	0.382	0.374	-0.76
Motherhood share: low skill	0.782	0.775	-0.61
Mean first-birth age: low skill	25.787	25.696	-0.45
One child: low skill	0.237	0.224	-1.07
Two children: low skill	0.418	0.433	0.98
2018 FS desired–realized fertility gradient	0.583	0.593	0.58
2018 FS religiosity–fertility gradient	0.039	0.032	-0.51
2018 FS norms–first-birth-age gradient	-0.816	-0.835	-0.19
1984 effect: capped children	-0.108	-0.034	1.16
1984 effect: motherhood	-0.041	-0.012	0.95
1984 effect: first-delivery age	0.412	0.076	-6.28
First birth up to 25: high skill	0.241	0.227	-1.32
First birth 26–31: high skill	0.427	0.473	2.97
First birth up to 25: low skill	0.500	0.496	-0.25
First birth 26–31: low skill	0.348	0.327	-1.50
Check: First birth up to 22: high skill	0.134	0.109	-3.19
Check: First birth 23–25: high skill	0.107	0.118	1.79
Check: First birth 26–28: high skill	0.196	0.230	3.05
Check: First birth 29–31: high skill	0.231	0.243	1.18
Check: First birth 32–34: high skill	0.188	0.146	-4.09
Check: First birth up to 22: low skill	0.298	0.339	2.68
Check: First birth 23–25: low skill	0.202	0.156	-3.51
Check: First birth 26–28: low skill	0.203	0.179	-2.27
Check: First birth 29–31: low skill	0.146	0.148	0.20
Check: First birth 32–34: low skill	0.083	0.082	-0.21

Notes: Motherhood and parity moments use all women, while first-birth timing conditions on motherhood. Rows marked "Check" divide the broader first-birth-age groups into finer cells and are excluded from the minimum-distance criterion. Preference moments correspond to a one-child increase in desired family size or a one-standard-deviation increase in the cultural measures.

Table H5 examines which moment blocks discipline the timing response, following the sensitivity logic of Andrews, Gentzkow and Shapiro (2017). Each diagnostic reestimates preferences after omitting the indicated moments and then predicts all original moments. Removing baseline timing brings the reform delay closer to its target but substantially distorts the baseline profile; removing reform timing does not predict the observed delay. The joint restrictions therefore matter for the shortfall. These exercises diagnose the benchmark rather than redefine its targets.

Table H5. Moment-block reestimation and omitted-moment predictions

Estimation	Original Q	Delay (yr)	Mothers (%)	First by 25 (%)	Bounds
Empirical target	–	0.412	78.2	50.0	–
Benchmark	80.36	0.076	77.5	49.6	0
Omit reform timing	92.95	-0.002	79.0	50.2	0
Omit baseline timing	1748.00	0.314	78.2	77.2	2
Omit fertility levels	4196.37	0.306	50.1	51.0	2
Omit preference gradients	2777.95	0.152	77.1	50.8	0
Omit reform quantities	81.06	0.081	77.3	49.1	0
Omit timing and norm gradient	1456.74	0.310	78.2	76.4	2

Notes: All cases retain the same ten preference parameters, bounds, labor inputs and observation equations. Estimation uses the inverse of the retained covariance submatrix, with the maintained regularization rule. Original Q evaluates every case on all eighteen original moments and their original weighting matrix; it is not the minimized criterion for an omitted-block fit. Delay uses the administrative first-delivery observation equation. Mothers and first births by 25 describe low-skilled women; the latter conditions on motherhood. Bounds counts active parameter bounds. The combined omission removes baseline timing and the gender-norm timing gradient. Three starts are used for each diagnostic.

Table H6 complements Figure 9 in the main text by reporting the exact completed-fertility shares. It shows how closely the model reproduces the distribution across zero, one, two, and three or more children, including the childlessness margin that is central for distinguishing delayed from forgone fertility.

Table H6. Completed-fertility distribution: data and model

Skill	Children	Data	Model
High	0	30.2%	29.7%
High	1	23.0%	24.7%
High	2	38.2%	37.4%
High	3+	8.7%	8.2%
Low	0	21.8%	22.5%
Low	1	23.7%	22.4%
Low	2	41.8%	43.3%
Low	3+	12.7%	11.8%

Notes: Survey-weighted shares for women aged 45–55. The model's parity-three state represents three or more biological children.

Birth spacing provides a final check that is not used to estimate the model. Table H7 compares the observed and model-implied time between the first and second birth by skill. Because these moments are held out from estimation, they provide information about whether the annual fertility structure generates reasonable spacing without introducing additional parity-specific timing parameters. The capped mean interval is close in both groups, while mean second-birth age and the share with intervals of four years or more remain higher in the model. These are distinct checks rather than evidence that every spacing feature is matched.

Table H7. Untargeted first-to-second-birth spacing

Skill	Moment	Role	Data	Model
High	Mean spacing (years, capped at four)	Held out	3.228	3.229
High	Mean age at second birth	Held out	32.631	33.227
High	Second birth within two years	Held out	0.257	0.261
High	Second birth after four or more years	Held out	0.560	0.626
Low	Mean spacing (years, capped at four)	Held out	3.360	3.363
Low	Mean age at second birth	Held out	29.969	30.997
Low	Second birth within two years	Held out	0.205	0.215
Low	Second birth after four or more years	Held out	0.650	0.690

Notes: Data use positive differences between the reported years of first and second births in the 2018 Fertility Survey. Same-year multiple births are excluded because the annual model permits at most one birth per period. None of these moments enters the minimum-distance criterion. *Source:* 2018 Spanish Fertility Survey and authors' model calculations.

I. MECHANISMS, SENSITIVITY, AND SCOPE

I.1. *Counterfactual uncertainty and the timing mechanism*

The counterfactual estimates combine uncertainty from the preference parameters, the mapping of the 1984 reform into labor-market conditions, and the MCVL first-step estimates of employment transitions and contribution-based income. Table I1 reports confidence intervals that account jointly for these sources of uncertainty. This matters because the same cohorts and survey respondents contribute to several of the labor and fertility moments used in estimation.[63]

[63] The covariance calculation retains the dependence among MCVL labor-market estimates and among overlapping 2018 Fertility Survey moments. Numerical derivatives also allow changes in the first-step inputs to affect the re-estimated preference parameters. Sampling errors from the MCVL, Fertility Survey, and birth records are treated as independent across data sources. Inference uses a first-order approximation; Appendix H describes the full covariance construction.

Table I1. Uncertainty in the reform and local counterfactuals

Estimand	Estimate	Standard error	95% interval
Historical: completed children	-0.034	0.011	[-0.055, -0.013]
Historical: motherhood	-0.012	0.004	[-0.020, -0.004]
Historical: first-birth age	+0.072	0.032	[+0.010, +0.134]
Local 1 pp: completed children	+0.007	0.002	[+0.003, +0.011]
Local 1 pp: motherhood	+0.002	0.001	[+0.001, +0.004]
Local 1 pp: first-birth age	-0.014	0.003	[-0.021, -0.008]

Notes: Standard errors incorporate uncertainty from the estimated preference parameters, the MCVL labor-market first step, and the empirical reform mapping. The full-reform estimates allow all estimated inputs to vary, while the one-percentage-point reduction in the share of the first decade spent in non-permanent employment fixes the size of the policy change by construction.

A full nonlinear check uses 199 fixed-seed positive cohort-multiplier draws, rebuilding the labor inputs, survey targets and covariance and reestimating preferences in every draw. Both the sampling and recentered-null runs converge, giving 398 verified fits with no discarded source-weight draw. The null 95th-percentile criterion is 16.69, compared with 80.36 in the data; the plus-one bootstrap probability is 0.005. This rejects the joint fertility restrictions under the maintained source-independence assumption. The paired model-minus-target timing gap has a 95% bootstrap interval of $[-0.478, -0.239]$ years. Timing therefore remains a quantitative overidentifying tension. The historical child-count interval is $[-0.0579, -0.0075]$, and the local child-count interval is $[0.0019, 0.0103]$. These nonlinear checks support the direction of the quantity responses conditional on the specification; Table I1 separately reports first-order intervals.

Parameter standard errors and first-order counterfactual intervals use the Gauss–Newton influence approximation. The local target-sensitivity matrix also incorporates residual curvature, holding the weighting matrix and first-step primitives fixed. The nonlinear bootstrap provides a complementary comparison by re-estimating the full procedure in each draw. For the insecurity coefficient, the bootstrap standard deviation is 0.347, compared with 0.303 under the first-order approximation, and the percentile interval is $[0.253, 1.586]$. One bootstrap estimate reaches the lower bound; the percentile interval allows for the resulting asymmetry.

An alternative covariance calculation aligns birth-cohort-quarter influence contributions across datasets to allow shared cohort shocks. The insecurity standard error is 0.307, compared with 0.303 under source independence. This parameter-precision check complements the nonlinear counterfactual intervals reported above, which retain independent sampling across sources.

The uncertainty estimates describe sampling variation conditional on the maintained specification. Under the full 1984 reform exposure, the 95 percent confidence interval for the change in completed fertility is $[-0.055, -0.013]$ children, while the interval for the change in first-birth age is $[+0.010, +0.134]$ years. For the local experiment that reduces the share of the first decade spent in non-permanent employment by one percentage point, the estimated change in completed fertility has a 95 percent interval of $[+0.0034, +0.0106]$ children. These intervals reflect sampling uncertainty around the maintained structural specification rather than uncertainty across alternative model specifications.

The model also allows us to distinguish two ways in which delayed motherhood can lower completed fertility. Women may make fewer birth attempts before the end of the reproductive horizon, or they may shift attempts toward older ages at which the probability of a successful birth is lower. Table I2 uses normalized attempt-age distributions. Within each observation cell j, let $A_j^r = \sum_a A_{ja}^r$ be total expected attempts and $\bar{q}_j^r = \sum_a q(a) A_{ja}^r / A_j^r$ their average success probability. The exact identity is

$$\Delta N_j = \tfrac{1}{2}(\bar{q}_j^1 + \bar{q}_j^0)\Delta A_j + \tfrac{1}{2}(A_j^1 + A_j^0)\Delta \bar{q}_j. \tag{I1}$$

The first term is attempt volume and the second age composition. Attempts lost because postponement reaches the age-40 boundary enter volume, including when the finite horizon is the economic reason they are not recovered. The code applies the same residualized DD weights used for the child-count effect after this cell-level decomposition. A proportional fall in attempts at every age has zero age-composition contribution.

Table I2. Accounting for the completed-fertility response

Component	Δ completed children
Attempt-volume component	-0.030
Attempt-age composition component	-0.004
Total completed-fertility effect	-0.034

Notes: The volume component values the change in total attempts at the average of the two regimes' attempt-weighted success probabilities. The composition component values the change in average success probability at average attempt volume. Components are calculated before applying the child-count DD observation weights and sum exactly to the reported full-reform effect.

The model's −0.0339 change in completed fertility combines a −0.0302 attempt-volume component and a −0.0037 age-composition component. This is exact accounting within the model, not an independently identified biological effect. Births not realized by age 40 need not all be permanently forgone after that age.

I.2. *Sensitivity to calibrated assumptions*

Table I3 examines whether the model's main counterfactual conclusions depend on the calibrated assumptions governing child-related resource costs, discounting, birth success at older ages, the size of the early-career employment shock, and the persistence of employment insecurity.

Most specifications vary one assumption while holding the estimated preference parameters fixed. The rows labeled *Re-estimated* instead re-estimate all ten preference parameters under the alternative calibration using the same empirical moments. This distinction separates changes that arise mechanically from a different calibration from those that remain after the model is allowed to adjust to the data.

Table I3. Sensitivity of the full-reform fertility response

Case	Children	Motherhood (pp)	First-birth age	Criterion
Baseline	-0.034	-1.20	+0.072	80.36
Double child-resource deduction	-0.028	-1.30	+0.083	–
Discount factor 0.94	-0.035	-1.15	+0.073	–
Discount factor 0.98	-0.033	-1.24	+0.072	–
Post-35 fecundity -20%	-0.033	-1.14	+0.081	–
Post-35 fecundity +20%	-0.034	-1.25	+0.064	–
Temporary first stage: lower 95% CI	-0.019	-0.65	+0.060	–
Temporary first stage: upper 95% CI	-0.047	-1.68	+0.081	–
Insecurity cost only while currently temporary	-0.004	-0.10	+0.034	–
Age-invariant conception schedule	-0.036	-1.39	+0.046	–
Biological decline delayed three years	-0.035	-1.28	+0.063	–
Re-estimated: double child resources	-0.040	-1.55	+0.075	104.02
Re-estimated: discount 0.94	-0.033	-1.14	+0.077	84.33
Re-estimated: discount 0.98	-0.037	-1.33	+0.069	77.31
Re-estimated: current temporary status	-0.014	-0.44	+0.129	63.64
Re-estimated: history cost while temporary	-0.014	-0.45	+0.129	63.73
Re-estimated: permanent employment relieves cost	-0.029	-1.04	+0.044	85.06

Notes: Changes are relative to the corresponding no-reform low-skilled baseline. Completed children are unconditional across women, while first-birth age is conditional on motherhood. Rows labeled *Re-estimated* jointly re-estimate all ten preference parameters under the alternative calibration; the remaining rows hold preference parameters at their benchmark values.

The exercises test the main margins that are calibrated rather than directly estimated. Changing the child-resource deduction alters the resources available after a birth, while changing the discount factor changes the weight placed on future utility. Varying birth success after age 35 tests the importance of biological timing, and changing the size of the early-career employment shock tests the dependence of the results on the reform mapping. The re-estimated current-status model replaces the prospective history cost by the current temporary-state indicator, using $c_T \log(2)$ for a temporary year. Two additional restriction checks use contemporaneous full-history costs, switched off outside temporary employment or in permanent employment, respectively. These differ from the benchmark, which values expected insecurity over the infant-care window. Each re-estimates the same ten preferences against the same targets and covariance. These

checks assess the history restriction; they are not separately identified policy experiments. The first-stage confidence-endpoint rows instead recalibrate the labor mapping at the stated exposure coefficient and hold preferences fixed; they are not full confidence intervals.

The completed nonlocal checks reoptimize the remaining preferences at fixed insecurity and high-skill career-cost coefficients, using multiple starts and the unchanged targets and weighting matrix. They confirm the numerical solution while showing that a stronger insecurity coefficient cannot close the timing gap without worsening the fertility-profile fit. A nested model with separate current-temporary and prospective-history costs yields a criterion of 63.622, compared with 63.645 for current status alone. The history and current-status coefficients are 0.045 and 10.166, respectively; adding history improves the criterion by only 0.023. Thus, local full rank and a significant history coefficient within the benchmark do not establish a uniquely identified causal history contribution. Desired fertility, religiosity and gender norms remain in all specifications.

Two further reestimations isolate child-cost duration while retaining the prospective insecurity mechanism, the discount factor and the annual resource scale. A fifteen-year per-birth utility obligation gives -0.0178 children and 0.023 years of woman-level first-birth delay, with criterion 98.25. An infant/school-age resource approximation gives -0.0119 and 0.010, with criterion 101.66. The infant and school phases follow Guner, Kaya and Sánchez-Marcos (2024); a new birth resets the phase approximation. Remaining costs beyond 40 use the age-40 labor/resource law, without extending fertility choices. The first alternative capitalizes additive utility liabilities rather than exact household consumption. These checks expose the incentive from costs ending at 40; poorer fit alone does not establish the benchmark accounting as correct. The 600-euro deduction remains an effective calibrated amount in the woman's contribution-resource budget, rather than a measured total household child cost. Across these reestimated alternatives, the local exposure reduction continues to raise children and motherhood and advance first births, with differing magnitudes.

The current-status model has a lower overall criterion and smaller fertility-quantity effects. The accumulated-history mechanism is therefore a maintained, quantitatively consequential channel whose empirical support is assessed alongside these alternatives. The contemporaneous temporary-only history check reaches a parameter bound and is retained as a diagnostic.

Table I4 checks the stability of the reform and local-counterfactual responses to alternative preference samples, residence at potential entry, and a labor process distinguishing first entry from reentry. The last extension improves cumulative-entry prediction in omitted birth-quarter cohorts, with slightly weaker employment-stock fit, while leaving the direction of the fertility and local-counterfactual responses unchanged. Its smaller timing response does not support replacing the benchmark. Moreover, the returner

association changes sign after adding person effects; dynamic spell selection and initial conditions preclude reading either association as causal experience dependence.

Table I4. Population and labor-process robustness

Specification/sample	Original Q	Children	Motherhood (pp)	Delay (yr)	Local children
Benchmark	80.36	-0.034	-1.20	0.076	0.0070
Preference gradients: ages 45–49	84.58	-0.032	-1.14	0.071	0.0067
Preference gradients: ages 50–55	85.63	-0.036	-1.29	0.082	0.0076
Resident by potential entry	121.58	-0.040	-1.45	0.100	0.0083
First entry versus reentry	84.46	-0.037	-1.32	0.062	0.0069
Observed contributions: outside years	80.16	-0.034	-1.20	0.076	0.0070
Observed contributions: all states	76.02	-0.034	-1.19	0.091	0.0069
Outside resources: 2,000-euro floor	128.79	-0.036	-1.41	-0.056	0.0075

Notes: Each diagnostic reestimates the same ten preferences from two starts. The preference-age rows change only the sample for preference gradients; the residence row changes the survey profile and reform samples and rebuilds their covariance. Effects use each row's stated population; original Q evaluates its parameters on the original population, targets and covariance and is not a like-for-like ranking of native estimation criteria. The contribution rows change the income measurement mapping; the floor row changes calibrated outside resources in utility and recalibrates the resource shift. Each re-estimates the same ten preferences using the original targets and covariance. The entry/reentry extension splits the outside state by prior annual employee main affiliation, adds no fertility preference coefficient, and retains the original fertility targets, first-decade exposure restriction and resource mapping. Delay is the matched administrative response, not the woman-level survey timing counterfactual. Local children is the response to the maintained local reduction in exposure. For the entry/reentry extension, joint first-order 95 percent intervals are [-0.0589, -0.0145] for historical children and [0.0034, 0.0104] for local children, propagating the additional transition-row uncertainty.

The contribution-measure checks retain the benchmark resource profiles in utility and re-estimate the ten preferences after recalibrating the within-state resource shift. Substituting observed contributions in outside years gives -0.0339 children and 0.0763 years of matched first-delivery delay; using observed contributions in all states gives -0.0339 and 0.0914, respectively. The criterion under the all-state contribution mapping is 76.02, and its quantity response remains close to the benchmark. Both alternatives transport observed contribution marks with pre-cutoff labor histories. Together with the benchmark resource index, they show how the measurement of resources affects fertility timing and the estimated employment channels.

A separate outside-resource check retains only the existing €2,000 floor, removing the permanent-income fraction, and re-estimates preferences. Its matched timing effect is -0.0563 years and its criterion is 128.79. This comparison illustrates the role of resources during nonemployment in the timing decision. Across these resource calibrations, the local reduction in temporary exposure continues to increase children by age 40. Each specification uses the same ten preferences, fertility targets and weighting matrix.

An untargeted first-job prediction exercise compares the history and current-status specifications for the same 1038 women. History has a slightly lower mean squared prediction error (0.04216 versus 0.04262), while current status better matches the average birth rate (5.03 percent, compared with 6.04 percent under history

and 4.18 percent observed). This is within-sample predictive evidence from selected reported first-job spells, assuming a constant recorded contract within each spell; it does not independently identify a causal history cost. Auxiliary checks of attachment weights, contract codes and child-resource units support the stated measurement definitions without identifying continuous reform exposure, historical missing contracts or exogenous late-life preferences. Pre-cutoff reweighting and the parity-three birth restriction remain enforced in the code.

The counterfactuals should be interpreted as partial-equilibrium changes in the employment conditions faced by women, holding preferences and the broader economic environment fixed. The model does not allow fertility decisions to alter subsequent employment transitions, nor does it model partner employment, household bargaining, or firms' responses to changes in contract regulation. The exercises therefore quantify how fertility responds to changes in early-career employment conditions through the mechanisms disciplined by the data, rather than providing a welfare or general-equilibrium evaluation of alternative labor-market policies.